

Trigonometry

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5.1 Introduction

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Trigonometry deals with the relationship between the angles and sides of a triangle. We will learn about trigonometric ratios in right-angled triangles, which form the basis of trigonometry.

There are many applications of trigonometry. Of particular value is the technique of triangulation, which is used in astronomy to measure the distances to nearby stars, in geography to measure distances between landmarks, and in satellite navigation systems. GPS (the global positioning system) would not be possible without trigonometry. Other fields which make use of trigonometry include acoustics, optics, analysis of financial markets, electronics, probability theory, statistics, biology, medical imaging (CAT scans and ultrasound), chemistry, cryptology, meteorology, oceanography, land surveying, architecture, phonetics, engineering, computer graphics and game development.



Figure 5.1: An artist's depiction of a GPS satellite orbiting the Earth. There are at least 24 GPS satellites operational at any one time. GPS uses an application of trigonometry, known as triangulation, to determine one's position. The accuracy of GPS is to within 15 metres.

VISIT:

The following video covers a brief history of trigonometry and some of the uses of trigonometry.

▶ See video: 2FN7 at www.everythingmaths.co.za

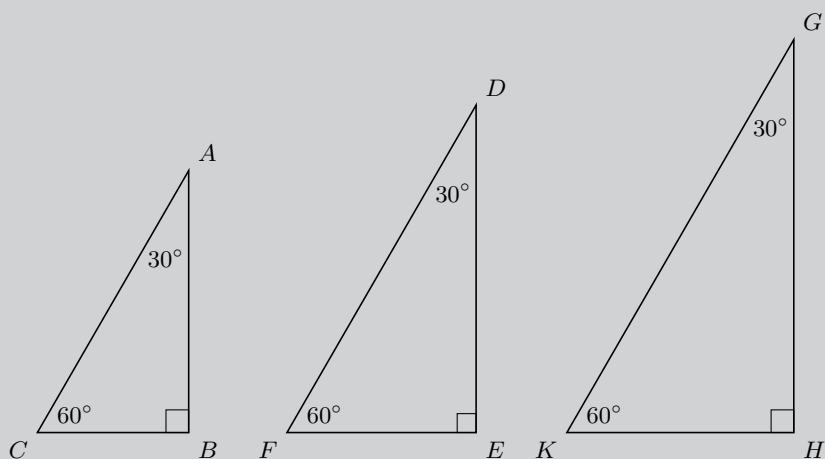
5.2 Similarity of triangles

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Before we delve into the theory of trigonometry, complete the following investigation to get a better understanding of the foundation of trigonometry.

Investigation: Ratios of similar triangles

Draw three similar triangles of different sizes using a protractor and a ruler, with each triangle having interior angles equal to 30° , 90° and 60° as shown below. Measure the angles and lengths accurately in order to fill in the table (leave your answers as a simplified fraction):

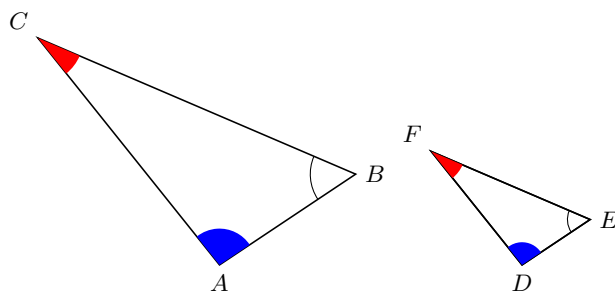


Dividing lengths of sides (ratios)		
$\frac{AB}{BC} =$	$\frac{AB}{AC} =$	$\frac{CB}{AC} =$
$\frac{DE}{EF} =$	$\frac{DE}{DF} =$	$\frac{FE}{DF} =$
$\frac{GH}{HK} =$	$\frac{GH}{GK} =$	$\frac{KH}{GK} =$

What observations can you make about the ratios of the sides?

Have you noticed that it does not matter what the lengths of the sides of the triangles are, if the angle remains constant, the ratio of the sides will always yield the same answer?

In the triangles below, $\triangle ABC$ is similar to $\triangle DEF$. This is written as: $\triangle ABC \sim \triangle DEF$



In similar triangles, it is possible to deduce ratios between corresponding sides:

$$\begin{aligned} \frac{AB}{BC} &= \frac{DE}{EF} \\ \frac{AB}{AC} &= \frac{DE}{DF} \\ \frac{BC}{AC} &= \frac{EF}{DF} \\ \frac{AB}{DE} &= \frac{BC}{EF} = \frac{AC}{DF} \end{aligned}$$

Another important fact about similar triangles ABC and DEF is that the angle at vertex A is equal to the angle at vertex D , the angle at vertex B is equal to the angle at vertex E , and the angle at vertex C is equal to the angle at vertex F .

$$\hat{A} = \hat{D}$$

$$\hat{B} = \hat{E}$$

$$\hat{C} = \hat{F}$$

NOTE:

The order of letters for similar triangles is very important. Always label similar triangles in corresponding order. For example,

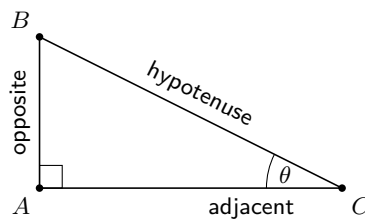
$\triangle ABC \parallel \triangle DEF$ is correct; but

$\triangle ABC \parallel \triangle DFE$ is incorrect.

5.3 Defining the trigonometric ratios

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The ratios of similar triangles are used to define the trigonometric ratios. Consider a right-angled triangle ABC with an angle marked θ (said 'theta').



In a right-angled triangle, we refer to the three sides according to how they are placed in relation to the angle θ . The side opposite to the right-angle is labelled the hypotenuse, the side opposite θ is labelled "opposite", the side next to θ is labelled "adjacent".

You can choose either non-90° internal angle and then define the adjacent and opposite sides accordingly. However, the hypotenuse remains the same regardless of which internal angle you are referring to because it is *always* opposite the right-angle and *always* the longest side.

We define the trigonometric ratios: sine (sin), cosine (cos) and tangent (tan), of an angle, as follows:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

These ratios, also known as trigonometric identities, relate the lengths of the sides of a right-angled triangle to its interior angles. These three ratios form the basis of trigonometry.

IMPORTANT!

The definitions of opposite, adjacent and hypotenuse are only applicable when working with right-angled triangles! Always check to make sure your triangle has a right-angle before you use them, otherwise you will get the wrong answer.

You may also hear people saying “Soh Cah Toa”. This is a mnemonic technique for remembering the trigonometric ratios:

$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$
$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

VISIT:

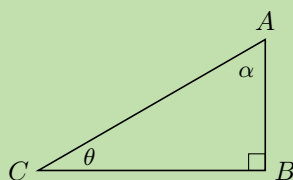
The following video introduces the three trigonometric ratios.

🎥 See video: [2FN8](#) at www.everythingmaths.co.za

Worked example 1: Trigonometric ratios

QUESTION

Given the following triangle:



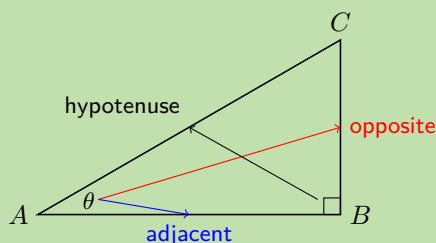
- Label the hypotenuse, opposite and adjacent sides of the triangle with respect to θ .
- State which sides of the triangle you would use to find $\sin \theta$, $\cos \theta$ and $\tan \theta$.
- Label the hypotenuse, opposite and adjacent sides of the triangle with respect to α .
- State which sides of the triangle you would use to find $\sin \alpha$, $\cos \alpha$ and $\tan \alpha$.

SOLUTION

Step 1: Label the triangle

First find the right angle, the hypotenuse is **always** directly opposite the right angle. The hypotenuse never changes position, it is always directly opposite the right angle and so we find this first.

The opposite and adjacent sides depend on the angle we are looking at. The opposite side relative to the angle θ is directly opposite (as the word opposite suggests) the angle θ . Finally the adjacent side is the remaining side of the triangle and must be one of the sides that forms the angle θ .



Step 2: Complete the trigonometric ratios

Now we can complete the trigonometric ratios for θ :

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{CB}{AC}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{AB}{AC}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{CB}{AB}$$

Therefore to find $\sin \theta$ we would use sides CB (opposite side to θ) and AC (hypotenuse). To find $\cos \theta$ we would use sides AB (adjacent side to θ) and AC (hypotenuse). To find $\tan \theta$ we would use sides CB (opposite side to θ) and AB (adjacent side to θ).

And then we can complete the trigonometric ratios for α . For angle α the opposite and adjacent sides switch places (redraw the triangle above to help you see this). Notice how the hypotenuse is still AC .

$$\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{AB}{AC}$$

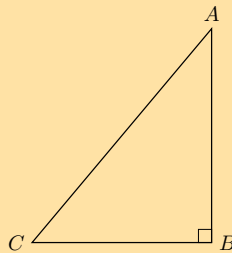
$$\cos \alpha = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{CB}{AC}$$

$$\tan \alpha = \frac{\text{opposite}}{\text{adjacent}} = \frac{AB}{CB}$$

Therefore to find $\sin \alpha$ we would use sides AB (opposite side to α) and AC (hypotenuse). To find $\cos \alpha$ we would use sides CB (adjacent side to α) and AC (hypotenuse). To find $\tan \alpha$ we would use sides AB (opposite side to α) and CB (adjacent side to α).

Exercise 5 – 1:

1. Complete each of the following:



a) $\sin \hat{A} =$

b) $\cos \hat{A} =$

c) $\tan \hat{A} =$

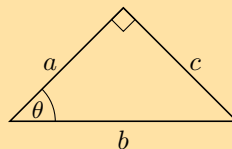
d) $\sin \hat{C} =$

e) $\cos \hat{C} =$

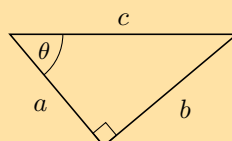
f) $\tan \hat{C} =$

2. In each of the following triangles, state whether a , b and c are the hypotenuse, opposite or adjacent sides of the triangle with respect to θ .

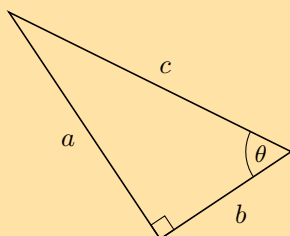
a)



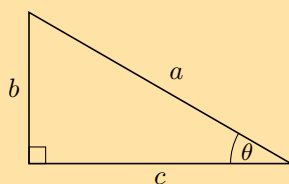
b)



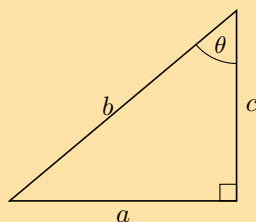
c)



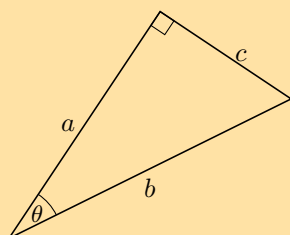
d)



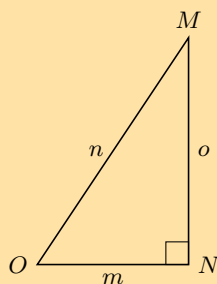
e)



f)



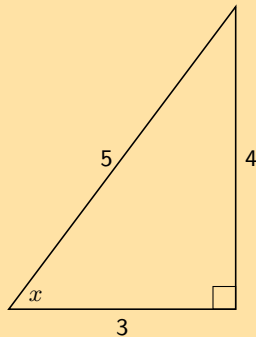
3. Consider the following diagram:



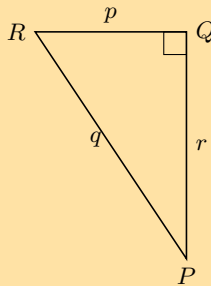
Without using a calculator, answer each of the following questions.

- Write down $\cos \hat{O}$ in terms of m , n and o .
- Write down $\tan \hat{M}$ in terms of m , n and o .
- Write down $\sin \hat{O}$ in terms of m , n and o .
- Write down $\cos \hat{M}$ in terms of m , n and o .

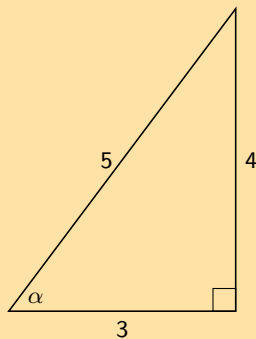
4. Find x in the diagram in three different ways. You do not need to calculate the value of x , just write down the appropriate ratio for x .



5. Which of these statements is true about $\triangle PQR$?

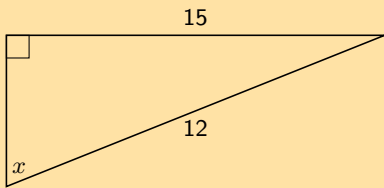


- a) $\sin \hat{R} = \frac{p}{q}$ b) $\tan \hat{Q} = \frac{r}{p}$ c) $\cos \hat{P} = \frac{r}{q}$ d) $\sin \hat{P} = \frac{p}{r}$
6. Sarah wants to find the value of α in the triangle below. Which statement is a correct line of working?

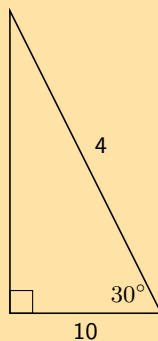


- a) $\sin \alpha = \frac{4}{5}$ b) $\cos \left(\frac{3}{5} \right) = \alpha$ c) $\tan \alpha = \frac{5}{4}$ d) $\cos 0,8 = \alpha$
7. Explain what is wrong with each of the following diagrams.

a)



b)



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1. [2FN9](#) 2a. [2FNB](#) 2b. [2FNC](#) 2c. [2FND](#) 2d. [2FNF](#) 2e. [2FNG](#) 2f. [2FNH](#) 3a. [2FNJ](#)
 3b. [2FNK](#) 3c. [2FNM](#) 3d. [2FNN](#) 4. [2FNP](#) 5. [2FNQ](#) 6. [2FNR](#) 7a. [2FNS](#) 7b. [2FNT](#)



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5.4 Reciprocal ratios

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Each of the three trigonometric ratios has a reciprocal. The reciprocals: cosecant (cosec), secant (sec) and cotangent (cot), are defined as follows:

$$\begin{aligned}\operatorname{cosec} \theta &= \frac{1}{\sin \theta} \\ \sec \theta &= \frac{1}{\cos \theta} \\ \cot \theta &= \frac{1}{\tan \theta}\end{aligned}$$

We can also define these reciprocals for any right-angled triangle:

$$\begin{aligned}\operatorname{cosec} \theta &= \frac{\text{hypotenuse}}{\text{opposite}} \\ \sec \theta &= \frac{\text{hypotenuse}}{\text{adjacent}} \\ \cot \theta &= \frac{\text{adjacent}}{\text{opposite}}\end{aligned}$$

Note that:

$$\begin{aligned}\sin \theta \times \operatorname{cosec} \theta &= 1 \\ \cos \theta \times \sec \theta &= 1 \\ \tan \theta \times \cot \theta &= 1\end{aligned}$$

VISIT:

This video covers the three reciprocal ratios for sin, cos and tan.

▶ See video: [2FNV](#) at www.everythingmaths.co.za

NOTE:

You may see cosecant abbreviated as csc.

In this section we will look at using a calculator to determine the values of the trigonometric ratios for any angle. For example we might want to know what the value of $\sin 55^\circ$ is or what the value of $\sec 34^\circ$ is.

When doing calculations involving the reciprocal ratios you need to convert the reciprocal ratio to one of the standard trigonometric ratios: $\sin$, $\cos$ and $\tan$ as this is the only way to calculate these ratios on your calculator.

IMPORTANT!

Most scientific calculators are quite similar but these steps might differ depending on the calculator you use. Make sure your calculator is in “degrees” mode.

NOTE:

Note that $\sin^2 \theta = (\sin \theta)^2$. This also applies for the other trigonometric ratios.

Worked example 2: Using your calculator

QUESTION

Use your calculator to calculate the following (correct to 2 decimal places):

1. $\cos 48^\circ$
2. $2 \sin 35^\circ$
3. $\tan^2 81^\circ$
4. $3 \sin^2 72^\circ$
5. $\frac{1}{4} \cos 27^\circ$
6. $\frac{5}{6} \tan 34^\circ$
7. $\sec 34^\circ$
8. $\cot 49^\circ$

SOLUTION

Step 1:

The following shows the keys to press on a Casio calculator. Other calculators work in a similar way. On a Casio calculator (is automatically added after pressing $\sin$, $\cos$ and $\tan$ so you just need to press $)$ after typing in the angle to close the brackets.

1. Press $\boxed{\cos} \boxed{48} \boxed{)} \boxed{=}$ 0,66913... $\approx 0,67$
2. Press $\boxed{2} \boxed{\sin} \boxed{35} \boxed{)} \boxed{=}$ 1,147152... $\approx 1,15$
3. Press $\boxed{(} \boxed{\tan} \boxed{81} \boxed{)} \boxed{)} \boxed{x^2} \boxed{=}$ 39,86345... $\approx 39,86$
OR
Press $\boxed{\tan} \boxed{81} \boxed{)} \boxed{=}$ $\boxed{\text{ANS}}$ $\boxed{x^2}$ $\boxed{\text{ANS}}$ $\boxed{=}$ 39,86345... $\approx 39,86$
4. Press $\boxed{3} \boxed{(} \boxed{\sin} \boxed{72} \boxed{)} \boxed{)} \boxed{x^2} \boxed{=}$ 2,71352... $\approx 2,71$
OR
Press $\boxed{\sin} \boxed{72} \boxed{)} \boxed{=}$ $\boxed{\text{ANS}}$ $\boxed{x^2} \boxed{=}$ $\boxed{\text{ANS}}$ $\boxed{\times} \boxed{3}$

$$y) \sqrt{4 \sec 99^\circ}$$

$$z) \sqrt{\frac{\cot 103^\circ + \sin 1090^\circ}{\sec 10^\circ + 5}}$$

2. If $x = 39^\circ$ and $y = 21^\circ$, use a calculator to determine whether the following statements are true or false:

a) $\cos x + 2 \cos x = 3 \cos x$

b) $\cos 2y = \cos y + \cos y$

c) $\tan x = \frac{\sin x}{\cos x}$

d) $\cos(x + y) = \cos x + \cos y$

3. Solve for x in $5^{\tan x} = 125$.

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1a. [2FNV](#)

1b. [2FNX](#)

1c. [2FNY](#)

1d. [2FNZ](#)

1e. [2FP2](#)

1f. [2FP3](#)

1g. [2FP4](#)

1h. [2FP5](#)

1i. [2FP6](#)

1j. [2FP7](#)

1k. [2FP8](#)

1l. [2FP9](#)

1m. [2FPB](#)

1n. [2FPC](#)

1o. [2FPD](#)

1p. [2FPF](#)

1q. [2FPG](#)

1r. [2FPH](#)

1s. [2FPJ](#)

1t. [2FPK](#)

1u. [2FPM](#)

1v. [2FPN](#)

1w. [2FPP](#)

1x. [2FPQ](#)

1y. [2FPR](#)

1z. [2FPS](#)

2. [2FPT](#)

3. [2FPV](#)



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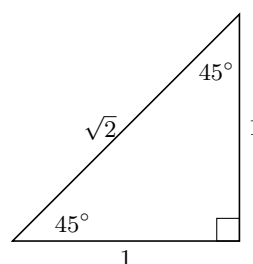
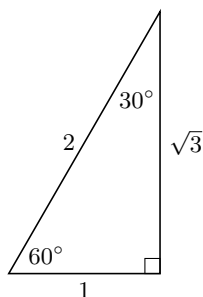
5.6 Special angles

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For most angles we need a calculator to calculate the values of $\sin$, $\cos$ and $\tan$. However, there are some angles we can easily work out the values for without a calculator as they produce simple ratios. The values of the trigonometric ratios for these special angles, as well as the triangles from which they are derived, are shown below.

NOTE:

Remember that the lengths of the sides of a right-angled triangle must obey the Theorem of Pythagoras: the square of the hypotenuse equals the sum of the squares of the two other sides.



θ	30°	45°	60°
$\cos \theta$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
$\sin \theta$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
$\tan \theta$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

These values are useful when we need to solve a problem involving trigonometric ratios without using a calculator.

Exercise 5 – 3:

1. For each expression select the closest answer from the list provided:

a) $\cos 45^\circ$

$$\frac{1}{2} \quad 1 \quad \sqrt{2} \quad \frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{2}}$$

b) $\sin 45^\circ$

$$\sqrt{2} \quad \frac{1}{2} \quad \frac{1}{\sqrt{2}} \quad \frac{\sqrt{3}}{1} \quad 1$$

c) $\tan 30^\circ$

$$\frac{1}{2} \quad \frac{\sqrt{3}}{1} \quad \frac{1}{\sqrt{2}} \quad \frac{\sqrt{3}}{2} \quad \frac{1}{\sqrt{3}}$$

d) $\tan 60^\circ$

$$\frac{\sqrt{3}}{2} \quad \frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{3}} \quad \frac{\sqrt{3}}{1} \quad \frac{1}{1}$$

e) $\cos 45^\circ$

$$\frac{\sqrt{3}}{2} \quad \frac{1}{2} \quad \frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{2}} \quad \sqrt{2}$$

f) $\tan 30^\circ$

$$\frac{1}{\sqrt{2}} \quad \frac{1}{2} \quad \frac{\sqrt{3}}{2} \quad \frac{1}{1} \quad \frac{1}{\sqrt{3}}$$

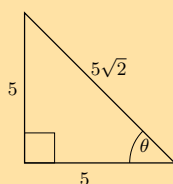
g) $\tan 30^\circ$

$$\frac{1}{\sqrt{3}} \quad \frac{1}{2} \quad \frac{1}{\sqrt{2}} \quad \frac{\sqrt{3}}{2} \quad \frac{\sqrt{3}}{1}$$

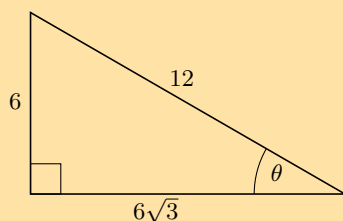
h) $\cos 60^\circ$

$$\frac{1}{\sqrt{3}} \quad \frac{1}{2} \quad \frac{\sqrt{3}}{2} \quad \frac{\sqrt{3}}{1} \quad \frac{1}{\sqrt{2}}$$

2. Solve for $\cos \theta$ in the following triangle, in surd form:



3. Solve for $\tan \theta$ in the following triangle, in surd form:



4. Calculate the value of the following without using a calculator:

a) $\sin 45^\circ \times \cos 45^\circ$

b) $\cos 60^\circ + \tan 45^\circ$

c) $\sin 60^\circ - \cos 60^\circ$

5. Evaluate the following without using a calculator. Select the closest answer from the list provided.

a) $\tan 45^\circ \div \sin 60^\circ$

$$\frac{2}{\sqrt{3}} \quad \frac{\sqrt{3}}{1} \quad \frac{\sqrt{2}}{\sqrt{3}} \quad \frac{1}{1} \quad \frac{1}{2}$$

b) $\tan 30^\circ - \sin 60^\circ$

$$0 \quad \frac{1}{2} \quad -\frac{\sqrt{3}}{2} \quad -\frac{1}{2\sqrt{3}} \quad \frac{\sqrt{3}}{2}$$

c) $\sin 30^\circ - \tan 45^\circ - \sin 30^\circ$

$$-\frac{\sqrt{3}}{2} \quad -1 \quad -\frac{1}{\sqrt{3}} \quad -\frac{\sqrt{3}}{1} \quad -\frac{7}{2\sqrt{3}}$$

d) $\tan 30^\circ \div \tan 30^\circ \div \sin 45^\circ$

$$\frac{\sqrt{3}}{1} \quad \frac{2\sqrt{3}}{\sqrt{2}} \quad \frac{2}{\sqrt{3}} \quad \frac{\sqrt{2}}{1} \quad \frac{2\sqrt{2}}{\sqrt{3}}$$

e) $\sin 45^\circ \div \sin 30^\circ \div \cos 45^\circ$

$$\frac{\sqrt{2}}{\sqrt{3}} \quad \frac{1}{\sqrt{2}} \quad \frac{4}{\sqrt{3}} \quad 2 \quad \frac{2\sqrt{2}}{\sqrt{3}}$$

f) $\tan 60^\circ - \tan 60^\circ - \sin 60^\circ$

$$-\frac{1}{\sqrt{3}} \quad -\frac{1}{2} \quad -\frac{1}{\sqrt{2}} \quad -\frac{1}{1} \quad -\frac{\sqrt{3}}{2}$$

g) $\cos 45^\circ - \sin 60^\circ - \sin 45^\circ$

$$-\frac{1}{2} \quad -\frac{1}{\sqrt{2}} \quad -\frac{7}{2\sqrt{3}} \quad -\frac{\sqrt{3}}{2} \quad -\frac{1}{\sqrt{3}}$$

6. Use special angles to show that:

a) $\frac{\sin 60^\circ}{\cos 60^\circ} = \tan 60^\circ$

b) $\sin^2 45^\circ + \cos^2 45^\circ = 1$

c) $\cos 30^\circ = \sqrt{1 - \sin^2 30^\circ}$

7. Use the definitions of the trigonometric ratios to answer the following questions:

- Explain why $\sin \alpha \leq 1$ for all values of α .
- Explain why $\cos \alpha$ has a maximum value of 1.
- Is there a maximum value for $\tan \alpha$?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1a. 2FPW | 1b. 2FPX | 1c. 2FPY | 1d. 2FPZ | 1e. 2FQ2 | 1f. 2FQ3 | 1g. 2FQ4 | 1h. 2FQ5 |
| 2. 2FQ6 | 3. 2FQ7 | 4a. 2FQ8 | 4b. 2FQ9 | 4c. 2FQB | 5a. 2FQC | 5b. 2FQD | 5c. 2FQF |
| 5d. 2FQG | 5e. 2FQH | 5f. 2FQJ | 5g. 2FQK | 6a. 2FQM | 6b. 2FQN | 6c. 2FQP | 7a. 2FQQ |
| 7b. 2FQR | 7c. 2FQS | | | | | | |



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5.7 Solving trigonometric equations

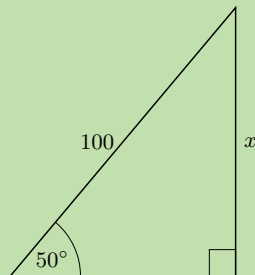
EMA3T

In this section we will first look at finding unknown lengths in right-angled triangles and then we will look at finding unknown angles in right-angled triangles. Finally we will look at how to solve more general trigonometric equations.

From the definitions of the trigonometric ratios and what we have learnt about determining the values of these ratios for any angle we can now use this to help us find unknown lengths in right-angled triangles. The following worked examples will show you how.

Worked example 4: Finding lengths**QUESTION**

Find the length of x in the following right-angled triangle using the appropriate trigonometric ratio (round your answer to two decimal places).

**SOLUTION****Step 1: Identify the opposite and adjacent sides and the hypotenuse with reference to the given angle**

Remember that the hypotenuse side is **always** opposite the right angle, it never changes position. The opposite side is opposite the angle we are interested in and the adjacent side is the remaining side.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$
$$\sin 50^\circ = \frac{x}{100}$$

Step 2: Rearrange the equation to solve for x

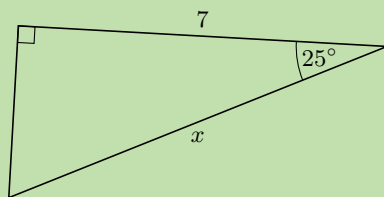
$$\sin 50^\circ \times 100 = \frac{x}{100} \times 100$$
$$\sin 50^\circ \times 100 = x$$
$$x = 100 \sin 50^\circ$$

Step 3: Use your calculator to find the answer

$$x = 76,60444\dots$$
$$x \approx 76,60$$

Worked example 5: Finding lengths**QUESTION**

Find the length of x in the following right-angled triangle using the appropriate trigonometric ratio (round your answer to two decimal places).



SOLUTION

Step 1: Identify the opposite and adjacent sides and the hypotenuse with reference to the given angle

Remember that the hypotenuse side is **always** opposite the right angle, it never changes position. The opposite side is opposite the angle we are interested in and the adjacent side is the remaining side.

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\cos 25^\circ = \frac{7}{x}$$

Step 2: Rearrange the equation to solve for x

$$\cos 25^\circ \times x = \frac{7}{x} \times x \quad \text{multiply both sides by } x$$

$$x \cos 25^\circ = 7$$

$$\frac{x \cos 25^\circ}{\cos 25^\circ} = \frac{7}{\cos 25^\circ} \quad \text{divide both sides by } \cos 25^\circ$$

$$x = \frac{7}{\cos 25^\circ}$$

Step 3: Use your calculator to find the answer

$$x = \frac{7}{0,90630...}$$

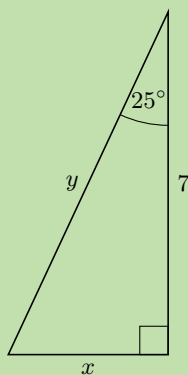
$$= 7,723645...$$

$$\approx 7,72$$

Worked example 6: Finding lengths

QUESTION

Find the length of x and y in the following right-angled triangle using the appropriate trigonometric ratio (round your answers to two decimal places).



SOLUTION

Step 1: Identify the opposite and adjacent sides and the hypotenuse with reference to the given angle

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\tan 25^\circ = \frac{x}{7}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\cos 25^\circ = \frac{7}{y}$$

Step 2: Rearrange the equations to solve for x and y

$$x = 7 \times \tan 25^\circ$$

$$y = \frac{7}{\cos 25^\circ}$$

Step 3: Use your calculator to find the answers

$$x = 3,26415\dots$$

$$x \approx 3,26$$

$$y = 7,72364\dots$$

$$y \approx 7,72$$

VISIT:

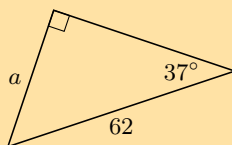
The following video shows an example of finding unknown lengths in a triangle using the trigonometric ratios.

► See video: [2FQT](https://www.youtube.com/watch?v=2FQT) at www.everythingmaths.co.za

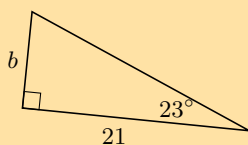
Exercise 5 – 4:

1. In each triangle find the length of the side marked with a letter. Give your answers correct to 2 decimal places.

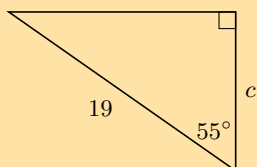
a)



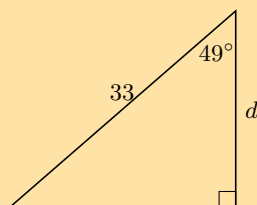
b)



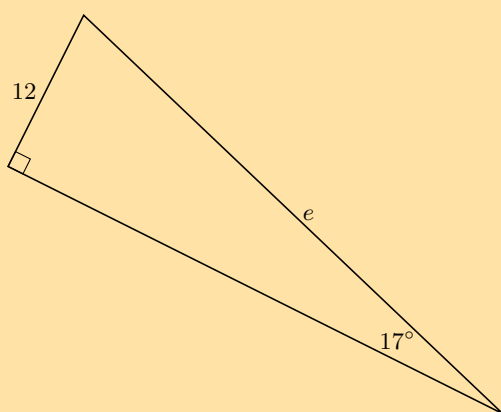
c)



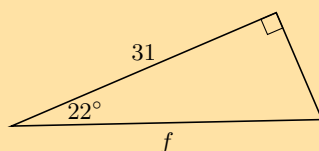
d)



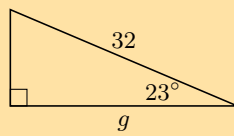
e)



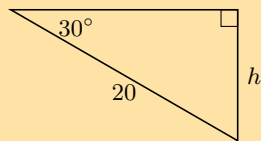
f)



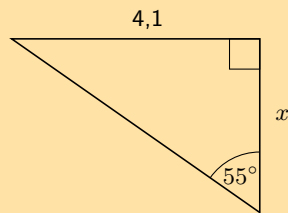
g)



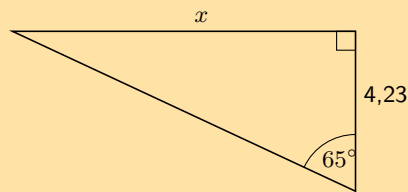
h)



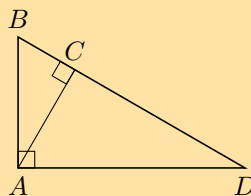
i)



j)



2. Write down two ratios for each of the following in terms of the sides: AB ; BC ; BD ; AD ; DC and AC .



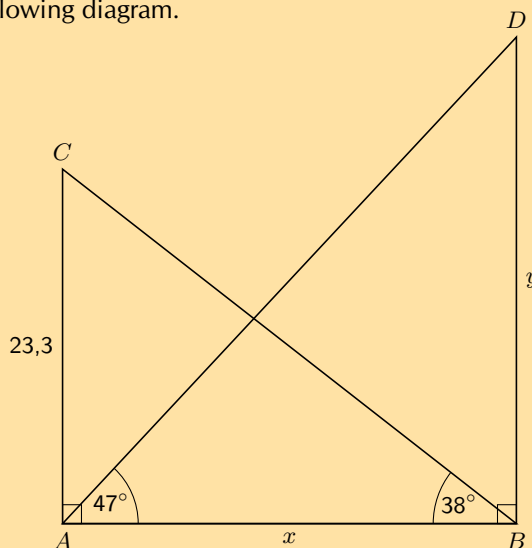
a) $\sin \hat{B}$

b) $\cos \hat{D}$

c) $\tan \hat{B}$

3. In $\triangle MNP$, $\hat{N} = 90^\circ$, $MP = 20$ and $\hat{P} = 40^\circ$. Calculate NP and MN (correct to 2 decimal places).

4. Calculate x and y in the following diagram.



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- 1a. [2FQV](#) 1b. [2FQW](#) 1c. [2FQX](#) 1d. [2FQY](#) 1e. [2FQZ](#) 1f. [2FR2](#) 1g. [2FR3](#)
 1h. [2FR4](#) 1i. [2FR5](#) 1j. [2FR6](#) 2. [2FR7](#) 3. [2FR8](#) 4. [2FR9](#)



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Finding an angle

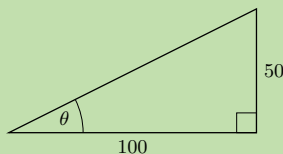
EMA3W

If the length of two sides of a triangle are known, the angles can be calculated using trigonometric ratios. In this section, we are finding angles inside right-angled triangles using the ratios of the sides.

Worked example 7: Finding angles

QUESTION

Find the value of θ in the following right-angled triangle using the appropriate trigonometric ratio.



SOLUTION

Step 1: Identify the opposite and adjacent sides with reference to the given angle and the hypotenuse

In this case you have the opposite side and the adjacent side for angle θ .

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\tan \theta = \frac{50}{100}$$

Step 2: Use your calculator to solve for θ

To solve for θ , you will need to use the inverse tangent function on your calculator. This works backwards by using the ratio of the sides to determine the angle which resulted in that ratio.

Press $\boxed{\text{SHIFT}} \boxed{\tan} \boxed{50} \boxed{\div} \boxed{100} \boxed{)} \boxed{=} 26,56505... \approx 26,6$

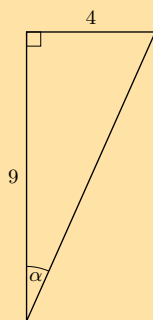
Step 3: Write the final answer

$$\theta \approx 26,6^\circ$$

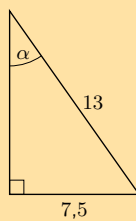
Exercise 5 – 5:

Determine α in the following right-angled triangles:

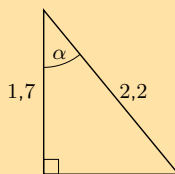
1.



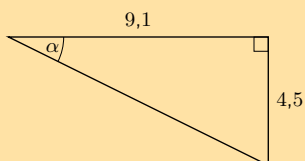
2.



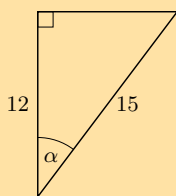
3.



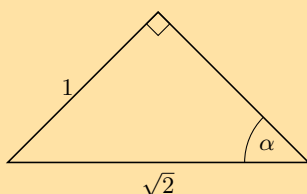
4.



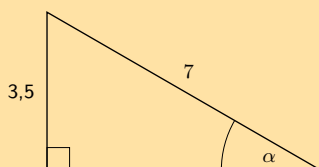
5.



6.



7.



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1. 2FRB 2. 2FRC 3. 2FRD 4. 2FRF 5. 2FRG 6. 2FRH 7. 2FRJ



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We have now seen how to solve trigonometric equations in right-angled triangles. We can use the same techniques to help us solve trigonometric equations when the triangle is not shown.

Worked example 8: Solving trigonometric equations

QUESTION

Find the value of θ if $\cos \theta = 0,2$.

SOLUTION

Step 1: Use your calculator to solve for θ

To solve for θ , you will need to use the inverse cosine function on your calculator. This works backwards by using the ratio of the sides to determine the angle which resulted in that ratio.

Press        $78,46304 \approx 78,46$

Step 2: Write the final answer

$\theta \approx 78,46^\circ$

Worked example 9: Solving trigonometric equations

QUESTION

Find the value of θ if $3 \sin \theta = 2,4$.

SOLUTION

Step 1: Rearrange the equation

We need to rearrange the equation so that $\sin \theta$ is on one side of the equation.

$$3 \sin \theta = 2,4$$

$$\sin \theta = \frac{2,4}{3}$$

Step 2: Use your calculator to solve for θ

To solve for θ , you will need to use the inverse sine function on your calculator. This works backwards by using the ratio of the sides to determine the angle which resulted in that ratio.

Press $\boxed{\text{SHIFT}} \boxed{\sin} \boxed{(} \boxed{2} \boxed{.} \boxed{4} \boxed{\div} \boxed{3} \boxed{)} \boxed{=}$ 53,1301... $\approx 53,13$.

Step 3: Write the final answer

$$\theta \approx 53,13^\circ$$

NOTE:

When you are solving trigonometric equations you might find that you get an error when you try to calculate $\sin$ or $\cos$ (remember that both the sine and cosine functions have a maximum value of 1). For these cases there is no solution to the equation.

Worked example 10: Solving trigonometric equations

QUESTION

Solve for α : $3 \sec \alpha = 1,4$.

SOLUTION

Step 1: Convert $\sec$ to $\cos$

There is no "sec" button on the calculator and so we need to convert $\sec$ to $\cos$ so we can find α .

$$3 \sec \alpha = 1,4$$

$$\frac{3}{\cos \alpha} = 1,4$$

Step 2: Rearrange the equation

We need to rearrange the equation so that we have $\cos \alpha$ on one side of the equation.

$$\frac{3}{\cos \alpha} = 1,4$$

$$3 = 1,4 \cos \alpha$$

$$\frac{3}{1,4} = \cos \alpha$$

Step 3: Use your calculator to solve for α

To solve for α , you will need to use the inverse cosine function on your calculator. This works backwards by using the ratio of the sides to determine the angle which resulted in that ratio.

Press          *math error*

In this case we get an error when we try to do the calculation. This is because $\frac{3}{1,4}$ is greater than 1 and the maximum value of the cosine function is 1. Therefore there is no solution. It is important in this case to write no solution and not math error.

Step 4: Write the final answer

There is no solution.

Exercise 5 – 6:

1. Determine the angle (correct to 1 decimal place):

- | | | | |
|---------------------------------|--|------------------------------------|-------------------------------|
| a) $\tan \theta = 1,7$ | b) $\sin \theta = 0,8$ | c) $\cos \alpha = 0,32$ | d) $\tan \beta = 4,2$ |
| e) $\tan \theta = 5\frac{3}{4}$ | f) $\sin \theta = \frac{2}{3}$ | g) $\cos \beta = 1,2$ | h) $4 \cos \theta = 3$ |
| i) $\cos 4\theta = 0,3$ | j) $\sin \beta + 2 = 2,65$ | k) $2 \sin \theta + 5 = 0,8$ | l) $3 \tan \beta = 1$ |
| m) $\sin 3\alpha = 1,2$ | n) $\tan \frac{\theta}{3} = \sin 48^\circ$ | o) $\frac{1}{2} \cos 2\beta = 0,3$ | p) $2 \sin 3\theta + 1 = 2,6$ |

2. If $x = 16^\circ$ and $y = 36^\circ$, use your calculator to evaluate each of the following, correct to 3 decimal places.

- | | | |
|-------------------------|-------------------------|----------------------------------|
| a) $\sin(x - y)$ | b) $3 \sin x$ | c) $\tan x - \tan y$ |
| d) $\cos x + \cos y$ | e) $\frac{1}{3} \tan y$ | f) $\operatorname{cosec}(x - y)$ |
| g) $2 \cos x + \cos 3y$ | h) $\tan(2x - 5y)$ | |

3. In each of the following find the value of x correct to two decimal places.

- | | | |
|-------------------------------|-----------------------------|---|
| a) $\sin x = 0,814$ | b) $\sin x = \tan 45^\circ$ | c) $\tan 2x = 3,123$ |
| d) $\tan x = 3 \sin 41^\circ$ | e) $\sin(2x + 45) = 0,123$ | f) $\sin(x - 10^\circ) = \cos 57^\circ$ |

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1a. 2FRK | 1b. 2FRM | 1c. 2FRN | 1d. 2FRP | 1e. 2FRQ | 1f. 2FRR | 1g. 2FRS | 1h. 2FRT |
| 1i. 2FRV | 1j. 2FRW | 1k. 2FRX | 1l. 2FRY | 1m. 2FRZ | 1n. 2FS2 | 1o. 2FS3 | 1p. 2FS4 |
| 2a. 2FS5 | 2b. 2FS6 | 2c. 2FS7 | 2d. 2FS8 | 2e. 2FS9 | 2f. 2FSB | 2g. 2FSC | 2h. 2FSD |
| 3a. 2FSF | 3b. 2FSG | 3c. 2FSH | 3d. 2FSJ | 3e. 2FSK | 3f. 2FSM | | |



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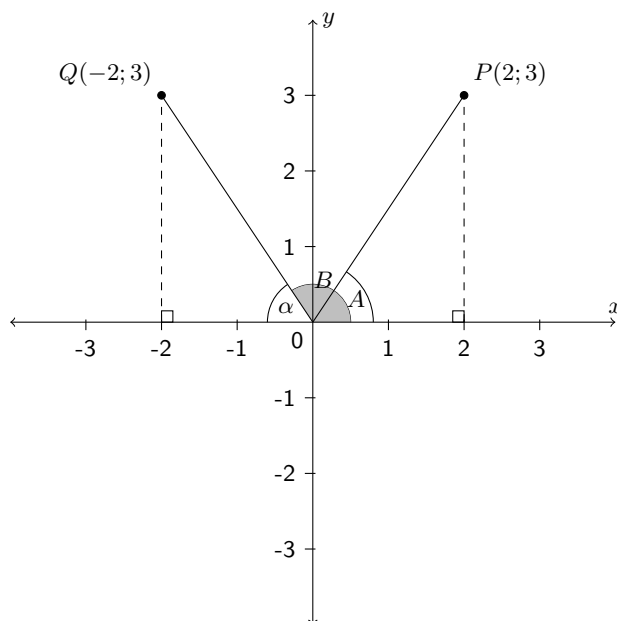


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We have defined the trigonometric ratios using right-angled triangles. We can extend these definitions to any angle, noting that the definitions do not rely on the lengths of the sides of the triangle, but on the size of the angle only. So if we plot any point on the Cartesian plane and then draw a line from the origin to that point, we can work out the angle between the positive x -axis and that line. We will first look at this for two specific points and then look at the more general case.

Finding an angle for specific points

In the figure below points P and Q have been plotted. A line from the origin (O) to each point is drawn. The dotted lines show how we can construct right-angled triangles for each point. The dotted line must always be drawn to the x -axis. Now we can find the angles A and B :



From the coordinates of $P(2; 3)$, we can see that $x = 2$ and $y = 3$. Therefore, we know the length of the side opposite $\hat{A}$ is 3 and the length of the adjacent side is 2. Using:

$$\tan \hat{A} = \frac{\text{opposite}}{\text{adjacent}} = \frac{3}{2}$$

we calculate that $\hat{A} = 56,3^\circ$.

We can also use the theorem of Pythagoras to calculate the hypotenuse of the triangle and then calculate $\hat{A}$ using:

$$\sin \hat{A} = \frac{\text{opposite}}{\text{hypotenuse}} \quad \text{or} \quad \cos \hat{A} = \frac{\text{adjacent}}{\text{hypotenuse}}$$

Consider point $Q(-2; 3)$. We define $\hat{B}$ as the angle formed between line OQ and the positive x -axis. This is called the standard position of an angle. Angles are always measured from the positive x -axis in an anti-clockwise direction. Let α be the angle formed between the line OQ and the negative x -axis such that $\hat{B} + \alpha = 180^\circ$.

From the coordinates of $Q(-2; 3)$, we know the length of the side opposite α is 3 and the length of the adjacent side is 2. Using:

$$\tan \alpha = \frac{\text{opposite}}{\text{adjacent}} = \frac{3}{2}$$

we calculate that $\alpha = 56,3^\circ$. Therefore $\hat{B} = 180^\circ - \alpha = 123,7^\circ$.

Similarly, an alternative method is to calculate the hypotenuse using the theorem of Pythagoras and calculate α using:

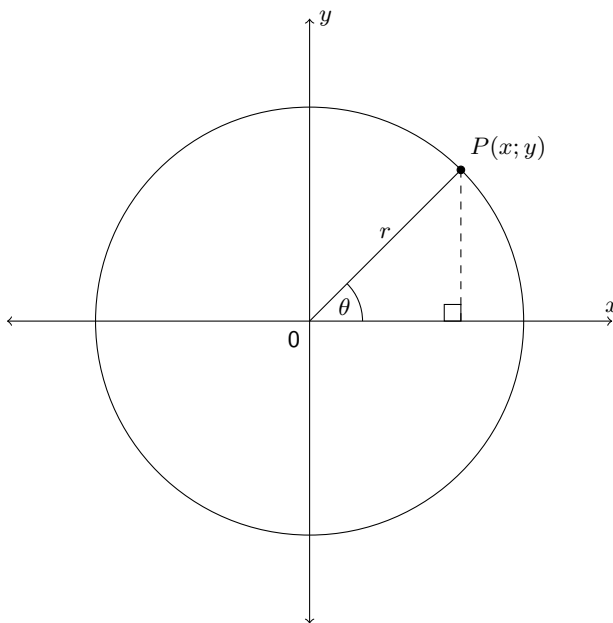
$$\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}} \quad \text{or} \quad \cos \alpha = \frac{\text{adjacent}}{\text{hypotenuse}}$$

NOTE:

We can also extend the definitions of the reciprocals in the same way.

Finding any angle

If we were to draw a circle centred on the origin (O) and passing through the point $P(x; y)$, then the length from the origin to point P is the radius of the circle, which we denote r . We denote the angle formed between the line OP and the x -axis as θ .



We can rewrite all the trigonometric ratios in terms of x , y and r . The general definitions for the trigonometric ratios are:

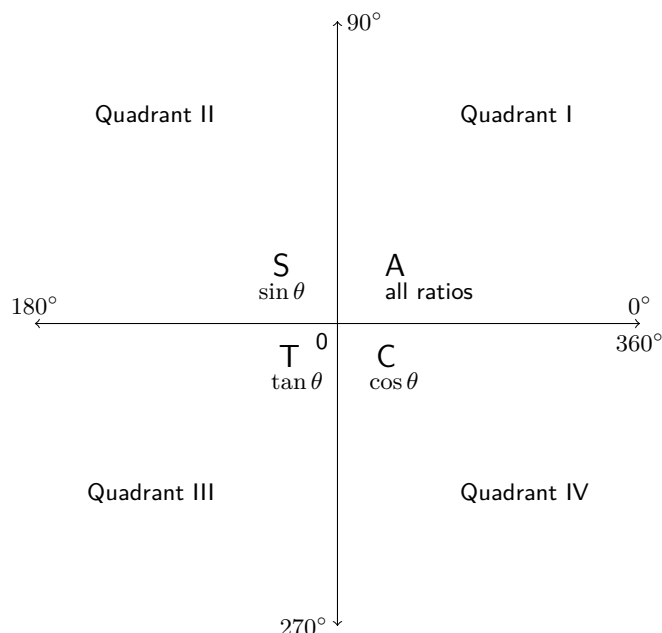
$$\sin \theta = \frac{y}{r} \quad \text{cosec } \theta = \frac{r}{y}$$

$$\cos \theta = \frac{x}{r} \quad \sec \theta = \frac{r}{x}$$

$$\tan \theta = \frac{y}{x} \quad \cot \theta = \frac{x}{y}$$

The CAST diagram

The Cartesian plane is divided into 4 quadrants in an anti-clockwise direction as shown in the diagram below. Notice that r is always positive but the values of x and y change depending on the position of the point in the Cartesian plane. As a result, the trigonometric ratios can be positive or negative. The letters C, A, S and T indicate which of the ratios are positive in each quadrant:



This diagram is known as the CAST diagram.

We note the following using the general definitions of the trigonometric ratios:

- Quadrant I
Both the x and y values are positive so all ratios are positive in this quadrant.
- Quadrant II
The y values are positive therefore $\sin$ and cosec are positive in this quadrant (recall that $\sin$ and cosec are defined in terms of y and r).
- Quadrant III
Both the x and the y values are negative therefore $\tan$ and $\cot$ are positive in this quadrant (recall that $\tan$ and $\cot$ are defined in terms of x and y).
- Quadrant IV
The x values are positive therefore $\cos$ and $\sec$ are positive in this quadrant (recall that $\cos$ and $\sec$ are defined in terms of x and r).

IMPORTANT!

The hypotenuse, r , is a length, and is therefore always positive.

VISIT:

The following video provides a summary of the trigonometric ratios in the Cartesian plane.

🔗 See video: [2FSN](https://www.everythingmaths.co.za) at www.everythingmaths.co.za

Special angles in the Cartesian plane

When working in the Cartesian plane we include two other special angles in right-angled triangles: 0° and 90° .

Notice that when $\theta = 0^\circ$ the length of the opposite side is equal to 0 and the length of the adjacent side is equal to the length of the hypotenuse.

Therefore:

$$\begin{aligned}\sin 0^\circ &= \frac{\text{opposite}}{\text{hypotenuse}} = \frac{0}{\text{hypotenuse}} = 0 \\ \cos 0^\circ &= \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\text{hypotenuse}}{\text{hypotenuse}} = 1 \\ \tan 0^\circ &= \frac{\text{opposite}}{\text{adjacent}} = \frac{0}{\text{adjacent}} = 0\end{aligned}$$

When $\theta = 90^\circ$ the length of the adjacent side is equal to 0 and the length of the opposite side is equal to the length of the hypotenuse. Therefore:

$$\begin{aligned}\sin 90^\circ &= \frac{\text{opposite}}{\text{hypotenuse}} = \frac{\text{hypotenuse}}{\text{hypotenuse}} = 1 \\ \cos 90^\circ &= \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{0}{\text{hypotenuse}} = 0 \\ \tan 90^\circ &= \frac{\text{opposite}}{\text{adjacent}} = \frac{\text{opposite}}{0} = \text{undefined}\end{aligned}$$

Now we can extend our knowledge of special angles.

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Worked example 11: Ratios in the Cartesian plane

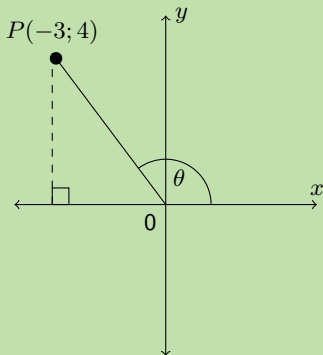
QUESTION

$P(-3; 4)$ is a point on the Cartesian plane with origin O . θ is the angle between OP and the positive x -axis. Without using a calculator, determine the value of:

- $\cos \theta$
- $3 \tan \theta$
- $\frac{1}{2} \operatorname{cosec} \theta$

SOLUTION

Step 1: Sketch point P in the Cartesian plane and label the angle θ



Step 2: Use the theorem of Pythagoras to calculate r

$$\begin{aligned}r^2 &= x^2 + y^2 \\&= (-3)^2 + (4)^2 \\&= 25 \\\therefore r &= 5\end{aligned}$$

Note r is positive as it is the radius of the circle.

Step 3: Substitute values for x , y and r into the required ratios

We note that $x = -3$, $y = 4$ and $r = 5$.

1. $\cos \theta = \frac{x}{r} = \frac{-3}{5}$
2. $3 \tan \theta = 3 \left(\frac{y}{x} \right) = 3 \left(\frac{4}{-3} \right) = -4$
3. $\frac{1}{2} \operatorname{cosec} \theta = \frac{1}{2} \left(\frac{r}{y} \right) = \frac{1}{2} \left(\frac{5}{4} \right) = \frac{5}{8}$

Worked example 12: Ratios in the Cartesian plane

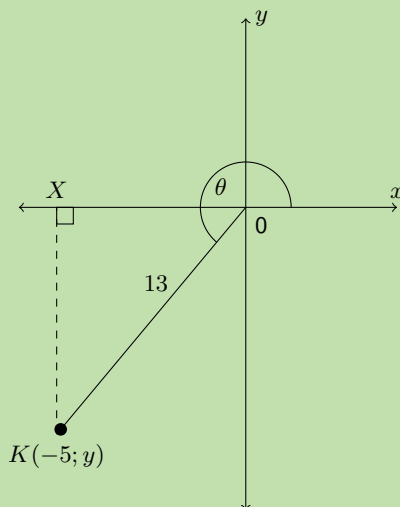
QUESTION

$X\hat{O}K = \theta$ is an angle in the third quadrant where X is a point on the positive x -axis and K is the point $(-5; y)$. OK is 13 units.

1. Determine, without using a calculator, the value of y .
2. Prove that $\tan^2 \theta + 1 = \sec^2 \theta$ without using a calculator.

SOLUTION

Step 1: Sketch point K in the Cartesian plane and label the angle θ



Step 2: Use the theorem of Pythagoras to calculate y

$$\begin{aligned}r^2 &= x^2 + y^2 \\y^2 &= r^2 - x^2 \\&= (13)^2 - (-5)^2 \\&= 169 - 25 \\&= 144 \\y &= \pm 12\end{aligned}$$

Given that θ lies in the third quadrant, y must be negative. Therefore $y = -12$

Step 3: Substitute values for x , y and r and simplify

$x = -5$, $y = -12$ and $r = 13$.

LHS

$$\begin{aligned}\tan^2 \theta + 1 &= \left(\frac{y}{x}\right)^2 + 1 \\&= \left(\frac{-12}{-5}\right)^2 + 1 \\&= \left(\frac{144}{25}\right) + 1 \\&= \frac{144 + 25}{25} \\&= \frac{169}{25}\end{aligned}$$

RHS

$$\begin{aligned}\sec^2 \theta &= \left(\frac{r}{x}\right)^2 \\&= \left(\frac{13}{-5}\right)^2 \\&= \frac{169}{25}\end{aligned}$$

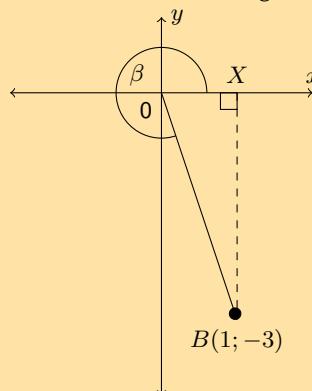
Therefore the LHS = RHS and we have proved that $\tan^2 \theta + 1 = \sec^2 \theta$.

NOTE:

Whenever you have to solve trigonometric problems without a calculator, it can be very helpful to make a sketch.

Exercise 5 – 7:

1. B is a point in the Cartesian plane. Determine without using a calculator:



a) OB

b) $\cos \beta$

c) $\operatorname{cosec} \beta$

d) $\tan \beta$

-
- A diagram illustrating a circle in the complex plane. The horizontal axis is labeled x and the vertical axis is labeled y . The origin is marked with 0 . A circle is centered at the origin. A point $A(a; b)$ is marked on the circle in the third quadrant. A line segment connects the origin to point A . The angle θ is indicated between the positive x -axis and the line segment connecting the origin to point A .

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10. If $\tan \alpha = \frac{5}{-12}$ and $0^\circ \leq \alpha \leq 180^\circ$, determine **without the use of a calculator** the value of $\frac{12}{\cos \alpha}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2FSP 2. 2FSQ 3. 2FSR 4. 2FSS 5. 2FST 6. 2FSV
7. 2FSW 8. 2FSX 9. 2FSY 10. 2FSZ



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5.9 Chapter summary

EMA3Y

► See presentation: 2FT2 at www.everythingmaths.co.za

- We can define three trigonometric ratios for right-angled triangles: sine (sin), cosine (cos) and tangent (tan)

These ratios can be defined as:

$$\begin{aligned} - \sin \theta &= \frac{\text{opposite}}{\text{hypotenuse}} = \frac{y}{r} \\ - \cos \theta &= \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{x}{r} \\ - \tan \theta &= \frac{\text{opposite}}{\text{adjacent}} = \frac{y}{x} \end{aligned}$$

- Each of these ratios have a reciprocal: cosecant (cosec), secant (sec) and cotangent (cot).

These ratios can be defined as:

$$\begin{aligned} - \operatorname{cosec} \theta &= \frac{\text{hypotenuse}}{\text{opposite}} = \frac{r}{y} \\ - \sec \theta &= \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{r}{x} \\ - \cot \theta &= \frac{\text{adjacent}}{\text{opposite}} = \frac{x}{y} \end{aligned}$$

- We can use the principles of solving equations and the trigonometric ratios to help us solve simple trigonometric equations.
- For some special angles (0° , 30° , 45° , 60° and 90°), we can easily find the values of sin, cos and tan without using a calculator.
- We can extend the definitions of the trigonometric ratios to any angle.

End of chapter Exercise 5 – 8:

1. State whether each of the following trigonometric ratios has been written correctly.

a) $\sin \theta = \frac{\text{hypotenuse}}{\text{adjacent}}$

b) $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

c) $\sec \theta = \frac{\text{hypotenuse}}{\text{adjacent}}$

2. Use your calculator to evaluate the following expressions to two decimal places:

a) $\tan 80^\circ$

b) $\cos 73^\circ$

c) $\sin 17^\circ$

d) $\tan 313^\circ$

e) $\cos 138^\circ$

f) $\sec 56^\circ$

g) $\cot 18^\circ$

h) $\operatorname{cosec} 37^\circ$

i) $\sec 257^\circ$

j) $\sec 304^\circ$

k) $3 \sin 51^\circ$

l) $4 \cot 54^\circ + 5 \tan 44^\circ$

m) $\frac{\cos 205^\circ}{4}$

n) $\sqrt{\sin 99^\circ}$

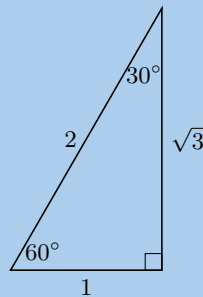
o) $\sqrt{\cos 687^\circ + \sin 120^\circ}$

p) $\frac{\tan 70^\circ}{\operatorname{cosec} 1^\circ}$

q) $\sec 84^\circ + 4 \sin 0,4^\circ \times 50 \cos 50^\circ$

r) $\frac{\cos 40^\circ}{\sin 35^\circ} + \tan 38^\circ$

3. Use the triangle below to complete the following:



a) $\sin 60^\circ =$

b) $\cos 60^\circ =$

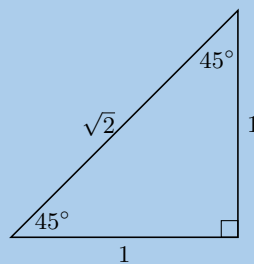
c) $\tan 60^\circ =$

d) $\sin 30^\circ =$

e) $\cos 30^\circ =$

f) $\tan 30^\circ =$

4. Use the triangle below to complete the following:



a) $\sin 45^\circ =$

b) $\cos 45^\circ =$

c) $\tan 45^\circ =$

5. Evaluate the following without using a calculator. Select the closest answer from the list provided.

a) $\sin 60^\circ - \tan 60^\circ$

$$0 \quad -\frac{1}{2} \quad \frac{2}{\sqrt{3}} \quad -\frac{\sqrt{3}}{2} \quad -\frac{2}{\sqrt{3}}$$

b) $\tan 30^\circ - \cos 30^\circ$

$$0 \quad -\frac{1}{2\sqrt{3}} \quad \frac{\sqrt{3}}{2} \quad -\frac{2}{\sqrt{3}} \quad -\frac{\sqrt{3}}{2}$$

c) $\tan 60^\circ - \sin 60^\circ - \tan 60^\circ$

$$-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{1} - \frac{1}{2} - \frac{1}{1} - \frac{1}{\sqrt{2}}$$

d) $\sin 30^\circ \times \sin 30^\circ \times \sin 30^\circ$

$$\frac{1}{2} \times \frac{1}{2\sqrt{3}} \times \frac{1}{8} \times \frac{1}{4} \times \frac{\sqrt{3}}{4\sqrt{2}}$$

e) $\sin 45^\circ \times \tan 45^\circ \times \tan 60^\circ$

$$\frac{3}{2\sqrt{2}} \times \frac{\sqrt{3}}{8} \times \frac{3}{4} \times \frac{\sqrt{3}}{\sqrt{2}} \times \frac{1}{4}$$

f) $\cos 60^\circ \times \cos 45^\circ \times \tan 60^\circ$

$$\frac{\sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{3}}{4} \times \frac{3}{4\sqrt{2}} \times \frac{1}{2} \times \frac{1}{4\sqrt{3}}$$

g) $\tan 45^\circ \times \sin 60^\circ \times \tan 45^\circ$

$$\frac{\sqrt{3}}{2} \times \frac{3}{8} \times \frac{1}{3} \times \frac{\sqrt{3}}{2\sqrt{2}} \times \frac{1}{4\sqrt{3}}$$

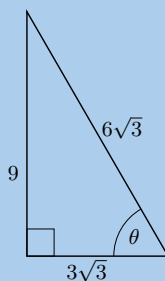
h) $\cos 30^\circ \times \cos 60^\circ \times \sin 60^\circ$

$$\frac{3}{8} \times \frac{3}{2\sqrt{2}} \times \frac{\sqrt{3}}{4\sqrt{2}} \times \frac{1}{2\sqrt{3}} \times \frac{1}{4\sqrt{3}}$$

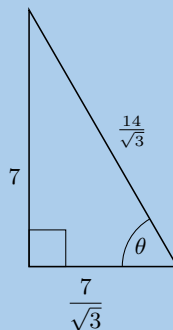
6. Without using a calculator, determine the value of:

$$\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ + \tan 45^\circ$$

7. Solve for $\sin \theta$ in the following triangle, in surd form:

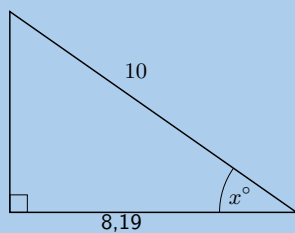


8. Solve for $\tan \theta$ in the following triangle, in surd form:

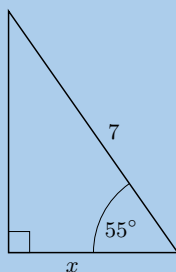


9. A right-angled triangle has hypotenuse 13 mm. Find the length of the other two sides if one of the angles of the triangle is 50° .
10. Solve for x to the nearest integer.

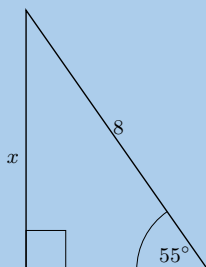
a)



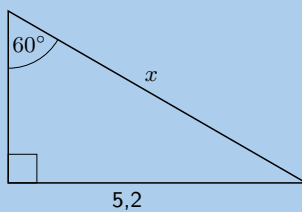
b)



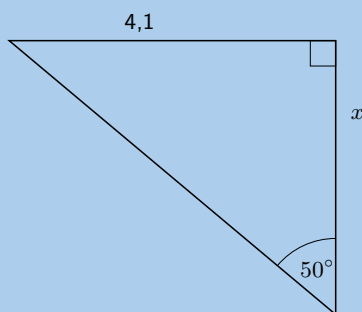
c)



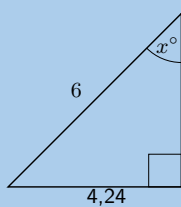
d)



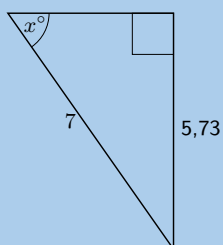
e)



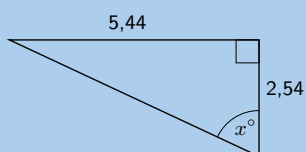
f)



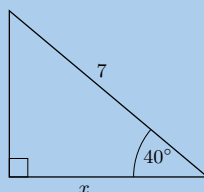
g)



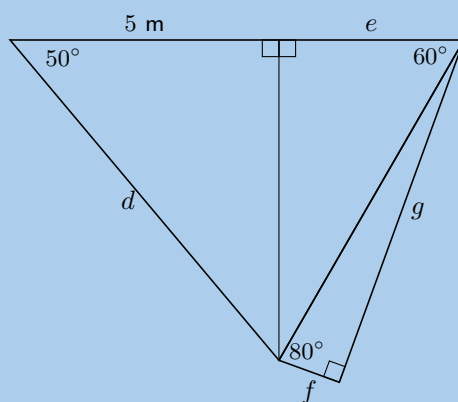
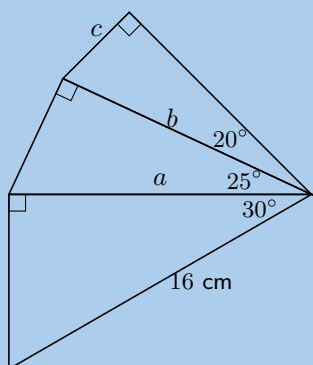
h)



i)



11. Calculate the unknown lengths in the diagrams below:



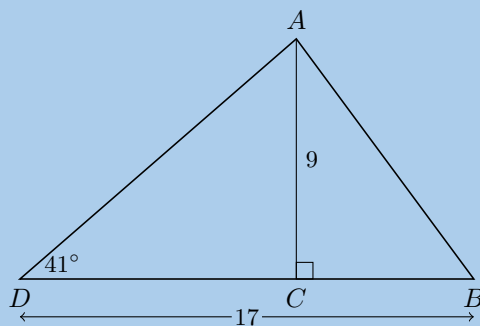
12. In $\triangle PQR$, $PR = 20$ cm, $QR = 22$ cm and $\hat{P}RQ = 30^\circ$. The perpendicular line from P to QR intersects QR at X . Calculate:

a) the length XR

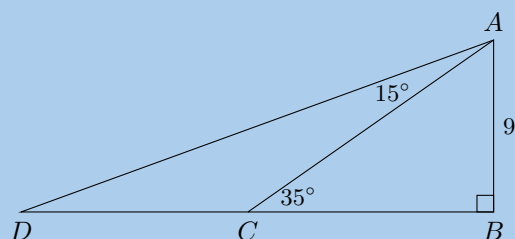
b) the length PX

c) the angle $\hat{Q}P\hat{X}$

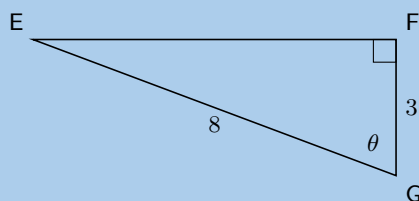
13. In the following triangle find the size of $\hat{ABC}$.



14. In the following triangle find the length of side CD :



15. In the following triangle determine:

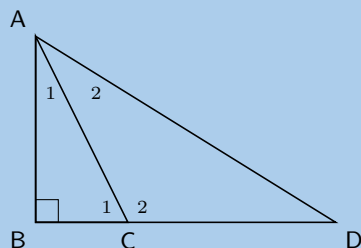


a) the length of EF

b) $\tan(90^\circ - \theta)$

c) the value of θ

16. Given that $\hat{D} = x$, $\hat{C}_1 = 2x$, $BC = 12,2$ cm, $AB = 24,6$ cm. Calculate CD .



17. Solve for θ if θ is a positive, acute angle:

a) $2 \sin \theta = 1,34$

b) $1 - \tan \theta = -1$

c) $\cos 2\theta = \sin 40^\circ$

d) $\sec \theta = 1,8$

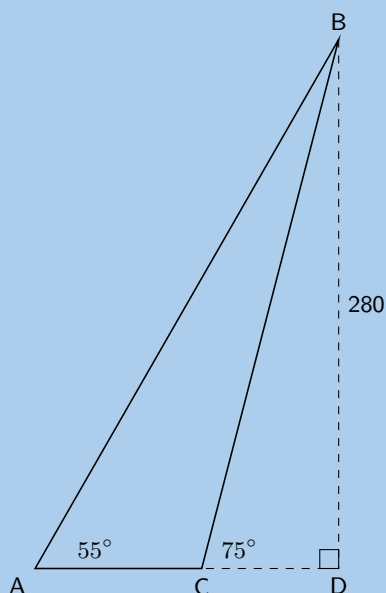
e) $\cot 4\theta = \sin 40^\circ$

f) $\sin 3\theta + 5 = 4$

g) $\cos(4 + \theta) = 0,45$

h) $\frac{\sin \theta}{\cos \theta} = 1$

18. If $a = 29^\circ$, $b = 38^\circ$ and $c = 47^\circ$, use your calculator to evaluate each of the following, correct to 2 decimal places.
- $\tan(a + c)$
 - $\operatorname{cosec}(c - b)$
 - $\sin(a \times b \times c)$
 - $\tan a + \sin b + \cos c$
19. If $3 \tan \alpha = -5$ and $0^\circ < \alpha < 270^\circ$, use a sketch to determine:
- $\cos \alpha$
 - $\tan^2 \alpha - \sec^2 \alpha$
20. Given $A(5; 0)$ and $B(11; 4)$, find the angle between the line through A and B and the x -axis.
21. Given $C(0; -13)$ and $D(-12; 14)$, find the angle between the line through C and D and the y -axis.
22. Given the points $E(5; 0)$, $F(6; 2)$ and $G(8; -2)$. Find the angle $\hat{FEG}$.
23. A triangle with angles 40° , 40° and 100° has a perimeter of 20 cm. Find the length of each side of the triangle.
24. Determine the area of $\triangle ABC$.



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- | | | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 1a. 2FT3 | 1b. 2FT4 | 1c. 2FT5 | 2a. 2FT6 | 2b. 2FT7 | 2c. 2FT8 |
| 2d. 2FT9 | 2e. 2FTB | 2f. 2FTC | 2g. 2FTD | 2h. 2FTF | 2i. 2FTG |
| 2j. 2FTH | 2k. 2FTJ | 2l. 2FTK | 2m. 2FTM | 2n. 2FTN | 2o. 2FTP |
| 2p. 2FTQ | 2q. 2FTR | 2r. 2FTS | 3. 2FTT | 4. 2FTV | 5a. 2FTW |
| 5b. 2FTX | 5c. 2FTY | 5d. 2FTZ | 5e. 2FV2 | 5f. 2FV3 | 5g. 2FV4 |
| 5h. 2FV5 | 6. 2FV6 | 7. 2FV7 | 8. 2FV8 | 9. 2FV9 | 10a. 2FVB |
| 10b. 2FVC | 10c. 2FVD | 10d. 2FVF | 10e. 2FVG | 10f. 2FVH | 10g. 2FVJ |
| 10h. 2FVK | 10i. 2FVM | 11. 2FVN | 12. 2FVP | 13. 2FVQ | 14. 2FVR |
| 15. 2FVS | 16. 2FVT | 17a. 2FVV | 17b. 2FVW | 17c. 2FVX | 17d. 2FVY |
| 17e. 2FVZ | 17f. 2FW2 | 17g. 2FW3 | 17h. 2FW4 | 18. 2FW5 | 19. 2FW6 |
| 20. 2FW7 | 21. 2FW8 | 22. 2FW9 | 23. 2FWB | 24. 2FWC | |



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Functions

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6.1 Introduction

EMA3Z

Functions are mathematical building blocks for designing machines, predicting natural disasters, curing diseases, understanding world economies and for keeping aeroplanes in the air. Functions can take input from many variables, but always give the same output, unique to that function.

Functions also allow us to visualise relationships in terms of graphs, which are much easier to read and interpret than lists of numbers.



Figure 6.1: A cricket player facing a delivery. If a cricket player is hit on his batting pads and the umpire thinks that the ball would have hit the stumps behind him, he is given out LBW (leg before wicket). At professional levels of the game, sophisticated software is used to determine if the ball will hit the stumps. The software uses functions to predict the flight of the ball if the cricket players leg had not been in the way.

Some examples of functions include:

- Money as a function of time. You never have more than one amount of money at any time because you can always add everything to give one total amount. By understanding how your money changes over time, you can plan to spend your money sensibly. Businesses find it very useful to plot the graph of their money over time so that they can see when they are spending too much.
- Temperature as a function of various factors. Temperature is a very complicated function because it has so many inputs, including: the time of day, the season, the amount of clouds in the sky, the strength of the wind, where you are and many more. But the important thing is that there is only one temperature output when you measure it in a specific place.
- Location as a function of time. You can never be in two places at the same time. If you were to plot the graphs of where two people are as a function of time, the place where the lines cross means that the two people meet each other at that time. This idea is used in logistics, an area of mathematics that tries to plan where people and items are for businesses.

DEFINITION: *Function*

A function is a mathematical relationship between two variables, where every input variable has one output variable.

Dependent and independent variables

EMA42

In functions, the x -variable is known as the input or independent variable, because its value can be chosen freely. The calculated y -variable is known as the output or dependent variable, because its value depends on the chosen input value.

Examples:

$\{x : x \in \mathbb{R}, x > 0\}$	The set of all x -values such that x is an element of the set of real numbers and is greater than 0.
$\{y : y \in \mathbb{N}, 3 < y \leq 5\}$	The set of all y -values such that y is a natural number, is greater than 3 and is less than or equal to 5.
$\{z : z \in \mathbb{Z}, z \leq 100\}$	The set of all z -values such that z is an integer and is less than or equal to 100.

It is important to note that this notation can only be used to represent an interval of real numbers.

Examples:

$(3; 11)$	Round brackets indicate that the number is not included. This interval includes all real numbers greater than but not equal to 3 and less than but not equal to 11.
$(-\infty; -2)$	Round brackets are always used for positive and negative infinity. This interval includes all real numbers less than, but not equal to -2 .
$[1; 9)$	A square bracket indicates that the number is included. This interval includes all real numbers greater than or equal to 1 and less than but not equal to 9.

This is a very useful way to express a function. Another way of writing $y = 2x + 1$ is $f(x) = 2x + 1$. We say “ f of x is equal to $2x + 1$ ”. Any letter can be used, for example, $g(x)$, $h(x)$, $p(x)$, etc.

1. **Determine the output value:**

“Find the value of the function for $x = -3$ ” can be written as: “find $f(-3)$ ”.

Replace x with -3 :

$$\begin{aligned} f(-3) &= 2(-3) + 1 = -5 \\ \therefore f(-3) &= -5 \end{aligned}$$

This means that when $x = -3$, the value of the function is -5 .

2. **Determine the input value:**

“Find the value of x that will give a y -value of 27” can be written as: “find x if $f(x) = 27$ ”.

We write the following equation and solve for x :

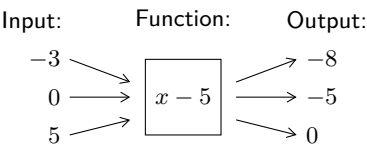
$$\begin{aligned} 2x + 1 &= 27 \\ \therefore x &= 13 \end{aligned}$$

This means that when $x = 13$ the value of the function is 27.

Functions can be expressed in many different ways for different purposes.

1. Words: "The relationship between two variables is such that one is always 5 less than the other."

2. Mapping diagram:



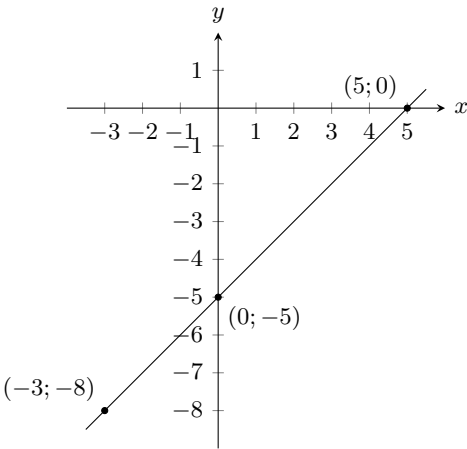
3. Table:

Input variable (x)	-3	0	5
Output variable (y)	-8	-5	0

4. Set of ordered number pairs: $(-3; -8), (0; -5), (5; 0)$

5. Algebraic formula: $f(x) = x - 5$

6. Graph:



Domain and range

EMA47

The domain of a function is the set of all independent x -values from which the function produces a single y -value for each x -value.

The range is the set of all dependent y -values which can be obtained using an independent x -value.

Exercise 6 – 1:

1. Write the following in set notation:

- a) $(-\infty; 7]$ b) $[-13; 4)$ c) $(35; \infty)$ d) $[\frac{3}{4}; 21)$ e) $[-\frac{1}{2}; \frac{1}{2}]$ f) $(-\sqrt{3}; \infty)$

2. Write the following in interval notation:

- a) $\{p : p \in \mathbb{R}, p \leq 6\}$ b) $\{k : k \in \mathbb{R}, -5 < k < 5\}$
c) $\{x : x \in \mathbb{R}, x > \frac{1}{5}\}$ d) $\{z : z \in \mathbb{R}, 21 \leq x < 41\}$

3. Complete the following tables and identify the function.

a)

x	1	2	3	4	5	6
y	5	10		20		

b)

x	1		3	4		6
y	5	5			5	5

c)

x	2			8	10	12
y	1	2	3			6

4. Plot the following points on a graph.

a)

x	1	2	3	4	5	6
y	0,1	0,2	0,3	0,4	0,5	0,6

b)

x	1	2	3	4	5	6
y	5	9	13	17	21	25

5. Create a table of values from the function given and then plot the function. Your table must have at least 5 ordered pairs.

a) $y = \frac{1}{2}x + 2$

b) $y = x - 3$

6. If the functions $f(x) = x^2 + 1$; $g(x) = x - 4$; $h(x) = 7 - x^2$; $k(x) = 3$ are given, find the value of the following:

a) $f(-1)$

b) $g(-7)$

c) $h(3)$

d) $k(100)$

e) $f(-2) + h(2)$

f) $k(-5) + h(3)$

g) $f(g(1))$

h) $k(f(6))$

7. The cost of petrol and diesel per litre are given by the functions P and D , where:

$$P = 13,61V \quad D = 12,46V$$

Use this information to answer the following:

a) Evaluate $P(8)$

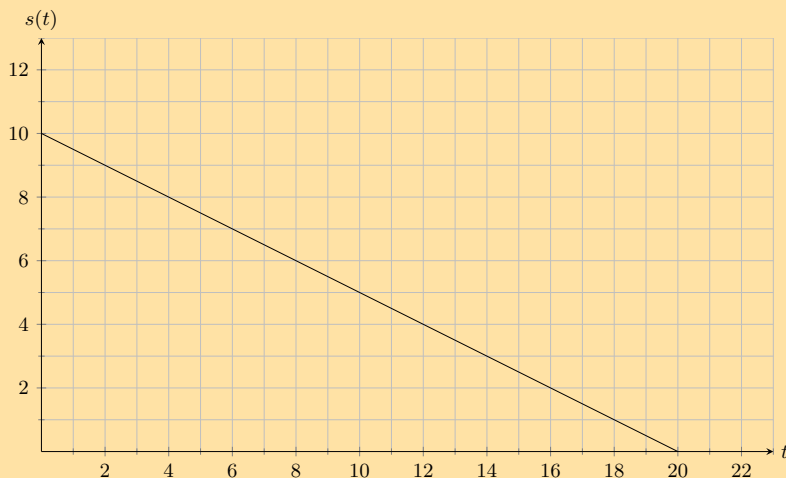
b) Evaluate $D(16)$

c) How many litres of petrol can you buy with R 300?

d) How many litres of petrol can you buy with R 275?

e) How much more expensive is petrol than diesel? Show your answer as a function.

8. A ball is rolling down a 10 m slope. The graph below shows the relationship between the distance and the time.



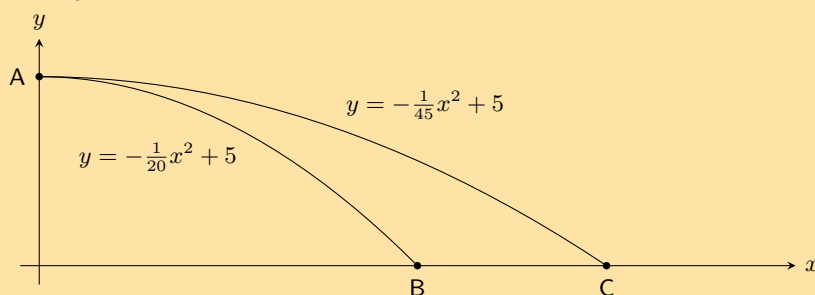
Use this information to answer the following:

a) After 6 s how much further does the ball have to roll?

b) What is the range of the function?

c) What is the domain of the function, and what does it represent?

9. James and Themba both throw a stone from the top of a building into a river. The path travelled by the stones can be described by quadratic equations. $y = -\frac{1}{20}x^2 + 5$ describes the path of the stone thrown by James and $y = -\frac{1}{45}x^2 + 5$ describes the path of Themba's stone.



- What is the height of the building that they stood on?
- How far did James throw his stone before it hit the river surface?
- How much farther did Themba throw his stone before it hit the river surface?

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- | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1a. 2FWD | 1b. 2FWF | 1c. 2FWG | 1d. 2FWH | 1e. 2FWJ | 1f. 2FWK | 2a. 2FWM |
| 2b. 2FWN | 2c. 2FWP | 2d. 2FWQ | 3a. 2FWR | 3b. 2FWS | 3c. 2FWT | 4a. 2FWV |
| 4b. 2FWW | 5a. 2FWX | 5b. 2FWY | 6. 2FWZ | 7. 2FX2 | 8. 2FX3 | 9. 2FX4 |



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6.2 Linear functions

EMA48

Functions of the form $y = x$

EMA49

Functions of the form $y = mx + c$ are called straight line functions. In the equation, $y = mx + c$, m and c are constants and have different effects on the graph of the function.

Worked example 1: Plotting a straight line graph

QUESTION

$$y = f(x) = x$$

Complete the following table for $f(x) = x$ and plot the points on a set of axes.

x	-2	-1	0	1	2
$f(x)$	-2				

- Join the points with a straight line.
- Determine the domain and range.
- About which line is f symmetrical?
- Using the graph, determine the value of x for which $f(x) = 4$. Confirm your answer graphically.
- Where does the graph cut the axes?

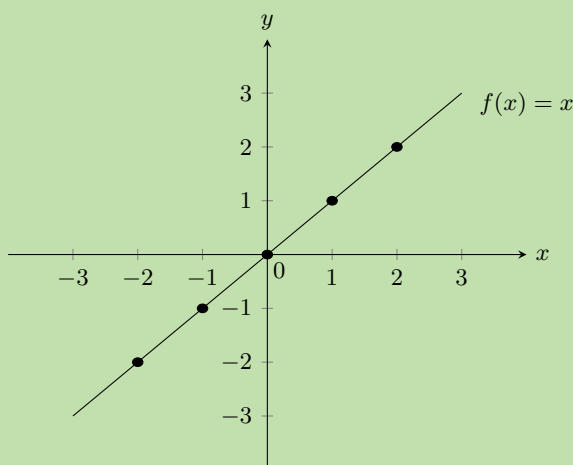
SOLUTION

Step 1: Substitute values into the equation

x	-2	-1	0	1	2
$f(x)$	-2	-1	0	1	2

Step 2: Plot the points and join with a straight line curve

From the table, we get the following points and the graph: $(-2; -2), (-1; -1), (0; 0), (1; 1), (2; 2)$



Step 3: Determine the domain and range

Domain: $x \in \mathbb{R}$

Range: $f(x) \in \mathbb{R}$

Step 4: Determine the value of x for which $f(x) = 4$

From the graph we see that when $f(x) = 4$, $x = 4$. This gives the point $(4; 4)$.

Step 5: Determine the intercept

The function f intercepts the axes at the origin $(0; 0)$.

Functions of the form $y = mx + c$

EMA4B

Investigation: The effects of m and c on a straight line graph

On the same set of axes, plot the following graphs:

1. $y = x - 2$
2. $y = x - 1$
3. $y = x$
4. $y = x + 1$
5. $y = x + 2$

Use your results to deduce the effect of different values of c on the graph.

On the same set of axes, plot the following graphs:

- $y = -2x$
- $y = -x$
- $y = x$
- $y = 2x$

Use your results to deduce the effect of different values of m on the graph.

The effect of m

We notice that the value of m affects the slope of the graph. As m increases, the gradient of the graph increases.

If $m > 0$ then the graph increases from left to right (slopes upwards).

If $m < 0$ then the graph increases from right to left (slopes downwards). For this reason, m is referred to as the gradient of a straight-line graph.

The effect of c

We also notice that the value of c affects where the graph cuts the y -axis. For this reason, c is known as the y -intercept.

If $c > 0$ the graph shifts vertically upwards.

If $c < 0$ the graph shifts vertically downwards.

	$m < 0$	$m = 0$	$m > 0$
$c > 0$			
$c = 0$			
$c < 0$			

Table 6.1: The effect of m and c on a straight line graph.

VISIT:
 You can use this [Phet simulation](#) to help you see the effects of changing m and c .

The standard form of a straight line graph is the equation $y = mx + c$.

Domain and range

The domain is $\{x : x \in \mathbb{R}\}$ because there is no value of x for which $f(x)$ is undefined.

The range of $f(x) = mx + c$ is also $\{f(x) : f(x) \in \mathbb{R}\}$ because $f(x)$ can take on any real value.

Intercepts

The y -intercept:

Every point on the y -axis has an x -coordinate of 0. Therefore to calculate the y -intercept, let $x = 0$.

For example, the y -intercept of $g(x) = x - 1$ is given by setting $x = 0$:

$$\begin{aligned} g(x) &= x - 1 \\ g(0) &= 0 - 1 \\ &= -1 \end{aligned}$$

This gives the point $(0; -1)$.

The x -intercept:

Every point on the x -axis has a y -coordinate of 0. Therefore to calculate the x -intercept, let $y = 0$.

For example, the x -intercept of $g(x) = x - 1$ is given by setting $y = 0$:

$$\begin{aligned} g(x) &= x - 1 \\ 0 &= x - 1 \\ \therefore x &= 1 \end{aligned}$$

This gives the point $(1; 0)$.

Sketching graphs of the form $y = mx + c$

In order to sketch graphs of the form, $f(x) = mx + c$, we need to determine three characteristics:

1. sign of m
2. y -intercept
3. x -intercept

Dual intercept method

Only two points are needed to plot a straight line graph. The easiest points to use are the x -intercept and the y -intercept.

QUESTION

Sketch the graph of $g(x) = x - 1$ using the dual intercept method.

SOLUTION

Step 1: Examine the standard form of the equation

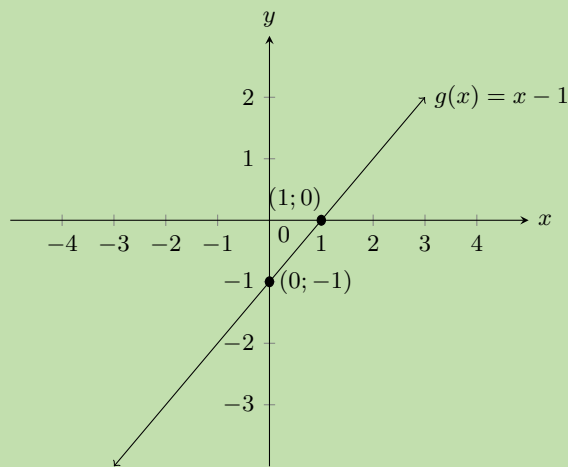
$m > 0$. This means that the graph increases as x increases.

Step 2: Calculate the intercepts

For the y -intercept, let $x = 0$; therefore $g(0) = -1$. This gives the point $(0; -1)$.

For the x -intercept, let $y = 0$; therefore $x = 1$. This gives the point $(1; 0)$.

Step 3: Plot the points and draw the graph



Gradient and y -intercept method

EMA4G

We can draw a straight line graph of the form $y = mx + c$ using the gradient (m) and the y -intercept (c).

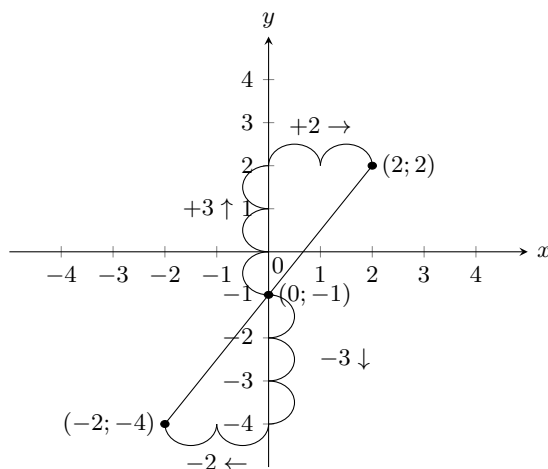
We calculate the y -intercept by letting $x = 0$. This gives us one point $(0; c)$ for drawing the graph and we use the gradient to calculate the second point.

The gradient of a line is the measure of steepness. Steepness is determined by the ratio of vertical change to horizontal change:

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{\text{vertical change}}{\text{horizontal change}}$$

For example, $y = \frac{3}{2}x - 1$, therefore $m > 0$ and the graph slopes upwards.

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{3 \uparrow}{2 \rightarrow} = \frac{-3 \downarrow}{-2 \leftarrow}$$



Worked example 3: Sketching a straight line graph using the gradient–intercept method

QUESTION

Sketch the graph of $p(x) = \frac{1}{2}x - 3$ using the gradient-intercept method.

SOLUTION

Step 1: Use the intercept

$c = -3$, which gives the point $(0; -3)$.

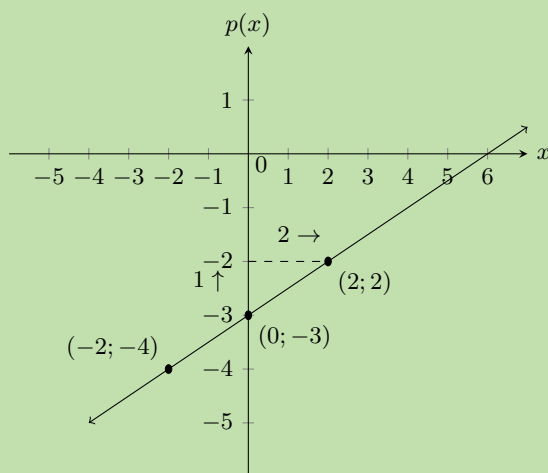
Step 2: Use the gradient

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{1 \uparrow}{2 \rightarrow} = \frac{-1 \downarrow}{-2 \leftarrow}$$

Start at $(0; -3)$. Move 1 unit up and 2 units to the right. This gives the second point $(2; -2)$.

Or start at $(0; -3)$, move 1 unit down and 2 units to the left. This gives the second point $(-2; -4)$.

Step 3: Plot the points and draw the graph



Always write the function in the form $y = mx + c$ and take note of m . After plotting the graph, make sure that the graph increases if $m > 0$ and that the graph decreases if $m < 0$.

Exercise 6 – 2:

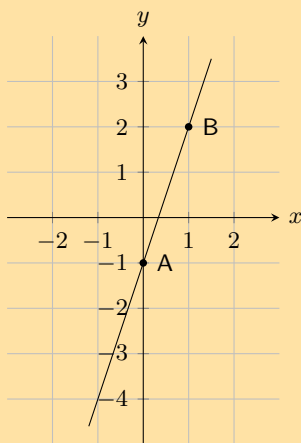
1. Determine the x -intercept and the y -intercept of the following equations.

a) $y = x - 1$

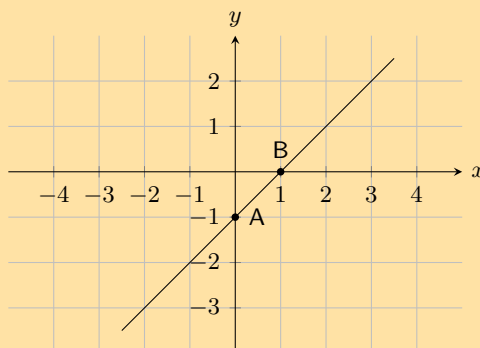
b) $y = x + 2$

c) $y = x - 3$

2. In the graph below there is a function with the equation $y = mx + c$. Determine the values of m (the gradient of the line) and c (the y -intercept of the line).



3. The graph below shows a function with the equation $y = mx + c$. Determine the values of m (the gradient of the line) and c (the y -intercept of the line).



4. List the x and y -intercepts for the following straight line graphs. Indicate whether the graph is increasing or decreasing:

a) $y = x + 1$

b) $y = x - 1$

c) $h(x) = 2x + 1$

d) $y + 3x = 1$

e) $3y - 2x = 6$

f) $k(x) = -3$

g) $x = 3y$

h) $\frac{x}{2} - \frac{y}{3} = 1$

5. State whether the following are true or not.

a) The gradient of $2y = 3x - 1$ is 3.

b) The y -intercept of $y = x + 4$ is 4.

c) The gradient of $2 - y = 2x - 1$ is -2 .

d) The gradient of $y = \frac{1}{2}x - 1$ is -1 .

e) The y -intercept of $2y = 3x - 6$ is 6.

6. Write the following in standard form ($y = mx + c$):

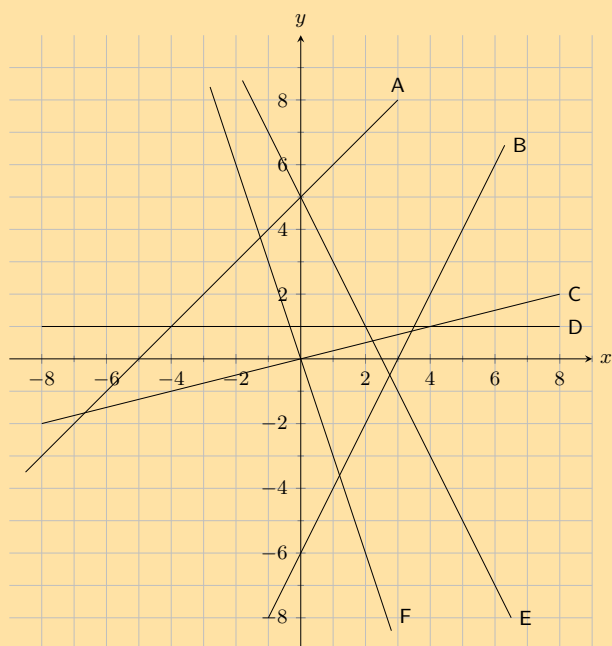
a) $2y + 3x = 1$

b) $3x - y = 5$

c) $3y - 4 = x$

d) $y + 2x - 3 = 1$

7. Look at the graphs below. Each graph is labelled with a letter. In the questions that follow, match any given equation with the label of a corresponding graph.



a) $y = 5 - 2x$

b) $x + 5$

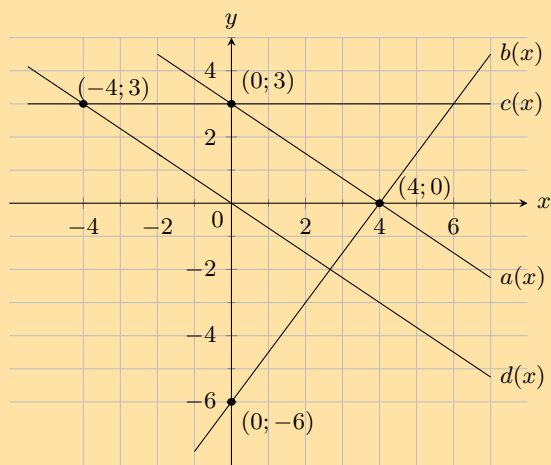
c) $y = 2x - 6$

d) $y = -3x$

e) $y = 1$

f) $y = \frac{1}{2}x$

8. For the functions in the diagram below, give the equation of each line:



a) $a(x)$

b) $b(x)$

c) $c(x)$

d) $d(x)$

9. Sketch the following functions on the same set of axes, using the dual intercept method. Clearly indicate the coordinates of the intercepts with the axes and the point of intersection of the two graphs: $x + 2y - 5 = 0$ and $3x - y - 1 = 0$.
10. On the same set of axes, draw the graphs of $f(x) = 3 - 3x$ and $g(x) = \frac{1}{3}x + 1$ using the gradient-intercept method.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1a. 2FX5 | 1b. 2FX6 | 1c. 2FX7 | 2. 2FX8 | 3. 2FX9 | 4a. 2FXB | 4b. 2FXC | 4c. 2FXD |
| 4d. 2FXF | 4e. 2FXG | 4f. 2FXH | 4g. 2FXJ | 4h. 2FXK | 5a. 2FXM | 5b. 2FXN | 5c. 2FXP |
| 5d. 2FXQ | 5e. 2FXR | 6a. 2FXS | 6b. 2FXT | 6c. 2FXV | 6d. 2FXW | 7. 2FXX | 8a. 2FXY |
| 8b. 2FXZ | 8c. 2FY2 | 8d. 2FY3 | 9. 2FY4 | 10. 2FY5 | | | |



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Functions of the general form $y = ax^2 + q$ are called parabolic functions. In the equation $y = ax^2 + q$, a and q are constants and have different effects on the parabola.

Worked example 4: Plotting a quadratic function

QUESTION

$$y = f(x) = x^2$$

Complete the following table for $f(x) = x^2$ and plot the points on a system of axes.

x	-3	-2	-1	0	1	2	3
$f(x)$	9						

1. Join the points with a smooth curve.
2. The domain of f is $x \in \mathbb{R}$. Determine the range.
3. About which line is f symmetrical?
4. Determine the value of x for which $f(x) = 6\frac{1}{4}$. Confirm your answer graphically.
5. Where does the graph cut the axes?

SOLUTION

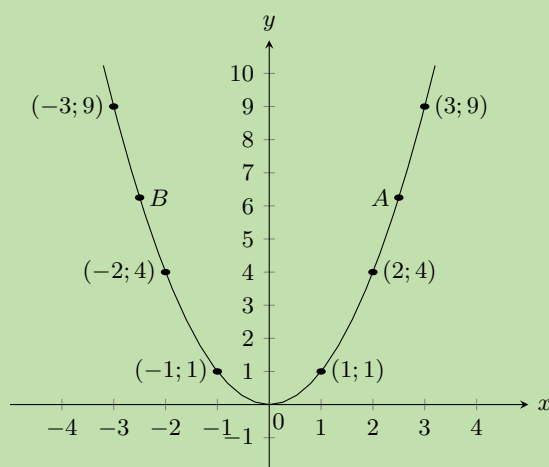
Step 1: Substitute values into the equation

$$\begin{aligned}
 f(x) &= x^2 \\
 f(-3) &= (-3)^2 = 9 \\
 f(-2) &= (-2)^2 = 4 \\
 f(-1) &= (-1)^2 = 1 \\
 f(0) &= (0)^2 = 0 \\
 f(1) &= (1)^2 = 1 \\
 f(2) &= (2)^2 = 4 \\
 f(3) &= (3)^2 = 9
 \end{aligned}$$

x	-3	-2	-1	0	1	2	3
$f(x)$	9	4	1	0	0	4	9

Step 2: Plot the points and join with a smooth curve

From the table, we get the following points: $(-3; 9)$, $(-2; 4)$, $(-1; 1)$, $(0; 0)$, $(1; 1)$, $(2; 4)$, $(3; 9)$



Step 3: Determine the domain and range

Domain: $x \in \mathbb{R}$

From the graph we see that for all values of x , $y \geq 0$.

Range: $\{y : y \in \mathbb{R}, y \geq 0\}$

Step 4: Find the axis of symmetry

f is symmetrical about the y -axis. Therefore the axis of symmetry of f is the line $x = 0$.

Step 5: Determine the x -value for which $f(x) = 6\frac{1}{4}$

$$\begin{aligned} f(x) &= \frac{25}{4} \\ \therefore \frac{25}{4} &= x^2 \\ x &= \pm \frac{5}{2} \\ &= \pm 2\frac{1}{2} \end{aligned}$$

See points A and B on the graph.

Step 6: Determine the intercept

The function f intercepts the axes at the origin $(0;0)$.

We notice that as the value of x increases from $-\infty$ to 0 , $f(x)$ decreases.

At the turning point $(0;0)$, $f(x) = 0$.

As the value of x increases from 0 to ∞ , $f(x)$ increases.

Investigation: The effects of a and q on a parabola.

Complete the table and plot the following graphs on the same system of axes:

1. $y_1 = x^2 - 2$
2. $y_2 = x^2 - 1$
3. $y_3 = x^2$
4. $y_4 = x^2 + 1$
5. $y_5 = x^2 + 2$

x	-2	-1	0	1	2
y_1					
y_2					
y_3					
y_4					
y_5					

Use your results to deduce the effect of q .

Complete the table and plot the following graphs on the same system of axes:

1. $y_6 = -2x^2$
2. $y_7 = -x^2$
3. $y_8 = x^2$
4. $y_9 = 2x^2$

x	-2	-1	0	1	2
y_6					
y_7					
y_8					
y_9					

Use your results to deduce the effect of a .

The effect of q

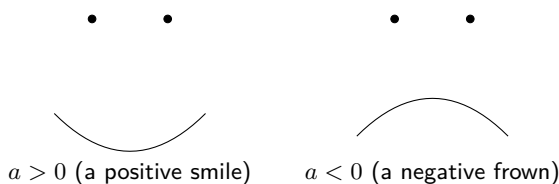
The effect of q is called a vertical shift because all points are moved the same distance in the same direction (it slides the entire graph up or down).

- For $q > 0$, the graph of $f(x)$ is shifted vertically upwards by q units. The turning point of $f(x)$ is above the y -axis.
- For $q < 0$, the graph of $f(x)$ is shifted vertically downwards by q units. The turning point of $f(x)$ is below the y -axis.

The effect of a

The sign of a determines the shape of the graph.

- For $a > 0$, the graph of $f(x)$ is a "smile" and has a minimum turning point at $(0; q)$. The graph of $f(x)$ is stretched vertically upwards; as a gets larger, the graph gets narrower.
For $0 < a < 1$, as a gets closer to 0, the graph of $f(x)$ gets wider.
- For $a < 0$, the graph of $f(x)$ is a "frown" and has a maximum turning point at $(0; q)$. The graph of $f(x)$ is stretched vertically downwards; as a gets smaller, the graph gets narrower.
For $-1 < a < 0$, as a gets closer to 0, the graph of $f(x)$ gets wider.



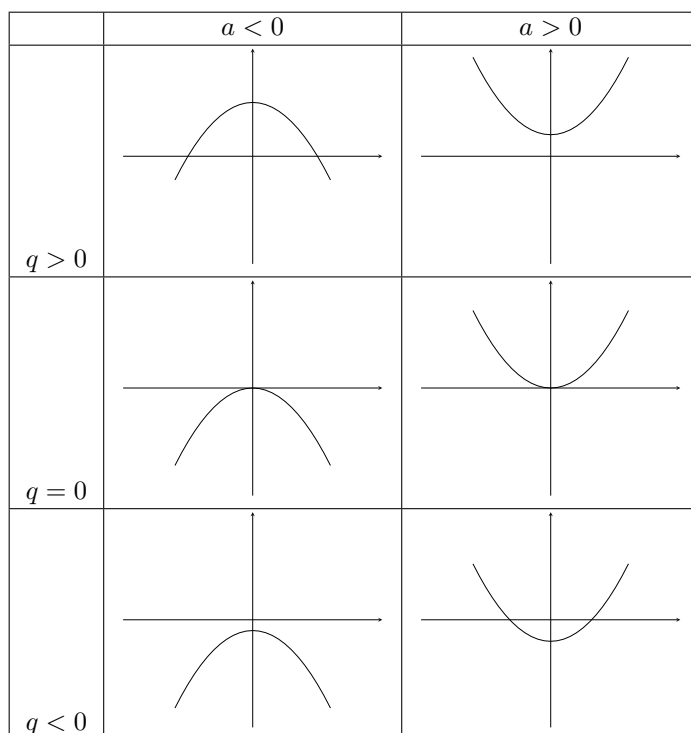


Table 6.2: The effect of a and q on a parabola.

VISIT:

You can use this [Phet simulation](#) to help you see the effects of changing a and q for a parabola.

Discovering the characteristics

EMA4M

The standard form of the equation of a parabola is $y = ax^2 + q$.

Domain and range

The domain is $\{x : x \in \mathbb{R}\}$ because there is no value for which $f(x)$ is undefined.

If $a > 0$ then we have:

$$\begin{aligned}
 x^2 &\geq 0 && \text{(Perfect square is always positive)} \\
 ax^2 &\geq 0 && \text{(since } a > 0) \\
 ax^2 + q &\geq q && \text{(add } q \text{ to both sides)} \\
 \therefore f(x) &\geq q
 \end{aligned}$$

Therefore if $a > 0$, the range is $[q; \infty)$. Similarly, if $a < 0$ then the range is $(-\infty; q]$.

Worked example 5: Domain and range of a parabola

QUESTION

If $g(x) = x^2 + 2$, determine the domain and range of the function.

SOLUTION

Step 1: Determine the domain

The domain is $\{x : x \in \mathbb{R}\}$ because there is no value for which $g(x)$ is undefined.

Step 2: Determine the range

The range of $g(x)$ can be calculated as follows:

$$\begin{aligned}x^2 &\geq 0 \\x^2 + 2 &\geq 2 \\g(x) &\geq 2\end{aligned}$$

Therefore the range is $\{g(x) : g(x) \geq 2\}$.

Intercepts

The y -intercept:

Every point on the y -axis has an x -coordinate of 0, therefore to calculate the y -intercept let $x = 0$.

For example, the y -intercept of $g(x) = x^2 + 2$ is given by setting $x = 0$:

$$\begin{aligned}g(x) &= x^2 + 2 \\g(0) &= 0^2 + 2 \\&= 2\end{aligned}$$

This gives the point $(0; 2)$.

The x -intercepts:

Every point on the x -axis has a y -coordinate of 0, therefore to calculate the x -intercept let $y = 0$.

For example, the x -intercepts of $g(x) = x^2 + 2$ are given by setting $y = 0$:

$$\begin{aligned}g(x) &= x^2 + 2 \\0 &= x^2 + 2 \\-2 &= x^2\end{aligned}$$

There is no real solution, therefore the graph of $g(x) = x^2 + 2$ does not have x -intercepts.

Turning points

The turning point of the function of the form $f(x) = ax^2 + q$ is determined by examining the range of the function.

- If $a > 0$, the graph of $f(x)$ is a “smile” and has a minimum turning point at $(0; q)$.
- If $a < 0$, the graph of $f(x)$ is a “frown” and has a maximum turning point at $(0; q)$.

Axes of symmetry

The axis of symmetry for functions of the form $f(x) = ax^2 + q$ is the y -axis, which is the line $x = 0$.

In order to sketch graphs of the form $f(x) = ax^2 + q$, we need to determine the following characteristics:

1. sign of a
2. y -intercept
3. x -intercept
4. turning point

Worked example 6: Sketching a parabola

QUESTION

Sketch the graph of $y = 2x^2 - 4$. Mark the intercepts and turning point.

SOLUTION

Step 1: Examine the standard form of the equation

We notice that $a > 0$. Therefore the graph is a “smile” and has a minimum turning point.

Step 2: Calculate the intercepts

For the y -intercept, let $x = 0$:

$$\begin{aligned} y &= 2x^2 - 4 \\ &= 2(0)^2 - 4 \\ &= -4 \end{aligned}$$

This gives the point $(0; -4)$.

For the x -intercepts, let $y = 0$:

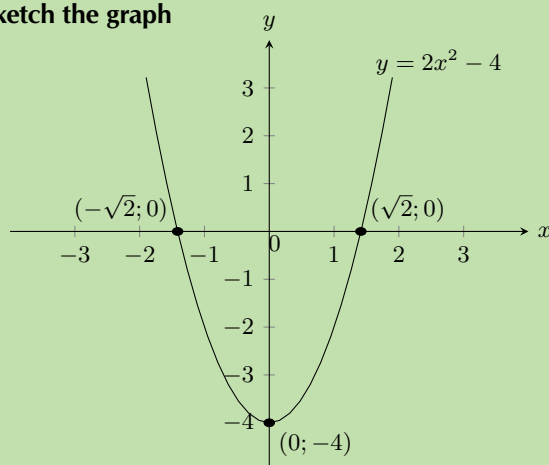
$$\begin{aligned} y &= 2x^2 - 4 \\ 0 &= 2x^2 - 4 \\ x^2 &= 2 \\ \therefore x &= \pm\sqrt{2} \end{aligned}$$

This gives the points $(-\sqrt{2}; 0)$ and $(\sqrt{2}; 0)$.

Step 3: Determine the turning point

From the standard form of the equation we see that the turning point is $(0; -4)$.

Step 4: Plot the points and sketch the graph



Domain: $\{x : x \in \mathbb{R}\}$

Range: $\{y : y \geq -4, y \in \mathbb{R}\}$

The axis of symmetry is the line $x = 0$.

QUESTION

Sketch the graph of $g(x) = -\frac{1}{2}x^2 - 3$. Mark the intercepts and the turning point.

SOLUTION

Step 1: Examine the standard form of the equation

We notice that $a < 0$. Therefore the graph is a “frown” and has a maximum turning point.

Step 2: Calculate the intercepts

For the y -intercept, let $x = 0$:

$$\begin{aligned} g(x) &= -\frac{1}{2}x^2 - 3 \\ g(0) &= -\frac{1}{2}(0)^2 - 3 \\ &= -3 \end{aligned}$$

This gives the point $(0; -3)$.

For the x -intercepts let $y = 0$:

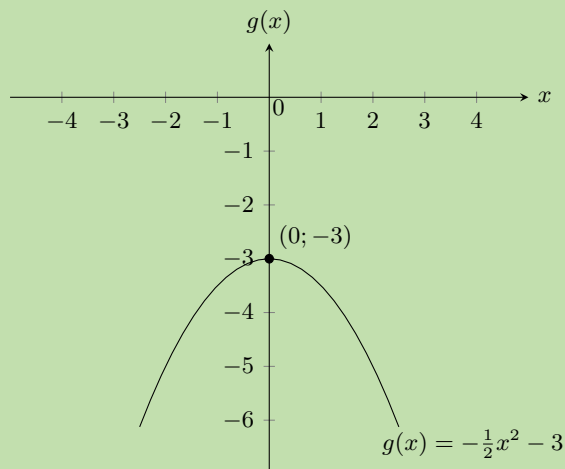
$$\begin{aligned} 0 &= -\frac{1}{2}x^2 - 3 \\ 3 &= -\frac{1}{2}x^2 \\ -2(3) &= x^2 \\ -6 &= x^2 \end{aligned}$$

There is no real solution, therefore there are no x -intercepts.

Step 3: Determine the turning point

From the standard form of the equation we see that the turning point is $(0; -3)$.

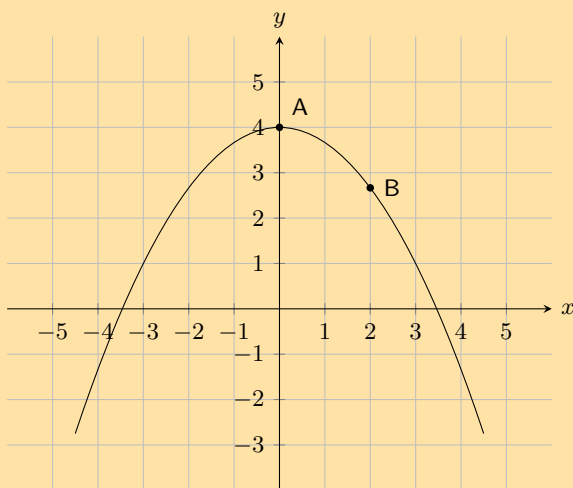
Step 4: Plot the points and sketch the graph



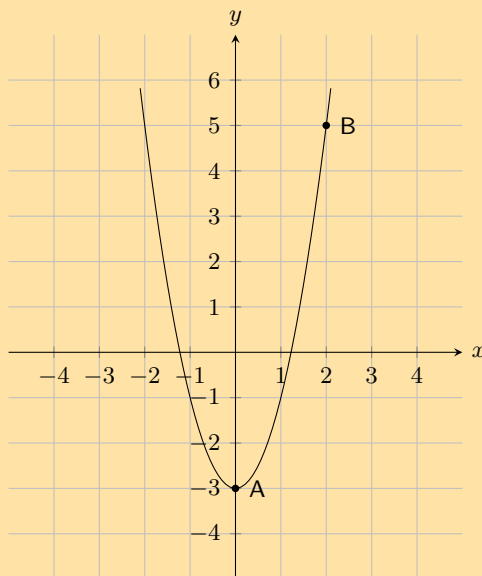
Domain: $x \in \mathbb{R}$. Range: $y \in (-\infty; -3]$. The axis of symmetry is the line $x = 0$.

Exercise 6 – 3:

1. The graph below shows a quadratic function with the following form: $y = ax^2 + q$.
Two points on the parabola are shown: **Point A**, the turning point of the parabola, at $(0; 4)$, and **Point B** is at $(2; \frac{8}{3})$. Calculate the values of a and q .

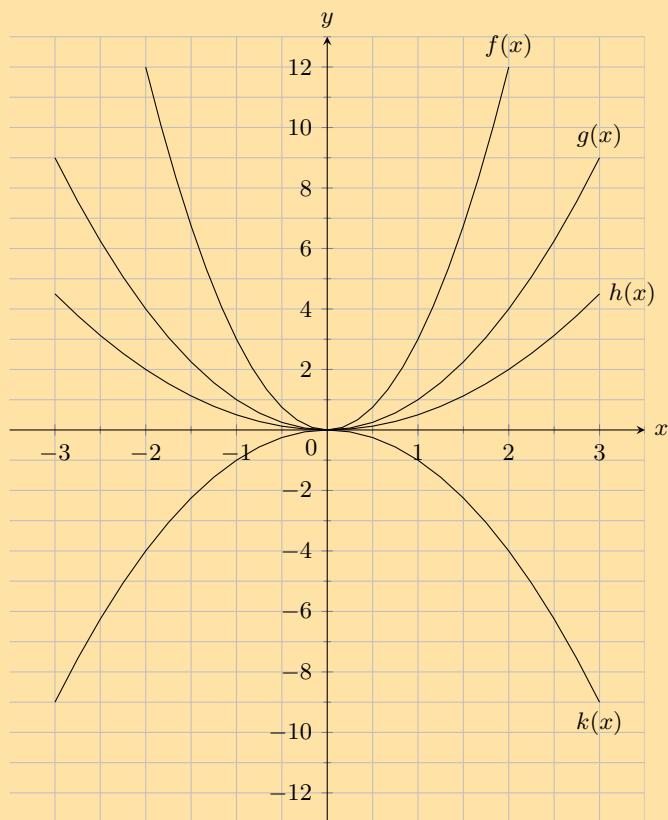


2. The graph below shows a quadratic function with the following form: $y = ax^2 + q$.
Two points on the parabola are shown: **Point A**, the turning point of the parabola, at $(0; -3)$, and **Point B** is at $(2; 5)$. Calculate the values of a and q .



3. Given the following equation:
 $y = 5x^2 - 2$
- Calculate the y -coordinate of the y -intercept.
 - Now calculate the x -intercepts. Your answer must be correct to 2 decimal places.
4. Given the following equation:
 $y = -2x^2 + 1$
- Calculate the y -coordinate of the y -intercept.
 - Now calculate the x -intercepts. Your answer must be correct to 2 decimal places.

5. Given the following graph, identify a function that matches each of the following equations:



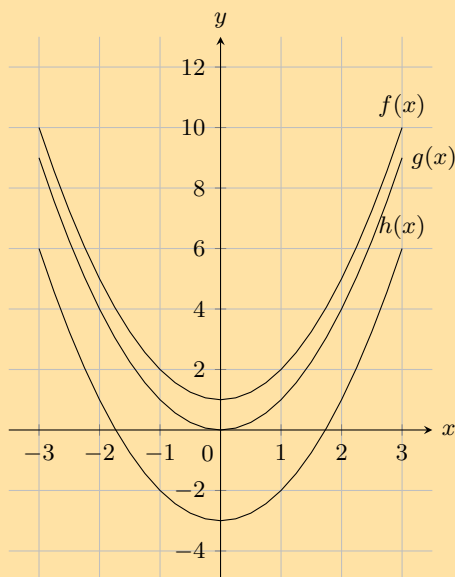
a) $y = 0,5x^2$

b) $y = x^2$

c) $y = 3x^2$

d) $y = -x^2$

6. Given the following graph, identify a function that matches each of the following equations:

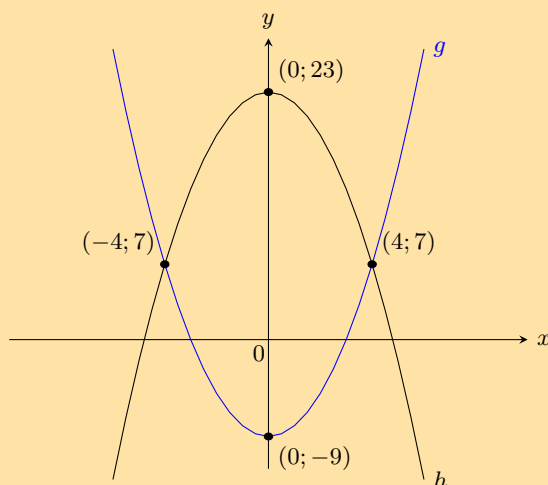


a) $y = x^2 - 3$

b) $y = x^2 + 1$

c) $y = x^2$

7. Two parabolas are drawn: $g : y = ax^2 + p$ and $h : y = bx^2 + q$.



- Find the values of a and p .
 - Find the values of b and q .
 - Find the values of x for which $g(x) \geq h(x)$.
 - For what values of x is g increasing?
- Show that if $a < 0$ the range of $f(x) = ax^2 + q$ is $\{f(x) : f(x) \leq q\}$.
 - Draw the graph of the function $y = -x^2 + 4$ showing all intercepts with the axes.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- 2FY6
- 2FY7
- 2FY8
- 2FY9
- 2FYB
- 2FYC
- 2FYD
- 2FYF
- 2FYG



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6.4 Hyperbolic functions

EMA4P

Functions of the form $y = \frac{1}{x}$

EMA4Q

Functions of the general form $y = \frac{a}{x} + q$ are called hyperbolic functions.

Worked example 8: Plotting a hyperbolic function

QUESTION

$$y = h(x) = \frac{1}{x}$$

Complete the following table for $h(x) = \frac{1}{x}$ and plot the points on a system of axes.

x	-3	-2	-1	$-\frac{1}{2}$	$-\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{2}$	1	2	3
$h(x)$	$-\frac{1}{3}$										

1. Join the points with smooth curves.
2. What happens if $x = 0$?
3. Explain why the graph consists of two separate curves.
4. What happens to $h(x)$ as the value of x becomes very small or very large?
5. The domain of $h(x)$ is $\{x : x \in \mathbb{R}, x \neq 0\}$. Determine the range.
6. About which two lines is the graph symmetrical?

SOLUTION

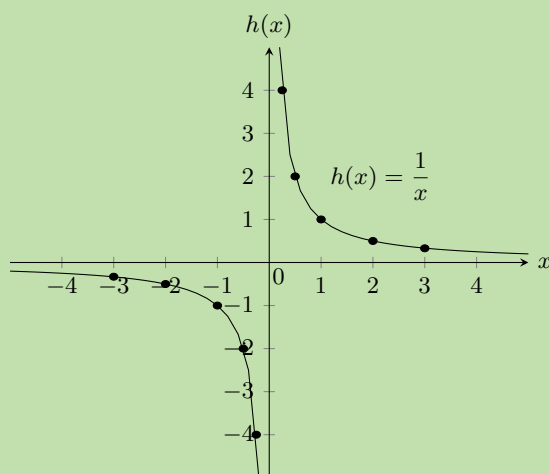
Step 1: Substitute values into the equation

$$\begin{aligned}
 h(x) &= \frac{1}{x} \\
 h(-3) &= \frac{1}{-3} = -\frac{1}{3} \\
 h(-2) &= \frac{1}{-2} = -\frac{1}{2} \\
 h(-1) &= \frac{1}{-1} = -1 \\
 h\left(-\frac{1}{2}\right) &= \frac{1}{-\frac{1}{2}} = -2 \\
 h\left(-\frac{1}{4}\right) &= \frac{1}{-\frac{1}{4}} = -4 \\
 h(0) &= \frac{1}{0} = \text{undefined} \\
 h\left(\frac{1}{4}\right) &= \frac{1}{\frac{1}{4}} = 4 \\
 h\left(\frac{1}{2}\right) &= \frac{1}{\frac{1}{2}} = 2 \\
 h(1) &= \frac{1}{1} = 1 \\
 h(2) &= \frac{1}{2} = \frac{1}{2} \\
 h(3) &= \frac{1}{3} = \frac{1}{3}
 \end{aligned}$$

x	-3	-2	-1	$-\frac{1}{2}$	$-\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{2}$	1	2	3
$h(x)$	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	-2	-4	undefined	4	2	1	$\frac{1}{2}$	$\frac{1}{3}$

Step 2: Plot the points and join with two smooth curves

From the table we get the following points: $(-3; -\frac{1}{3})$, $(-2; -\frac{1}{2})$, $(-1; -1)$, $(-\frac{1}{2}; -2)$, $(-\frac{1}{4}; -4)$, $(\frac{1}{4}; 4)$, $(\frac{1}{2}; 2)$, $(1; 1)$, $(2; \frac{1}{2})$, $(3; \frac{1}{3})$



For $x = 0$ the function h is undefined. This is called a discontinuity at $x = 0$.

$y = h(x) = \frac{1}{x}$ therefore we can write that $x \times y = 1$. Since the product of two positive numbers **and** the product of two negative numbers can be equal to 1, the graph lies in the first and third quadrants.

Step 3: Determine the asymptotes

As the value of x gets larger, the value of $h(x)$ gets closer to, but does not equal 0. This is a horizontal asymptote, the line $y = 0$. The same happens in the third quadrant; as x gets smaller $h(x)$ also approaches the negative x -axis asymptotically.

We also notice that there is a vertical asymptote, the line $x = 0$; as x gets closer to 0, $h(x)$ approaches the y -axis asymptotically.

Step 4: Determine the range

Domain: $\{x : x \in \mathbb{R}, x \neq 0\}$

From the graph, we see that y is defined for all values except 0.

Range: $\{y : y \in \mathbb{R}, y \neq 0\}$

Step 5: Determine the lines of symmetry

The graph of $h(x)$ has two axes of symmetry: the lines $y = x$ and $y = -x$. About these two lines, one half of the hyperbola is a mirror image of the other half.

Functions of the form $y = \frac{a}{x} + q$

EMA4R

Investigation: The effects of a and q on a hyperbola.

On the same set of axes, plot the following graphs:

1. $y_1 = \frac{1}{x} - 2$
2. $y_2 = \frac{1}{x} - 1$

$$3. y_3 = \frac{1}{x}$$

$$4. y_4 = \frac{1}{x} + 1$$

$$5. y_5 = \frac{1}{x} + 2$$

Use your results to deduce the effect of q .
 On the same set of axes, plot the following graphs:

$$1. y_6 = \frac{-2}{x}$$

$$2. y_7 = \frac{-1}{x}$$

$$3. y_8 = \frac{1}{x}$$

$$4. y_9 = \frac{2}{x}$$

Use your results to deduce the effect of a .

The effect of q

The effect of q is called a vertical shift because all points are moved the same distance in the same direction (it slides the entire graph up or down).

- For $q > 0$, the graph of $f(x)$ is shifted vertically upwards by q units.
- For $q < 0$, the graph of $f(x)$ is shifted vertically downwards by q units.

The horizontal asymptote is the line $y = q$ and the vertical asymptote is always the y -axis, the line $x = 0$.

The effect of a

The sign of a determines the shape of the graph.

- If $a > 0$, the graph of $f(x)$ lies in the first and third quadrants.

For $a > 1$, the graph of $f(x)$ will be further away from the axes than $y = \frac{1}{x}$.

For $0 < a < 1$, as a tends to 0, the graph moves closer to the axes than $y = \frac{1}{x}$.

- If $a < 0$, the graph of $f(x)$ lies in the second and fourth quadrants.

For $a < -1$, the graph of $f(x)$ will be further away from the axes than $y = -\frac{1}{x}$.

For $-1 < a < 0$, as a tends to 0, the graph moves closer to the axes than $y = -\frac{1}{x}$.

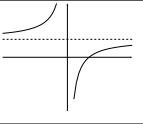
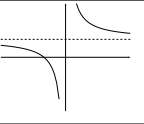
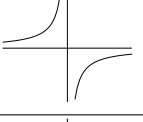
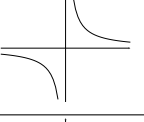
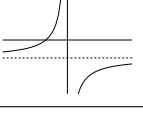
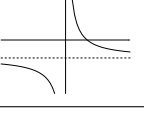
	$a < 0$	$a > 0$
$q > 0$		
$q = 0$		
$q < 0$		

Table 6.3: The effects of a and q on a hyperbola.

The standard form of a hyperbola is the equation $y = \frac{a}{x} + q$.

Domain and range

For $y = \frac{a}{x} + q$, the function is undefined for $x = 0$. The domain is therefore $\{x : x \in \mathbb{R}, x \neq 0\}$.

We see that $y = \frac{a}{x} + q$ can be rewritten as:

$$\begin{aligned} y &= \frac{a}{x} + q \\ y - q &= \frac{a}{x} \\ \text{If } x \neq 0 \text{ then: } (y - q)x &= a \\ x &= \frac{a}{y - q} \end{aligned}$$

This shows that the function is undefined only at $y = q$.

Therefore the range is $\{f(x) : f(x) \in \mathbb{R}, f(x) \neq q\}$

Worked example 9: Domain and range of a hyperbola

QUESTION

If $g(x) = \frac{2}{x} + 2$, determine the domain and range of the function.

SOLUTION

Step 1: Determine the domain

The domain is $\{x : x \in \mathbb{R}, x \neq 0\}$ because $g(x)$ is undefined only at $x = 0$.

Step 2: Determine the range

We see that $g(x)$ is undefined only at $y = 2$. Therefore the range is $\{g(x) : g(x) \in \mathbb{R}, g(x) \neq 2\}$

Intercepts

The y -intercept:

Every point on the y -axis has an x -coordinate of 0, therefore to calculate the y -intercept let $x = 0$.

For example, the y -intercept of $g(x) = \frac{2}{x} + 2$ is given by setting $x = 0$:

$$\begin{aligned} y &= \frac{2}{x} + 2 \\ y &= \frac{2}{0} + 2 \end{aligned}$$

which is undefined, therefore there is no y -intercept.

The x -intercept:

Every point on the x -axis has a y -coordinate of 0, therefore to calculate the x -intercept, let $y = 0$.

For example, the x -intercept of $g(x) = \frac{2}{x} + 2$ is given by setting $y = 0$:

$$\begin{aligned}y &= \frac{2}{x} + 2 \\0 &= \frac{2}{x} + 2 \\\frac{2}{x} &= -2 \\x &= \frac{2}{-2} \\&= -1\end{aligned}$$

This gives the point $(-1; 0)$.

Asymptotes

There are two asymptotes for functions of the form $y = \frac{a}{x} + q$.

The horizontal asymptote is the line $y = q$ and the vertical asymptote is always the y -axis, the line $x = 0$.

Axes of symmetry

There are two lines about which a hyperbola is symmetrical: $y = x + q$ and $y = -x + q$.

Sketching graphs of the form $y = \frac{a}{x} + q$

EMA4T

In order to sketch graphs of functions of the form, $y = f(x) = \frac{a}{x} + q$, we need to determine four characteristics:

1. sign of a
2. y -intercept
3. x -intercept
4. asymptotes

Worked example 10: Sketching a hyperbola

QUESTION

Sketch the graph of $g(x) = \frac{2}{x} + 2$. Mark the intercepts and the asymptotes.

SOLUTION

Step 1: Examine the standard form of the equation

We notice that $a > 0$ therefore the graph of $g(x)$ lies in the first and third quadrant.

Step 2: Calculate the intercepts

For the y -intercept, let $x = 0$:

$$g(x) = \frac{2}{x} + 2$$

$$g(0) = \frac{2}{0} + 2$$

This is undefined, therefore there is no y -intercept.

For the x -intercept, let $y = 0$:

$$g(x) = \frac{2}{x} + 2$$

$$0 = \frac{2}{x} + 2$$

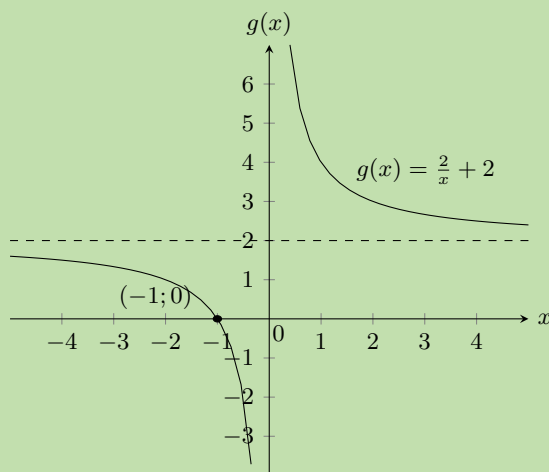
$$\frac{2}{x} = -2$$

$$\therefore x = -1$$

This gives the point $(-1; 0)$.

Step 3: Determine the asymptotes

The horizontal asymptote is the line $y = 2$. The vertical asymptote is the line $x = 0$.

Step 4: Sketch the graph

Domain: $\{x : x \in \mathbb{R}, x \neq 0\}$

Range: $\{y : y \in \mathbb{R}, y \neq 2\}$

Worked example 11: Sketching a hyperbola**QUESTION**

Sketch the graph of $y = \frac{-4}{x} + 7$.

SOLUTION

Step 1: Examine the standard form of the equation

We see that $a < 0$ therefore the graph lies in the second and fourth quadrants.

Step 2: Calculate the intercepts

For the y -intercept, let $x = 0$:

$$\begin{aligned}y &= \frac{-4}{x} + 7 \\&= \frac{-4}{0} + 7\end{aligned}$$

This is undefined, therefore there is no y -intercept.

For the x -intercept, let $y = 0$:

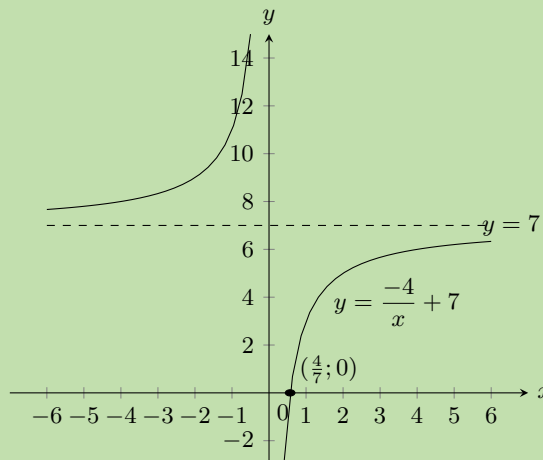
$$\begin{aligned}y &= \frac{-4}{x} + 7 \\0 &= \frac{-4}{x} + 7 \\ \frac{-4}{x} &= -7 \\\therefore x &= \frac{4}{7}\end{aligned}$$

This gives the point $\left(\frac{4}{7}; 0\right)$.

Step 3: Determine the asymptotes

The horizontal asymptote is the line $y = 7$. The vertical asymptote is the line $x = 0$.

Step 4: Sketch the graph



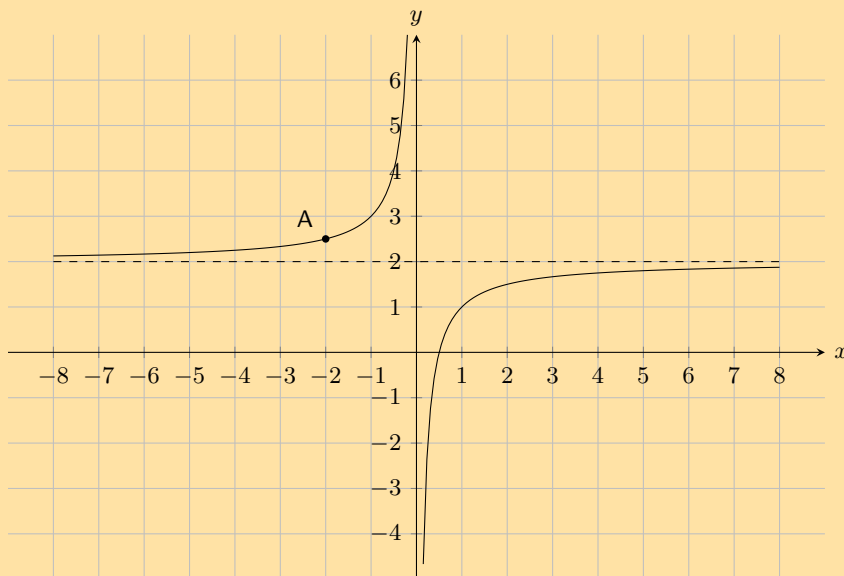
Domain: $\{x : x \in \mathbb{R}, x \neq 0\}$

Range: $\{y : y \in \mathbb{R}, y \neq 7\}$

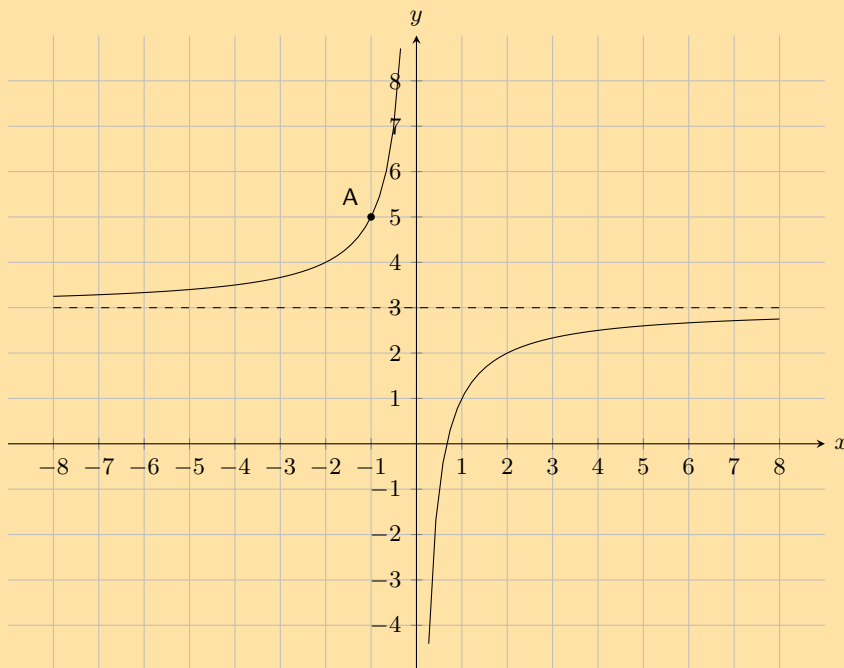
Axis of symmetry: $y = x + 7$ and $y = -x + 7$

Exercise 6 – 4:

1. The following graph shows a hyperbolic equation of the form $y = \frac{a}{x} + q$. **Point A** is shown at $\left(-2; \frac{5}{2}\right)$. Calculate the values of a and q .



2. The following graph shows a hyperbolic equation of the form $y = \frac{a}{x} + q$. **Point A** is shown at $(-1; 5)$. Calculate the values of a and q .



3. Given the following equation:

$$y = \frac{3}{x} + 2$$

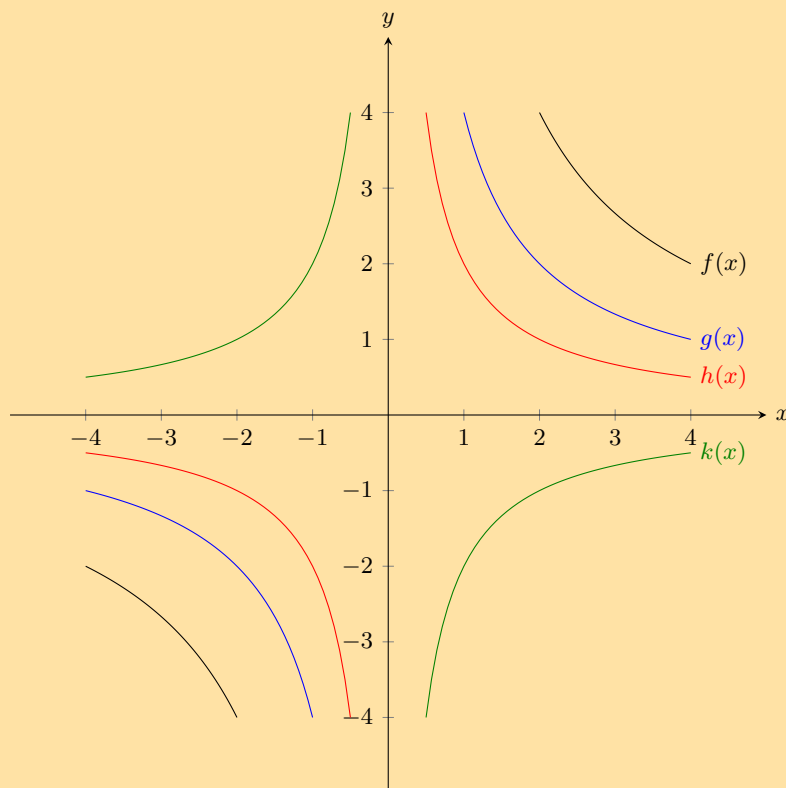
- Determine the location of the y -intercept.
- Determine the location of the x -intercept. Give your answer as a fraction.

4. Given the following equation:

$$y = -\frac{2}{x} - 2$$

- Determine the location of the y -intercept.
- Determine the location of the x -intercept.

5. Given the following graph, identify a function that matches each of the following equations:



a) $y = \frac{2}{x}$

b) $y = \frac{4}{x}$

c) $y = -\frac{2}{x}$

d) $y = \frac{8}{x}$

6. Given the function: $xy = -6$.

- Draw the graph.
- Does the point $(-2; 3)$ lie on the graph? Give a reason for your answer.
- If the x -value of a point on the graph is 0,25 what is the corresponding y -value?
- What happens to the y -values as the x -values become very large?
- Give the equation of the asymptotes.
- With the line $y = -x$ as a line of symmetry, what is the point symmetrical to $(-2; 3)$?

7. Given the function: $h(x) = \frac{8}{x}$.

- Draw the graph.
- How would the graph of $g(x) = \frac{8}{x} + 3$ compare with that of $h(x) = \frac{8}{x}$? Explain your answer fully.
- Draw the graph of $y = \frac{8}{x} + 3$ on the same set of axes, showing asymptotes, axes of symmetry and the coordinates of one point on the graph.

8. Sketch the functions given and describe the transformation performed on the first function to obtain the second function. Show all asymptotes.

a) $y = \frac{1}{x}$ and $\frac{3}{x}$

b) $y = \frac{6}{x}$ and $\frac{6}{x} - 1$

c) $y = \frac{5}{x}$ and $-\frac{5}{x}$

d) $y = \frac{1}{x}$ and $\frac{1}{2x}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2FYH 2. 2FYJ 3. 2FYK 4. 2FYM 5. 2FYN 6. 2FYP

7. 2FYQ 8a. 2FYR 8b. 2FYS 8c. 2FYT 8d. 2FV



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6.5 Exponential functions

EMA4V

Functions of the form $y = b^x$

EMA4W

Functions of the general form $y = ab^x + q$ are called exponential functions. In the equation a and q are constants and have different effects on the function.

Worked example 12: Plotting an exponential function

QUESTION

$y = f(x) = b^x$ for $b > 0$ and $b \neq 1$

Complete the following table for each of the functions and draw the graphs on the same system of axes: $f(x) = 2^x$, $g(x) = 3^x$, $h(x) = 5^x$.

	-2	-1	0	1	2
$f(x) = 2^x$					
$g(x) = 3^x$					
$h(x) = 5^x$					

- At what point do these graphs intersect?
- Explain why they do not cut the x -axis.
- Give the domain and range of $h(x)$.
- As x increases, does $h(x)$ increase or decrease?
- Which of these graphs increases at the slowest rate?
- For $y = k^x$ and $k > 1$, the greater the value of k , the steeper the curve of the graph. True or false?

Complete the following table for each of the functions and draw the graphs on the same system of axes:
 $F(x) = \left(\frac{1}{2}\right)^x$, $G(x) = \left(\frac{1}{3}\right)^x$, $H(x) = \left(\frac{1}{5}\right)^x$

	-2	-1	0	1	2
$F(x) = \left(\frac{1}{2}\right)^x$					
$G(x) = \left(\frac{1}{3}\right)^x$					
$H(x) = \left(\frac{1}{5}\right)^x$					

1. Give the y -intercept for each function.
2. Describe the relationship between the graphs $f(x)$ and $F(x)$.
3. Describe the relationship between the graphs $g(x)$ and $G(x)$.
4. Give the domain and range of $H(x)$.
5. For $y = \left(\frac{1}{k}\right)^x$ and $k > 1$, the greater the value of k , the steeper the curve of the graph. True or false?
6. Give the equation of the asymptote for the functions.

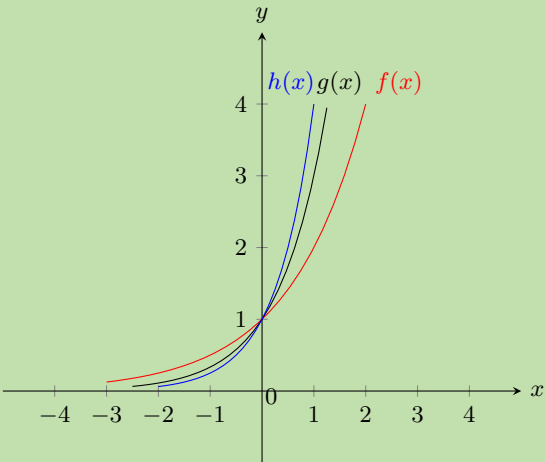
SOLUTION

Step 1: Substitute values into the equations

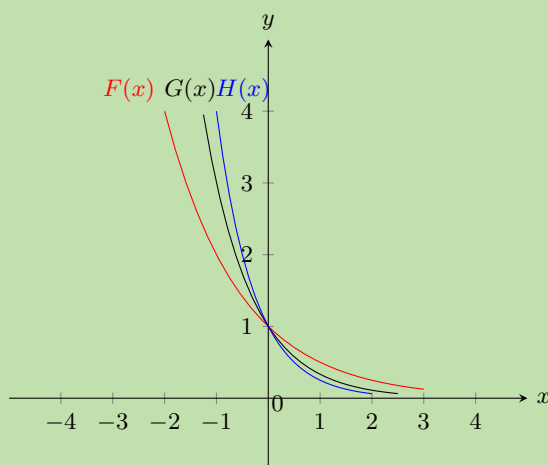
	-2	-1	0	1	2
$f(x) = 2^x$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
$g(x) = 3^x$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9
$h(x) = 5^x$	$\frac{1}{25}$	$\frac{1}{5}$	1	5	25

	-2	-1	0	1	2
$F(x) = \left(\frac{1}{2}\right)^x$	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$
$G(x) = \left(\frac{1}{3}\right)^x$	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$
$H(x) = \left(\frac{1}{5}\right)^x$	25	5	1	$\frac{1}{5}$	$\frac{1}{25}$

Step 2: Plot the points and join with a smooth curve



1. We notice that all graphs pass through the point $(0; 1)$. Any number with exponent 0 is equal to 1.
2. The graphs do not cut the x -axis because you can never get 0 by raising any non-zero number to the power of any other number.
3. Domain: $\{x : x \in \mathbb{R}\}$
Range: $\{y : y \in \mathbb{R}, y > 0\}$
4. As x increases, $h(x)$ increases.
5. $f(x) = 2^x$ increases at the slowest rate because it has the smallest base.
6. True: the greater the value of k ($k > 1$), the steeper the graph of $y = k^x$.



1. The y -intercept is the point $(0; 1)$ for all graphs. For any real number z : $z^0 = 1$ $z \neq 0$.
2. $F(x)$ is the reflection of $f(x)$ about the y -axis.
3. $G(x)$ is the reflection of $g(x)$ about the y -axis.
4. Domain: $\{x : x \in \mathbb{R}\}$
Range: $\{y : y \in \mathbb{R}, y > 0\}$
5. True: the greater the value of k ($k > 1$), the steeper the graph of $y = \left(\frac{1}{k}\right)^x$.
6. The equation of the horizontal asymptote is $y = 0$, the x -axis.

Functions of the form $y = ab^x + q$

EMA4X

Investigation: The effects of a , q and b on an exponential graph.

On the same set of axes, plot the following graphs ($a = 1$, $q = 0$ and b changes):

1. $y_1 = 2^x$
2. $y_2 = \left(\frac{1}{2}\right)^x$
3. $y_3 = 6^x$
4. $y_4 = \left(\frac{1}{6}\right)^x$

	-2	-1	0	1	2
$y_1 = 2^x$					
$y_2 = \left(\frac{1}{2}\right)^x$					
$y_3 = 6^x$					
$y_4 = \left(\frac{1}{6}\right)^x$					

Use your results to deduce the effect of b .

On the same set of axes, plot the following graphs ($b = 2$, $a = 1$ and q changes):

1. $y_5 = 2^x - 2$
2. $y_6 = 2^x - 1$
3. $y_7 = 2^x$
4. $y_8 = 2^x + 1$
5. $y_9 = 2^x + 2$

	-2	-1	0	1	2
$y_5 = 2^x - 2$					
$y_6 = 2^x - 1$					
$y_7 = 2^x$					
$y_8 = 2^x + 1$					
$y_9 = 2^x + 2$					

Use your results to deduce the effect of q .

On the same set of axes, plot the following graphs ($b = 2$, $q = 0$ and a changes).

1. $y_{10} = 1 \times 2^x$
2. $y_{11} = 2 \times 2^x$
3. $y_{12} = -1 \times 2^x$
4. $y_{13} = -2 \times 2^x$

	-2	-1	0	1	2
$y_{10} = 1 \times 2^x$					
$y_{11} = 2 \times 2^x$					
$y_{12} = -1 \times 2^x$					
$y_{13} = -2 \times 2^x$					

Use your results to deduce the effect of a .

The effect of q

The effect of q is called a vertical shift because all points are moved the same distance in the same direction (it slides the entire graph up or down).

- For $q > 0$, the graph is shifted vertically upwards by q units.
- For $q < 0$, the graph is shifted vertically downwards by q units.

The horizontal asymptote is shifted by q units and is the line $y = q$.

The effect of a

The sign of a determines whether the graph curves upwards or downwards.

For $0 < b < 1$:

- For $a > 0$, the graph curves downwards. It reflects the graph about the horizontal asymptote.
- For $a < 0$, the graph curves upwards.

For $b > 1$:

- For $a > 0$, the graph curves upwards.
- For $a < 0$, the graph curves downwards. It reflects the graph about the horizontal asymptote.

$b > 1$	$a < 0$	$a > 0$
$q > 0$		
$q < 0$		

Table 6.4: The effect of a and q on an exponential graph when $b > 1$.

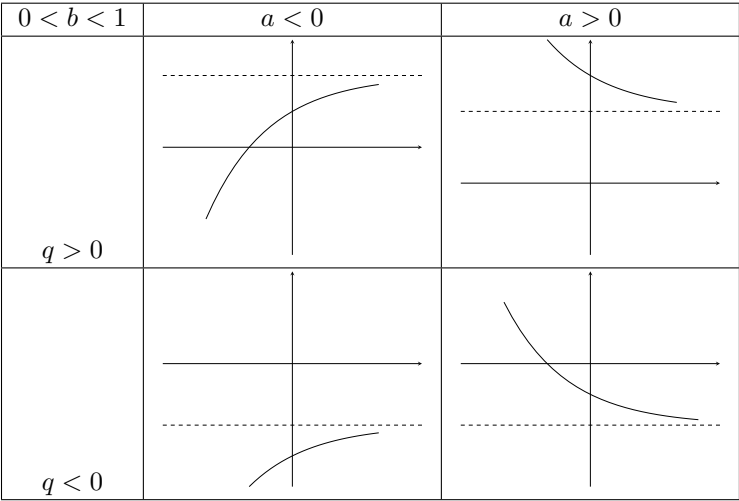


Table 6.5: The effect of a and q on an exponential graph when $0 < b < 1$.

The standard form of an exponential function is $y = ab^x + q$.

Domain and range

For $y = ab^x + q$, the function is defined for all real values of x . Therefore the domain is $\{x : x \in \mathbb{R}\}$.

The range of $y = ab^x + q$ is dependent on the sign of a .

For $a > 0$:

$$\begin{aligned} b^x &> 0 \\ ab^x &> 0 \\ ab^x + q &> q \\ f(x) &> q \end{aligned}$$

Therefore, for $a > 0$ the range is $\{f(x) : f(x) > q\}$.

For $a < 0$:

$$\begin{aligned} b^x &> 0 \\ ab^x &< 0 \\ ab^x + q &< q \\ f(x) &< q \end{aligned}$$

Therefore, for $a < 0$ the range is $\{f(x) : f(x) < q\}$.

QUESTION

Find the domain and range of $g(x) = 5 \cdot 2^x + 1$

SOLUTION

Step 1: Find the domain

The domain of $g(x) = 5 \times 2^x + 1$ is $\{x : x \in \mathbb{R}\}$.

Step 2: Find the range

$$\begin{aligned} 2^x &> 0 \\ 5 \times 2^x &> 0 \\ 5 \times 2^x + 1 &> 1 \end{aligned}$$

Therefore the range is $\{g(x) : g(x) > 1\}$.

Intercepts

The y -intercept.

For the y -intercept, let $x = 0$:

$$\begin{aligned} y &= ab^x + q \\ &= ab^0 + q \\ &= a(1) + q \\ &= a + q \end{aligned}$$

For example, the y -intercept of $g(x) = 5 \times 2^x + 1$ is given by setting $x = 0$:

$$\begin{aligned} y &= 5 \times 2^x + 1 \\ &= 5 \times 2^0 + 1 \\ &= 5 + 1 \\ &= 6 \end{aligned}$$

This gives the point $(0; 6)$.

The x -intercept:

For the x -intercept, let $y = 0$.

For example, the x -intercept of $g(x) = 5 \times 2^x + 1$ is given by setting $y = 0$:

$$\begin{aligned} y &= 5 \times 2^x + 1 \\ 0 &= 5 \times 2^x + 1 \\ -1 &= 5 \times 2^x \\ 2^x &= -\frac{1}{5} \end{aligned}$$

There is no real solution. Therefore, the graph of $g(x)$ does not have any x -intercepts.

Asymptotes

Exponential functions of the form $y = ab^x + q$ have a single horizontal asymptote, the line $x = q$.

Sketching graphs of the form $y = ab^x + q$

EMA4Z

In order to sketch graphs of functions of the form, $y = ab^x + q$, we need to determine four characteristics:

1. sign of a
2. y -intercept
3. x -intercept
4. asymptote

VISIT:

The following video shows some examples of sketching exponential functions.

▶ See video: [2FYW](https://www.everythingmaths.co.za) at www.everythingmaths.co.za

Worked example 14: Sketching an exponential function

QUESTION

Sketch the graph of $g(x) = 3 \times 2^x + 2$. Mark the intercept and the asymptote.

SOLUTION

Step 1: Examine the standard form of the equation

From the equation we see that $a > 1$, therefore the graph curves upwards. $q > 0$ therefore the graph is shifted vertically upwards by 2 units.

Step 2: Calculate the intercepts

For the y -intercept, let $x = 0$:

$$\begin{aligned}y &= 3 \times 2^x + 2 \\&= 3 \times 2^0 + 2 \\&= 3 + 2 \\&= 5\end{aligned}$$

This gives the point $(0; 5)$.

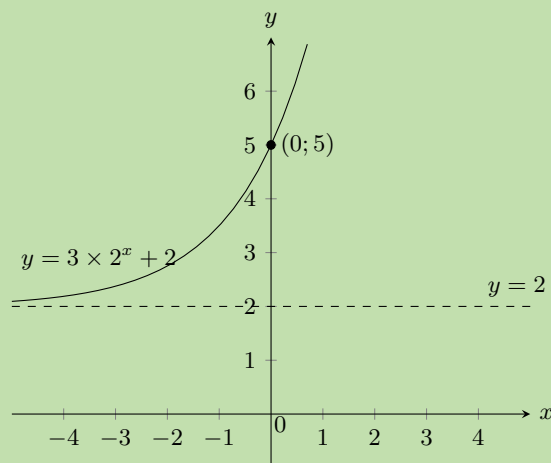
For the x -intercept, let $y = 0$:

$$\begin{aligned}y &= 3 \times 2^x + 2 \\0 &= 3 \times 2^x + 2 \\-2 &= 3 \times 2^x \\2^x &= -\frac{2}{3}\end{aligned}$$

There is no real solution, therefore there is no x -intercept.

Step 3: Determine the asymptote

The horizontal asymptote is the line $y = 2$.

Step 4: Plot the points and sketch the graph

Domain: $\{x : x \in \mathbb{R}\}$ Range: $\{g(x) : g(x) > 2\}$

Note that there is no axis of symmetry for exponential functions.

Worked example 15: Sketching an exponential graph**QUESTION**

Sketch the graph of $y = -2 \times 3^x + 6$

SOLUTION**Step 1: Examine the standard form of the equation**

From the equation we see that $a < 0$ therefore the graph curves downwards. $q > 0$ therefore the graph is shifted vertically upwards by 6 units.

Step 2: Calculate the intercepts

For the y -intercept, let $x = 0$:

$$\begin{aligned} y &= -2 \times 3^x + 6 \\ &= -2 \times 3^0 + 6 \\ &= 4 \end{aligned}$$

This gives the point $(0; 4)$.

For the x -intercept, let $y = 0$:

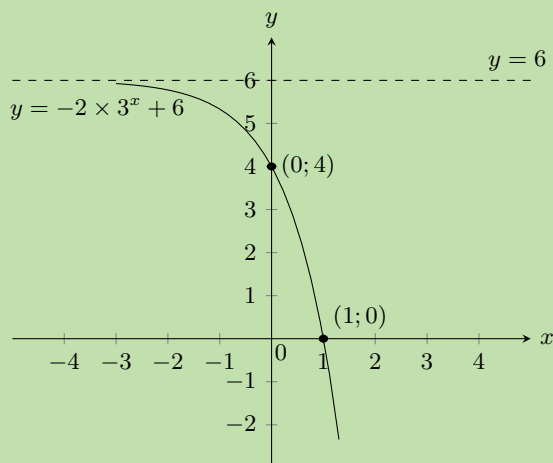
$$\begin{aligned} y &= -2 \times 3^x + 6 \\ 0 &= -2 \times 3^x + 6 \\ -6 &= -2 \times 3^x \\ 3^1 &= 3^x \\ \therefore x &= 1 \end{aligned}$$

This gives the point $(1; 0)$.

Step 3: Determine the asymptote

The horizontal asymptote is the line $y = 6$.

Step 4: Plot the points and sketch the graph



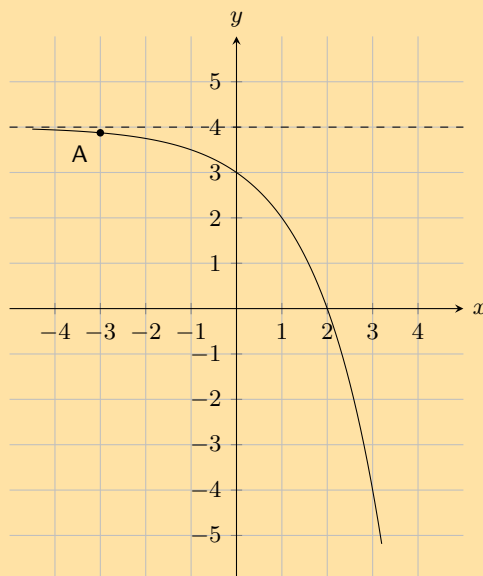
Domain: $\{x : x \in \mathbb{R}\}$ Range: $\{g(x) : g(x) < 6\}$

Exercise 6 – 5:

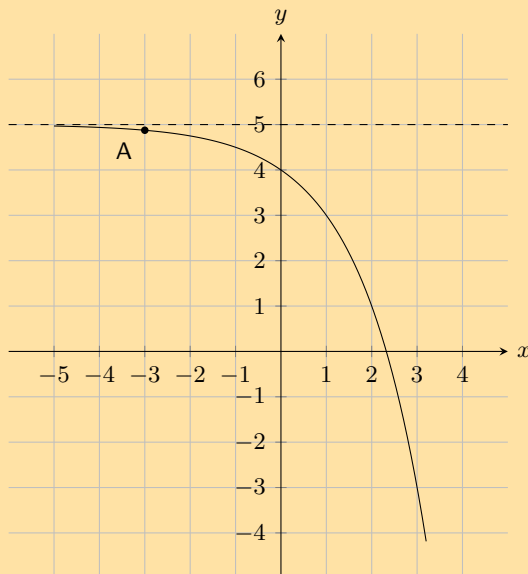
1. Given the following equation:

$$y = -\frac{2}{3} \cdot (3)^x + 1$$

- Calculate the y -intercept. Your answer must be correct to 2 decimal places.
 - Now calculate the x -intercept. Estimate your answer to one decimal place if necessary.
2. The graph here shows an exponential function with the equation $y = a \cdot 2^x + q$. One point is given on the curve: **Point A** is at $(-3; 3,875)$. Determine the values of a and q , correct to the nearest integer.



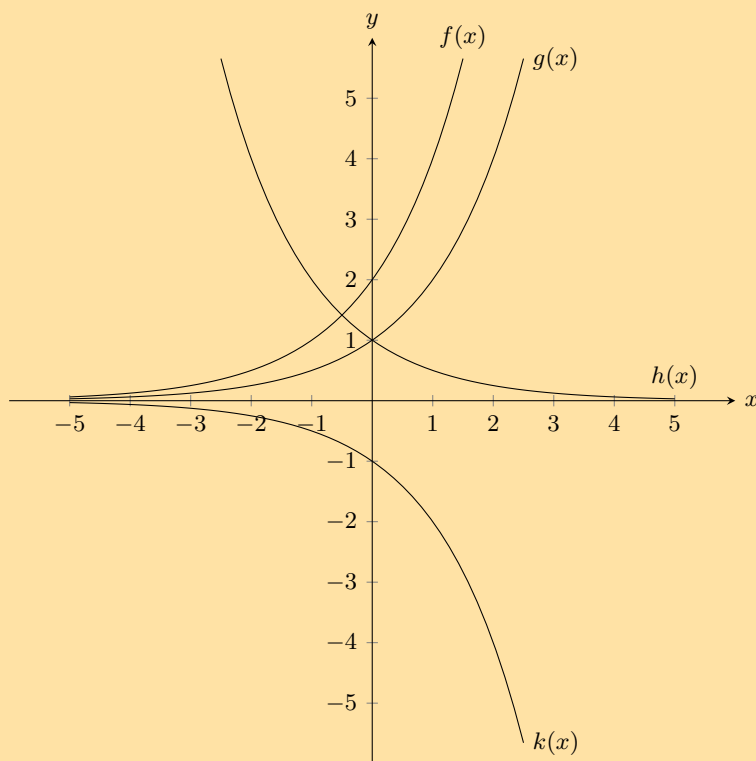
3. Below you see a graph of an exponential function with the equation $y = a \cdot 2^x + q$. One point is given on the curve: **Point A** is at $(-3; 4,875)$. Calculate the values of a and q , correct to the nearest integer.



4. Given the following equation:

$$y = \frac{1}{4} \cdot (4)^x - 1$$

- Calculate the y -intercept. Your answer must be correct to 2 decimal places.
 - Now calculate the x -intercept.
5. Given the following graph, identify a function that matches each of the following equations:



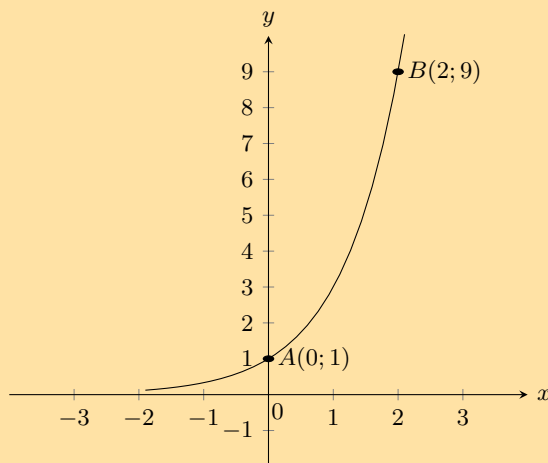
a) $y = 2^x$

b) $y = -2^x$

c) $y = 2 \cdot 2^x$

d) $y = \left(\frac{1}{2}\right)^x$

6. Given the functions $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$.
- Draw the graphs on the same set of axes.
 - Is the x -axis an asymptote or an axis of symmetry to both graphs? Explain your answer.
 - Which graph can be described by the equation $y = 2^{-x}$? Explain your answer.
 - Solve the equation $2^x = \left(\frac{1}{2}\right)^x$ graphically and check your answer is correct by using substitution.
7. The curve of the exponential function f in the accompanying diagram cuts the y -axis at the point $A(0; 1)$ and passes through the point $B(2; 9)$.



- Determine the equation of the function f .
- Determine the equation of the function $h(x)$, the reflection of $f(x)$ in the x -axis.
- Determine the range of $h(x)$.
- Determine the equation of the function $g(x)$, the reflection of $f(x)$ in the y -axis.
- Determine the equation of the function $j(x)$ if $j(x)$ is a vertical stretch of $f(x)$ by $+2$ units.
- Determine the equation of the function $k(x)$ if $k(x)$ is a vertical shift of $f(x)$ by -3 units.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2FYX 2. 2FYY 3. 2FYZ 4. 2FZ2 5. 2FZ3 6. 2FZ4 7. 2FZ5



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6.6 Trigonometric functions

EMA52

This section describes the graphs of trigonometric functions.

Sine function

EMA53

Functions of the form $y = \sin \theta$

EMA54

Worked example 16: Plotting a sine graph

QUESTION

$$y = f(\theta) = \sin \theta \quad [0^\circ \leq \theta \leq 360^\circ]$$

Use your calculator to complete the following table.

Choose an appropriate scale and plot the values of θ on the x -axis and of $\sin \theta$ on the y -axis. Round answers to 2 decimal places.

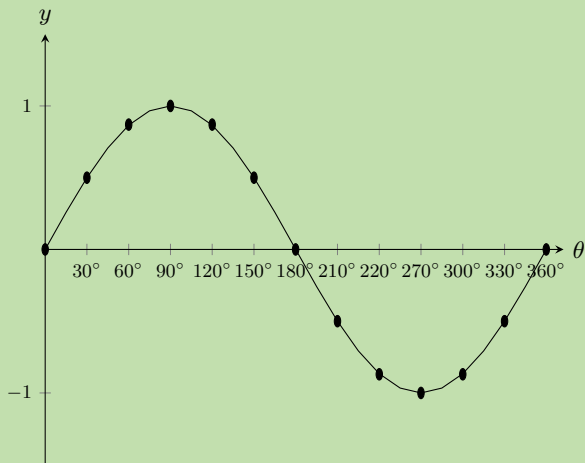
θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\sin \theta$													

SOLUTION

Step 1: Substitute values for θ

θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\sin \theta$	0	0,5	0,87	1	0,87	0,5	0	-0,5	-0,87	-1	-0,87	-0,5	0

Step 2: Plot the points and join with a smooth curve



Notice the wave shape of the graph. Each complete wave takes 360° to complete. This is called the period. The height of the wave above and below the x -axis is called the graph's amplitude. The maximum value of $y = \sin \theta$ is 1 and the minimum value is -1 .

Domain: $[0^\circ; 360^\circ]$

Range: $[-1; 1]$

x -intercepts: $(0^\circ; 0)$, $(180^\circ; 0)$, $(360^\circ; 0)$

y -intercept: $(0^\circ; 0)$

Maximum turning point: $(90^\circ; 1)$

Minimum turning point: $(270^\circ; -1)$

Investigation: The effects of a and q on a sine graph

In the equation, $y = a \sin \theta + q$, a and q are constants and have different effects on the graph. On the same set of axes, plot the following graphs for $0^\circ \leq \theta \leq 360^\circ$:

1. $y_1 = \sin \theta - 2$
2. $y_2 = \sin \theta - 1$
3. $y_3 = \sin \theta$
4. $y_4 = \sin \theta + 1$
5. $y_5 = \sin \theta + 2$

Use your results to deduce the effect of q .

On the same set of axes, plot the following graphs for $0^\circ \leq \theta \leq 360^\circ$:

1. $y_6 = -2 \sin \theta$
2. $y_7 = -\sin \theta$
3. $y_8 = \sin \theta$
4. $y_9 = 2 \sin \theta$

Use your results to deduce the effect of a .

The effect of q

The effect of q is called a vertical shift because the whole sine graph shifts up or down by q units.

- For $q > 0$, the graph is shifted vertically upwards by q units.
- For $q < 0$, the graph is shifted vertically downwards by q units.

The effect of a

The value of a affects the amplitude of the graph; the height of the peaks and the depth of the troughs.

- For $a > 1$, there is a vertical stretch and the amplitude increases.
For $0 < a < 1$, the amplitude decreases
- For $a < 0$, there is a reflection about the x -axis.
For $-1 < a < 0$, there is a reflection about the x -axis and the amplitude decreases.
For $a < -1$, there is a reflection about the x -axis and the amplitude increases.

Note that amplitude is always positive.

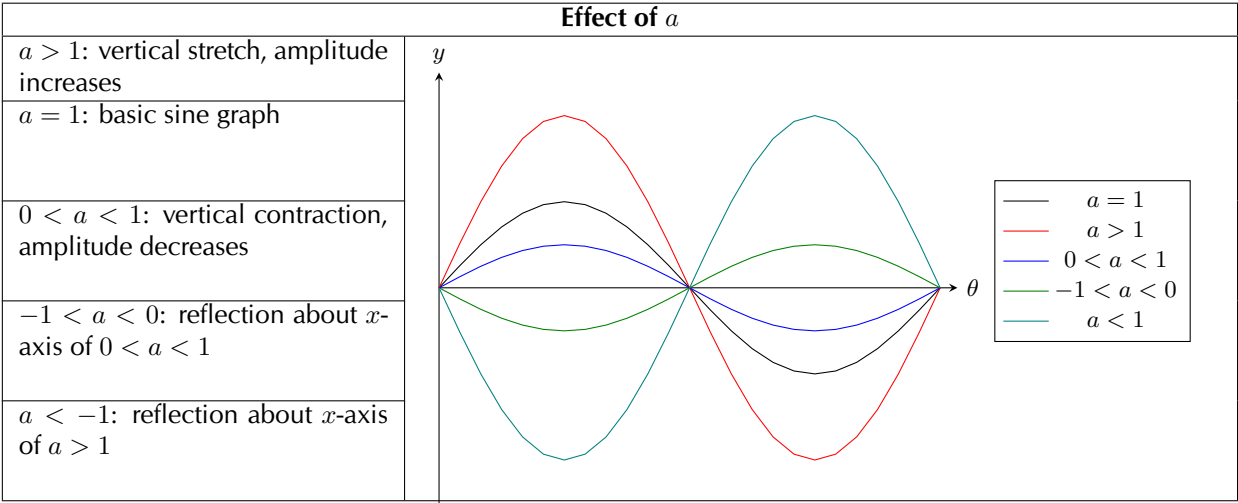


Table 6.6: The effect of a on a sine graph.

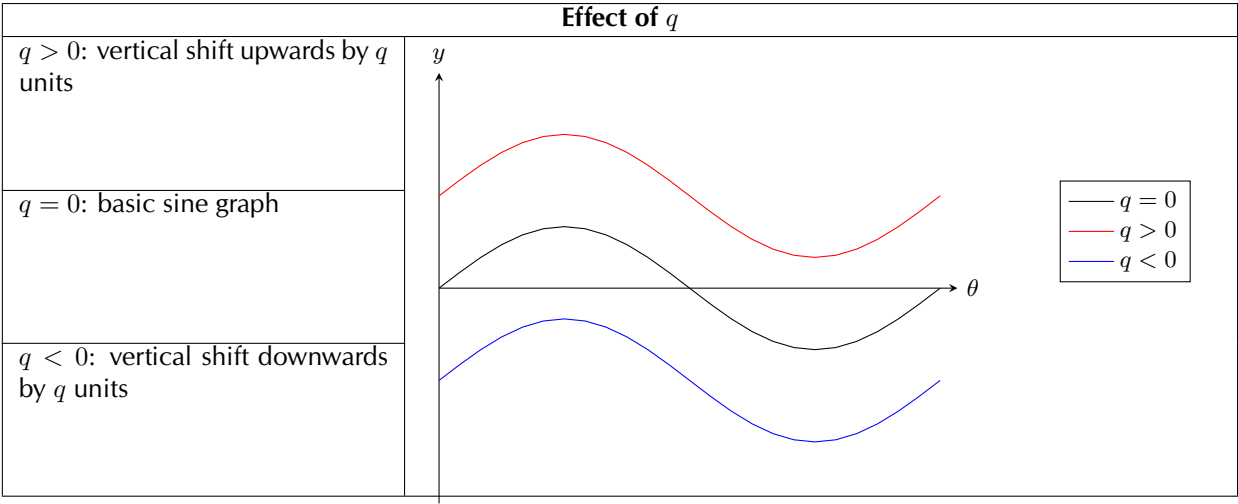


Table 6.7: The effect of q on a sine graph.

Discovering the characteristics

EMA56

Domain and range

For $f(\theta) = a \sin \theta + q$, the domain is $[0^\circ; 360^\circ]$

The range of $f(\theta) = a \sin \theta + q$ depends on the values of a and q .

For $a > 0$:

$$\begin{aligned}
 -1 &\leq \sin \theta \leq 1 \\
 -a &\leq a \sin \theta \leq a \\
 -a + q &\leq a \sin \theta + q \leq a + q \\
 -a + q &\leq f(\theta) \leq a + q
 \end{aligned}$$

For all values of θ , $f(\theta)$ is always between $-a + q$ and $a + q$.

Therefore for $a > 0$, the range of $f(\theta) = a \sin \theta + q$ is $\{f(\theta) : f(\theta) \in [-a + q, a + q]\}$

Similarly, for $a < 0$, the range of $f(\theta) = a \sin \theta + q$ is $\{f(\theta) : f(\theta) \in [a + q, -a + q]\}$

Period

The period of $y = a \sin \theta + q$ is 360° . This means that one sine wave is completed in 360° .

Intercepts

The y -intercept of $f(\theta) = a \sin \theta + q$ is simply the value of $f(\theta)$ at $\theta = 0^\circ$

$$\begin{aligned}y &= f(0^\circ) \\&= a \sin 0^\circ + q \\&= a(0) + q \\&= q\end{aligned}$$

This gives the point $(0; q)$

Important: when sketching trigonometric graphs, always start with the basic graph and then consider the effects of a and q .

Worked example 17: Sketching a sine graph

QUESTION

Sketch the graph of $f(\theta) = 2 \sin \theta + 3$ for $\theta \in [0^\circ; 360^\circ]$.

SOLUTION

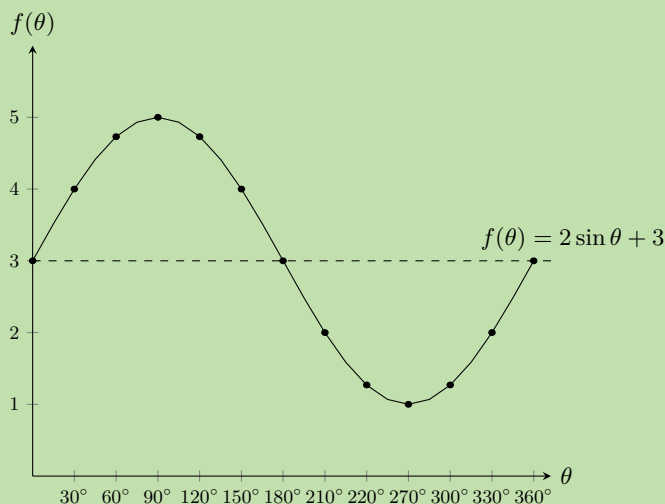
Step 1: Examine the standard form of the equation

From the equation we see that $a > 1$ so the graph is stretched vertically. We also see that $q > 0$ so the graph is shifted vertically upwards by 3 units.

Step 2: Substitute values for θ

θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$f(\theta)$	3	4	4,73	5	4,73	4	3	2	1,27	1	1,27	2	3

Step 3: Plot the points and join with a smooth curve



Domain: $[0^\circ; 360^\circ]$ Range: $[1; 5]$

x -intercepts: none y -intercepts: $(0^\circ; 3)$

Maximum turning point: $(90^\circ; 5)$ Minimum turning point: $(270^\circ; 1)$

Worked example 18: Plotting a cosine graph

QUESTION

$$y = f(\theta) = \cos \theta \quad [0^\circ \leq \theta \leq 360^\circ]$$

Use your calculator to complete the following table.

Choose an appropriate scale and plot the values of θ on the x -axis and $\cos \theta$ on the y -axis. Round your answers to 2 decimal places.

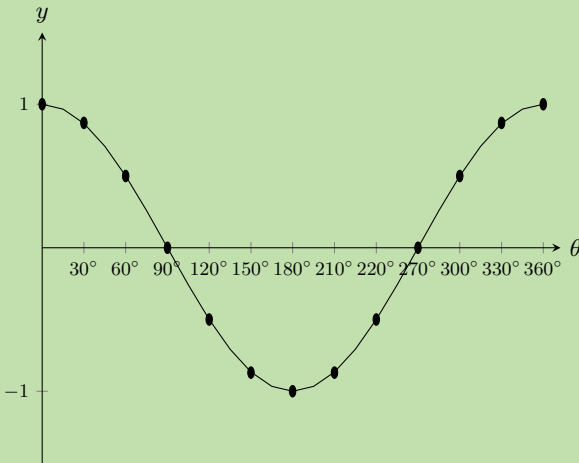
θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\cos \theta$													

SOLUTION

Step 1: Substitute values for θ

θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\cos \theta$	1	0,87	0,5	0	-0,5	-0,87	-1	-0,87	-0,5	0	0,5	0,87	1

Step 2: Plot the points and join with a smooth curve



Notice the similar wave shape of the graph. The period is also 360° and the amplitude is 1. The maximum value of $y = \cos \theta$ is 1 and the minimum value is -1 .

Domain: $[0^\circ; 360^\circ]$ Range: $[-1; 1]$

x -intercepts: $(90^\circ; 0)$, $(270^\circ; 0)$ y -intercepts: $(0^\circ; 1)$

Maximum turning points: $(0^\circ; 1)$, $(360^\circ; 1)$ Minimum turning point: $(180^\circ; -1)$

Investigation: The effects of a and q on a cosine graph

In the equation, $y = a \cos \theta + q$, a and q are constants and have different effects on the graph.

On the same set of axes, plot the following graphs for $0^\circ \leq \theta \leq 360^\circ$:

1. $y_1 = \cos \theta - 2$
2. $y_2 = \cos \theta - 1$
3. $y_3 = \cos \theta$
4. $y_4 = \cos \theta + 1$
5. $y_5 = \cos \theta + 2$

Use your results to deduce the effect of q .

On the same set of axes, plot the following graphs for $0^\circ \leq \theta \leq 360^\circ$:

1. $y_6 = -2 \cos \theta$
2. $y_7 = -\cos \theta$
3. $y_8 = \cos \theta$
4. $y_9 = 2 \cos \theta$

Use your results to deduce the effect of a .

The effect of q

The effect of q is called a vertical shift because the whole cosine graph shifts up or down by q units.

- For $q > 0$, the graph is shifted vertically upwards by q units.
- For $q < 0$, the graph is shifted vertically downwards by q units.

The effect of a

The value of a affects the amplitude of the graph; the height of the peaks and the depth of the troughs.

- For $a > 0$, there is a vertical stretch and the amplitude increases.
For $0 < a < 1$, the amplitude decreases.
- For $a < 0$, there is a reflection about the x -axis.
For $-1 < a < 0$, there is a reflection about the x -axis and the amplitude decreases.
For $a < -1$, there is a reflection about the x -axis and the amplitude increases.

Note that amplitude is always positive.

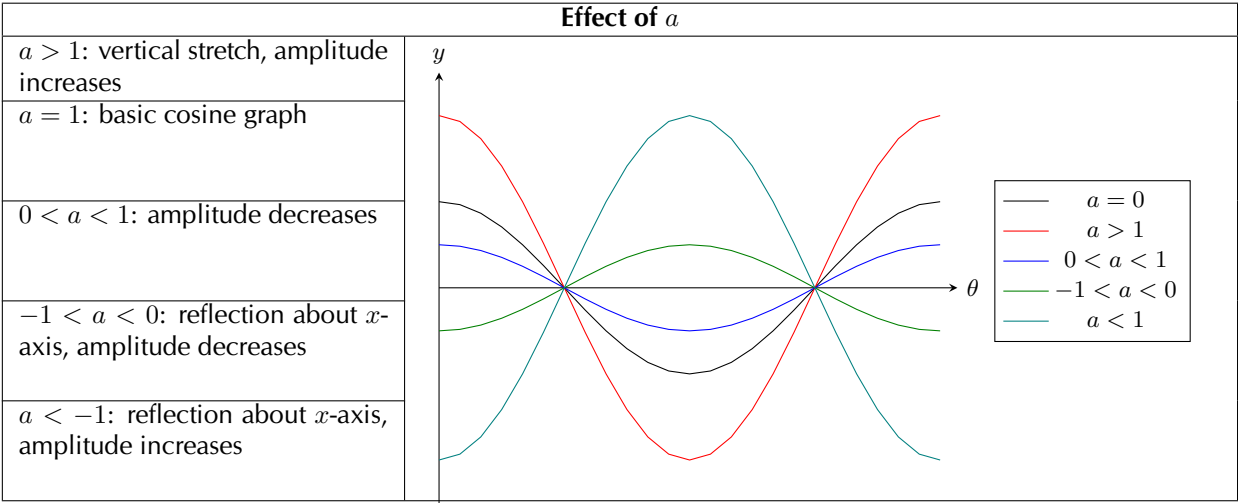


Table 6.8: The effect of a on a cosine graph.

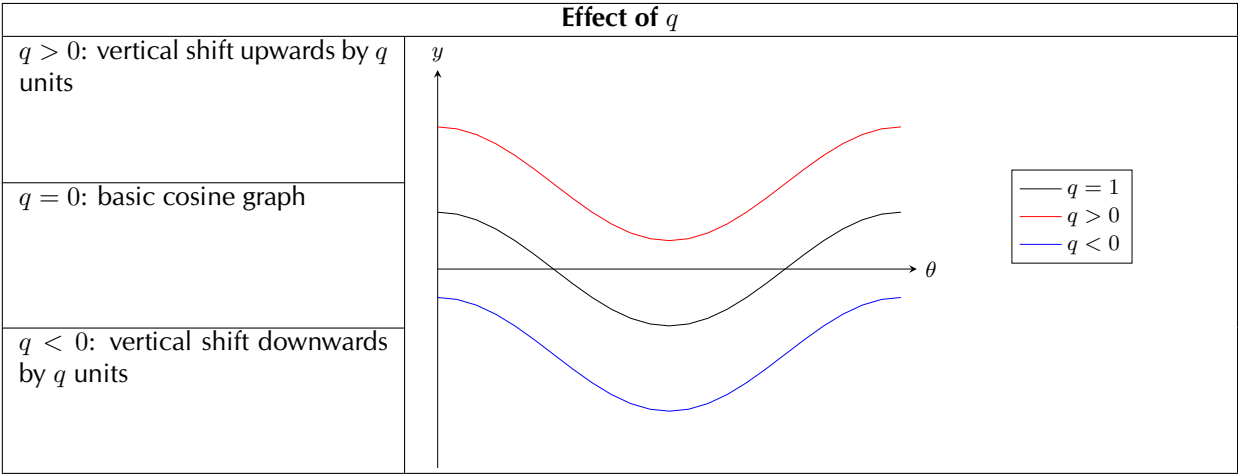


Table 6.9: The effect of q on a cosine graph.

Discovering the characteristics

EMA5B

Domain and range

For $f(\theta) = a \cos \theta + q$, the domain is $[0^\circ; 360^\circ]$

It is easy to see that the range of $f(\theta)$ will be the same as the range of $a \sin \theta + q$. This is because the maximum and minimum values of $a \cos \theta + q$ will be the same as the maximum and minimum values of $a \sin \theta + q$.

For $a > 0$ the range of $f(\theta) = a \cos \theta + q$ is $\{f(\theta) : f(\theta) \in [-a + q; a + q]\}$

For $a < 0$ the range of $f(\theta) = a \cos \theta + q$ is $\{f(\theta) : f(\theta) \in [a + q; -a + q]\}$

Period

The period of $y = a \cos \theta + q$ is 360° . This means that one cosine wave is completed in 360° .

Intercepts

The y -intercept of $f(\theta) = a \cos \theta + q$ is calculated in the same way as for sine.

$$\begin{aligned}
 y &= f(0^\circ) \\
 &= a \cos 0^\circ + q \\
 &= a(1) + q \\
 &= a + q
 \end{aligned}$$

This gives the point $(0^\circ; a + q)$.

Worked example 19: Sketching a cosine graph

QUESTION

Sketch the graph of $f(\theta) = 2 \cos \theta + 3$ for $\theta \in [0^\circ; 360^\circ]$.

SOLUTION

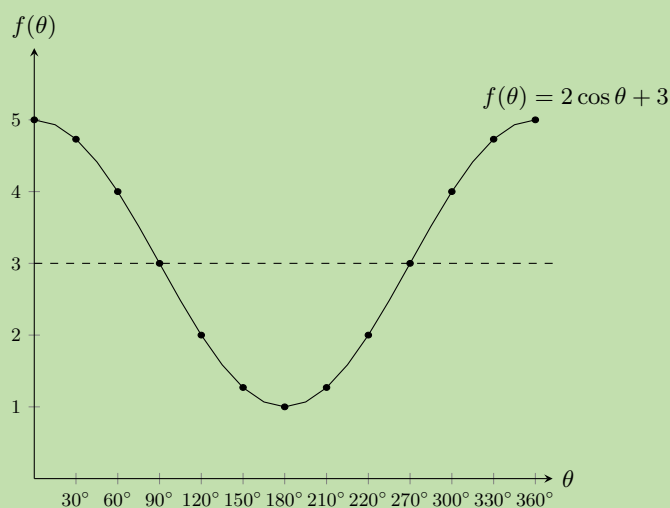
Step 1: Examine the standard form of the equation

From the equation we see that $a > 1$ so the graph is stretched vertically. We also see that $q > 0$ so the graph is shifted vertically upwards by 3 units.

Step 2: Substitute values for θ

θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$f(\theta)$	5	4,73	4	3	2	1,27	1	1,27	2	3	4	4,73	5

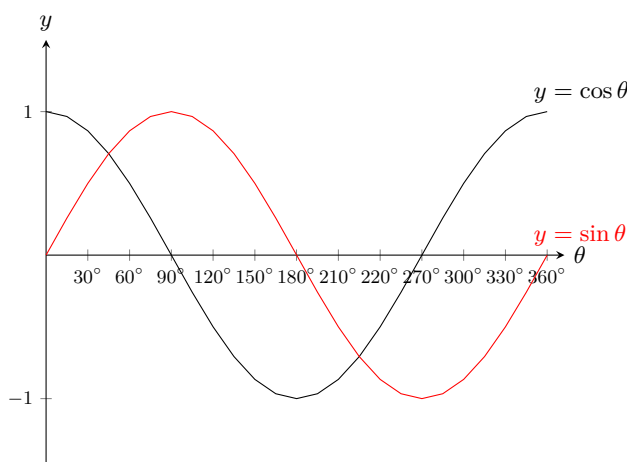
Step 3: Plot the points and join with a smooth curve



Domain: $[0^\circ; 360^\circ]$ Range: $[1; 5]$

x -intercepts: none y -intercept: $(0^\circ; 5)$

Maximum turning points: $(0^\circ; 5)$, $(360^\circ; 5)$ Minimum turning point: $(180^\circ; 1)$



Notice that the two graphs look very similar. Both waves move up and down along the x -axis. The distances between the peaks for each graph is the same. The height of the peaks and the depths of the troughs are also the same.

If you shift the whole cosine graph to the right by 90° it will overlap perfectly with the sine graph. If you shift the sine graph 90° to the left it would overlap perfectly with the cosine graph. This means that:

$$\begin{aligned}\sin \theta &= \cos (\theta - 90^\circ) && \text{(shift the cosine graph to the right)} \\ \cos \theta &= \sin (\theta + 90^\circ) && \text{(shift the sine graph to the left)}\end{aligned}$$

Tangent function

EMA5D

Functions of the form $y = \tan \theta$

EMA5F

Worked example 20: Plotting a tangent graph

QUESTION

$$y = f(\theta) = \tan \theta \quad [0^\circ \leq \theta \leq 360^\circ]$$

Use your calculator to complete the following table.

Choose an appropriate scale and plot the values with θ on the x -axis and $\tan \theta$ on the y -axis. Round your answers to 2 decimal places.

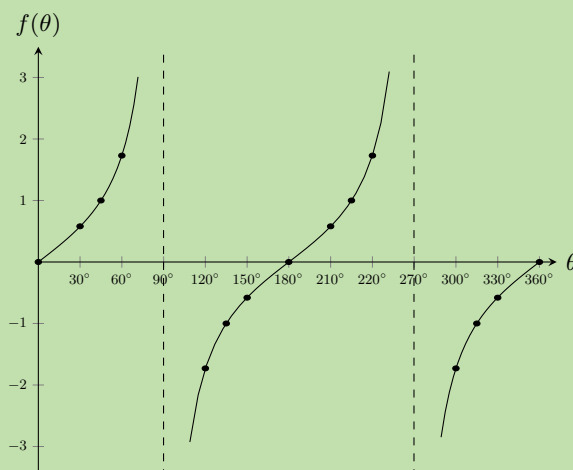
θ	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\tan \theta$									
θ	210°	235°	240°	270°	300°	315°	330°	360°	
$\tan \theta$									

SOLUTION

Step 1: Substitute values for θ

θ	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\tan \theta$	0	0,58	1	1,73	undef	-1,73	-1	-0,58	0
θ	210°	235°	240°	270°	300°	315°	330°	360°	
$\tan \theta$	0,58	1	1,73	undef	-1,73	-1	-0,58	0	

Step 2: Plot the points and join with a smooth curve



There is an easy way to visualise the tangent graph. Consider our definitions of $\sin \theta$ and $\cos \theta$ for right-angled triangles:

$$\begin{aligned}\frac{\sin \theta}{\cos \theta} &= \frac{\left(\frac{\text{opposite}}{\text{hypotenuse}}\right)}{\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right)} \\ &= \frac{\text{opposite}}{\text{hypotenuse}} \times \frac{\text{hypotenuse}}{\text{adjacent}} \\ &= \frac{\text{opposite}}{\text{adjacent}} \\ &= \tan \theta\end{aligned}$$

So for any value of θ : $\tan \theta = \frac{\sin \theta}{\cos \theta}$

So we know that for values of θ for which $\sin \theta = 0$, we must also have $\tan \theta = 0$. Also if $\cos \theta = 0$ the value of $\tan \theta$ is undefined as we cannot divide by 0. The dashed vertical lines are at the values of θ where $\tan \theta$ is not defined and are called the asymptotes.

Asymptotes: the lines $\theta = 90^\circ$ and $\theta = 270^\circ$

Period: 180°

Domain: $\{\theta : 0^\circ \leq \theta \leq 360^\circ, \theta \neq 90^\circ; 270^\circ\}$ Range: $\{f(\theta) : f(\theta) \in \mathbb{R}\}$

x -intercepts: $(0^\circ; 0)$, $(180^\circ; 0)$, $(360^\circ; 0)$ y -intercept: $(0^\circ; 0)$

Functions of the form $y = a \tan \theta + q$

EMA5G

Investigation: The effects of a and q on a tangent graph

On the same set of axes, plot the following graphs for $0^\circ \leq \theta \leq 360^\circ$:

1. $y_1 = \tan \theta - 2$
2. $y_2 = \tan \theta - 1$
3. $y_3 = \tan \theta$
4. $y_4 = \tan \theta + 1$
5. $y_5 = \tan \theta + 2$

Use your results to deduce the effect of q .

On the same set of axes, plot the following graphs for $0^\circ \leq \theta \leq 360^\circ$:

1. $y_6 = -2 \tan \theta$
2. $y_7 = -\tan \theta$
3. $y_8 = \tan \theta$
4. $y_9 = 2 \tan \theta$

Use your results to deduce the effect of a .

The effect of q

The effect of q is called a vertical shift because the whole tangent graph shifts up or down by q units.

- For $q > 0$, the graph is shifted vertically upwards by q units.
- For $q < 0$, the graph is shifted vertically downwards by q units.

The effect of a

The value of a affects the steepness of each of the branches of the graph. The greater the value of a , the quicker the branches of the graph approach the asymptotes.

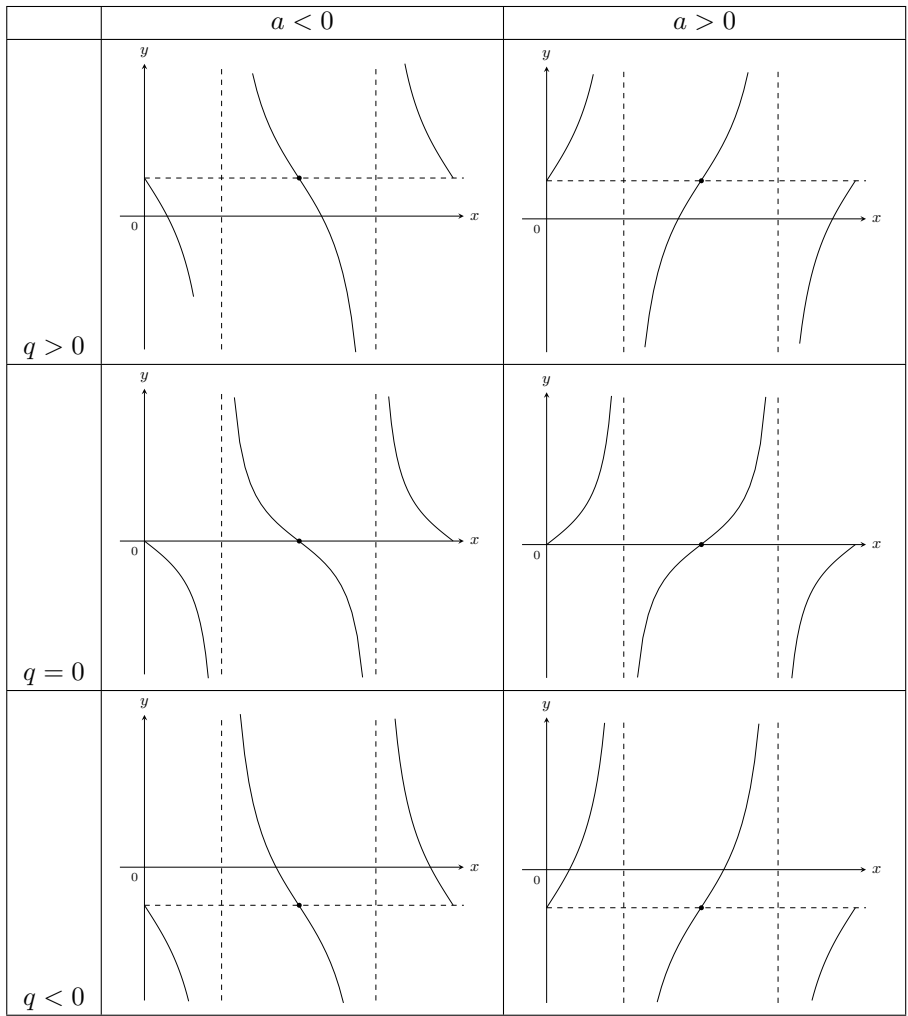


Table 6.10: The effect of a and q on a tangent graph.

Domain and range

From the graph we see that $\tan \theta$ is undefined at $\theta = 90^\circ$ and $\theta = 270^\circ$.

Therefore the domain is $\{\theta : 0^\circ \leq \theta \leq 360^\circ, \theta \neq 90^\circ; 270^\circ\}$.

The range is $\{f(\theta) : f(\theta) \in \mathbb{R}\}$.

Period

The period of $y = a \tan \theta + q$ is 180° . This means that one tangent cycle is completed in 180° .

Intercepts

The y -intercept of $f(\theta) = a \tan \theta + q$ is simply the value of $f(\theta)$ at $\theta = 0^\circ$.

$$\begin{aligned} y &= f(0^\circ) \\ &= a \tan 0^\circ + q \\ &= a(0) + q \\ &= q \end{aligned}$$

This gives the point $(0^\circ; q)$.

Asymptotes

The graph has asymptotes at $\theta = 90^\circ$ and $\theta = 270^\circ$.

Worked example 21: Sketching a tangent graph**QUESTION**

Sketch the graph of $y = 2 \tan \theta + 1$ for $\theta \in [0^\circ; 360^\circ]$.

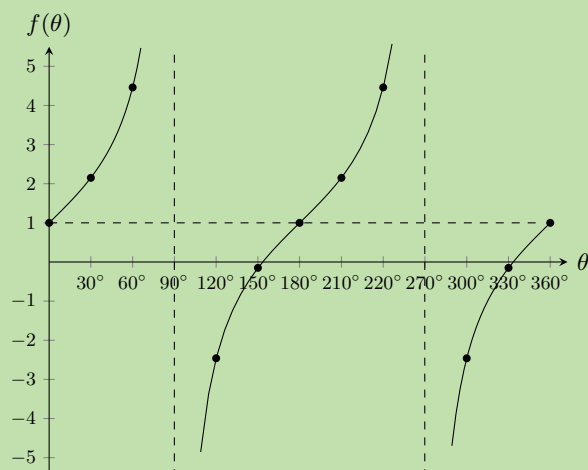
SOLUTION**Step 1: Examine the standard form of the equation**

We see that $a > 1$ so the branches of the curve will be steeper. We also see that $q > 0$ so the graph is shifted vertically upwards by 1 unit.

Step 2: Substitute values for θ

θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
y	1	2,15	4,46	–	–2,46	–0,15	1	2,15	4,46	–	–2,46	–0,15	1

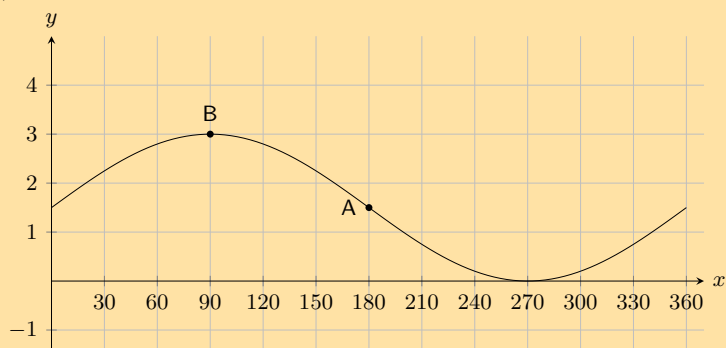
Step 3: Plot the points and join with a smooth curve



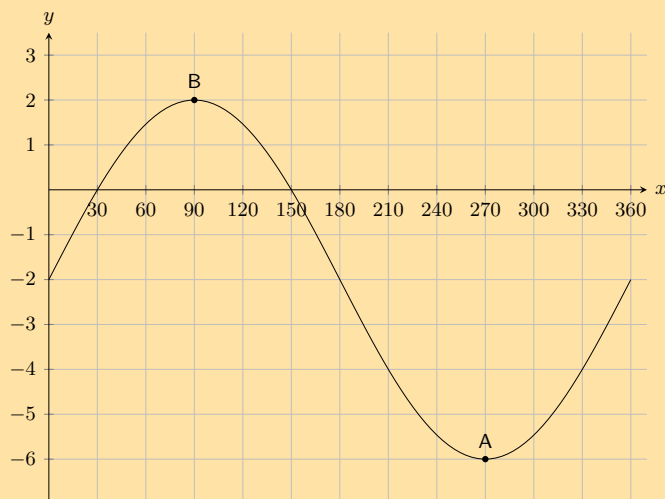
Domain: $\{\theta : 0^\circ \leq \theta \leq 360^\circ, \theta \neq 90^\circ; 270^\circ\}$ Range: $\{f(\theta) : f(\theta) \in \mathbb{R}\}$

Exercise 6 – 6:

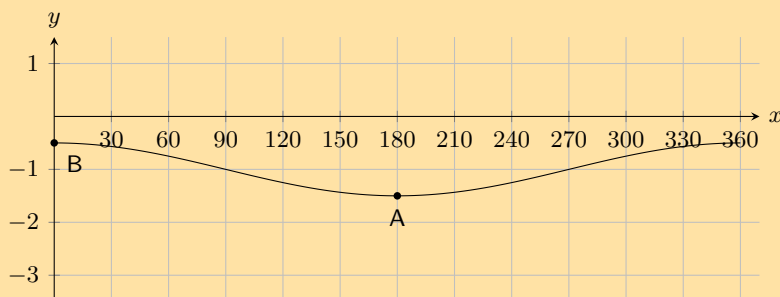
1. Shown the following graph of the following form: $y = a \sin \theta + q$ where **Point A** is at $(180^\circ; 1,5)$, and **Point B** is at $(90^\circ; 3)$, find the values of a and q .



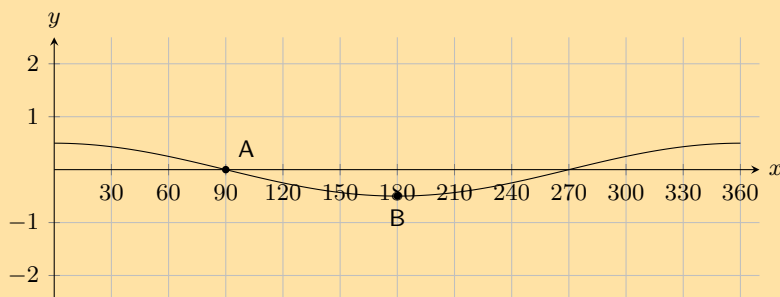
2. Shown the following graph of the following form: $y = a \sin \theta + q$ where **Point A** is at $(270^\circ; -6)$, and **Point B** is at $(90^\circ; 2)$, determine the values of a and q .



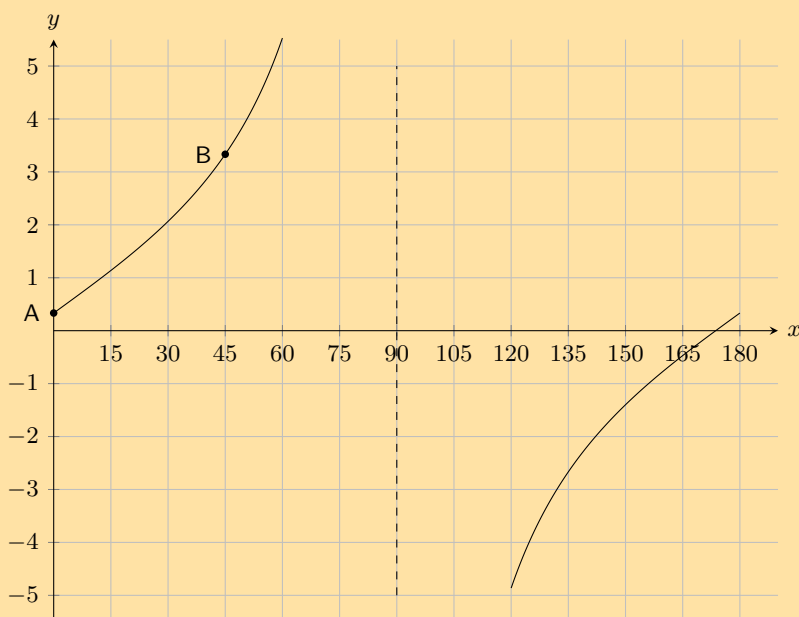
3. The graph below shows a trigonometric equation of the following form: $y = a \cos \theta + q$. Two points are shown on the graph: **Point A** at $(180^\circ; -1,5)$, and **Point B**: $(0^\circ; -0,5)$. Calculate the values of a (the amplitude of the graph) and q (the vertical shift of the graph).



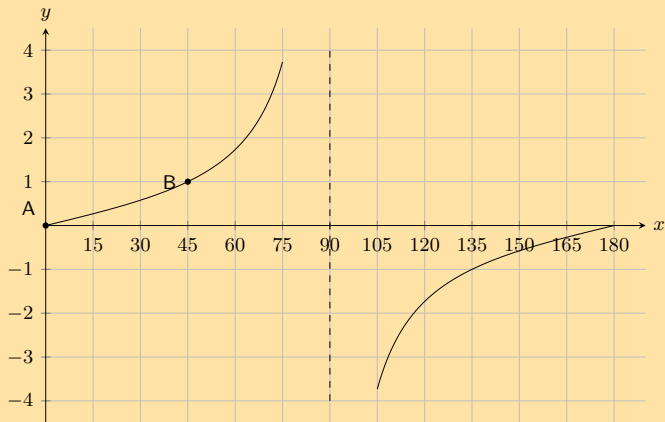
4. The graph below shows a trigonometric equation of the following form: $y = a \cos \theta + q$. Two points are shown on the graph: **Point A** at $(90^\circ; 0,0)$, and **Point B**: $(180^\circ; -0,5)$. Calculate the values of a (the amplitude of the graph) and q (the vertical shift of the graph).



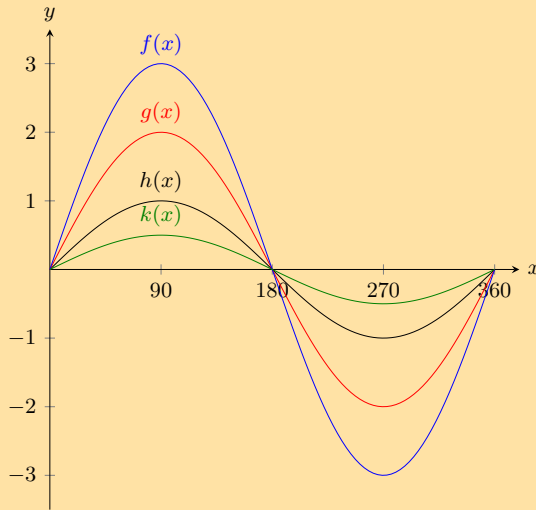
5. On the graph below you see a tangent curve of the following form: $y = a \tan \theta + q$. Two points are labelled on the curve: **Point A** is at $(0^\circ; \frac{1}{3})$, and **Point B** is at $(45^\circ; \frac{10}{3})$. Calculate, or otherwise determine, the values of a and q .



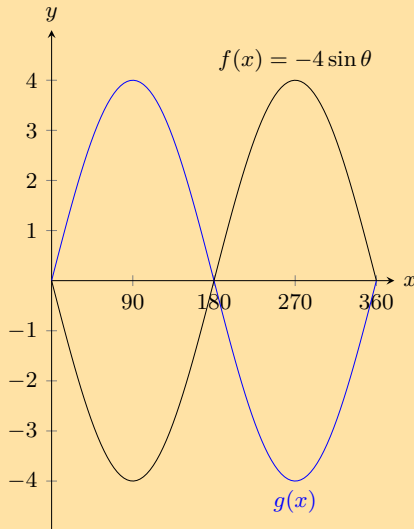
6. The graph below shows a tangent curve with an equation of the form $y = a \tan \theta + q$. Two points are labelled on the curve: **Point A** is at $(0^\circ; 0)$, and **Point B** is at $(45^\circ; 1)$. Find a and q .



7. Given the following graph, identify a function that matches each of the following equations:



- a) $y = \sin \theta$ b) $y = \frac{1}{2} \sin \theta$ c) $y = 3 \sin \theta$ d) $y = 2 \sin \theta$
8. The graph below shows functions $f(x)$ and $g(x)$



What is the equation for $g(x)$?

9. With the assistance of the table below sketch the three functions on the same set of axes.

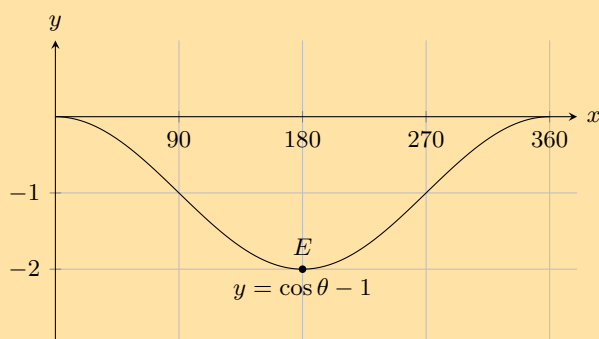
θ	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\tan \theta$	0	1	undefined	-1	0	1	undefined	-1	0
$3 \tan \theta$	0	3	undefined	-3	0	3	undefined	-3	0
$\frac{1}{2} \tan \theta$	0	$\frac{1}{2}$	undefined	$-\frac{1}{2}$	0	$\frac{1}{2}$	undefined	$-\frac{1}{2}$	0

10. With the assistance of the table below sketch the three functions on the same set of axes.

θ	0°	90°	180°	270°	360°
$\cos \theta - 2$	-1	-2	-3	-2	-1
$\cos \theta + 4$	5	4	2	4	5
$\cos \theta + 2$	3	2	1	2	3

11. State the coordinates at E and the range of the function.

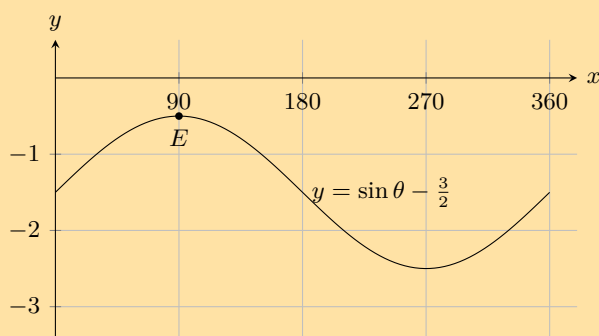
a)



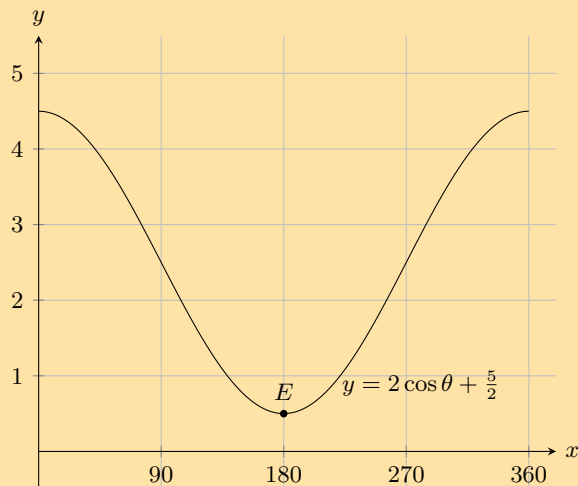
b)



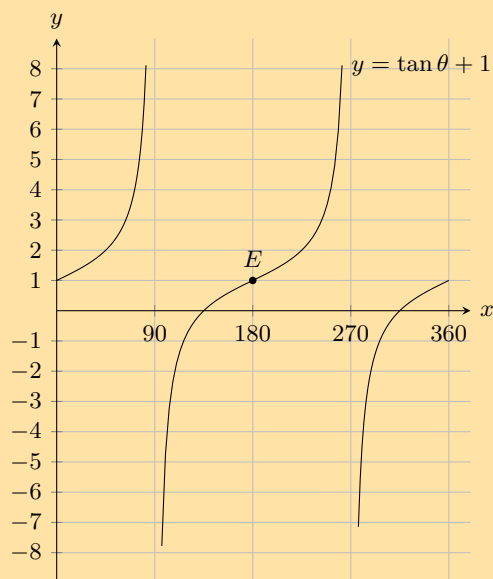
c)



d)



12. State the coordinates at E and the domain and range of the function in the interval shown.

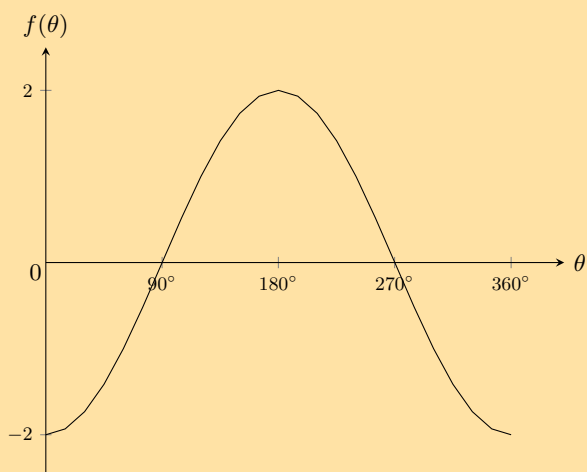


13. Using your knowledge of the effects of a and q , sketch each of the following graphs, without using a table of values, for $\theta \in [0^\circ; 360^\circ]$

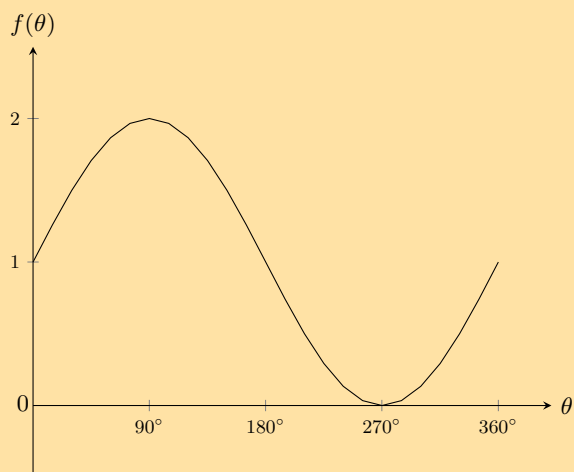
- $y = 2 \sin \theta$
- $y = -4 \cos \theta$
- $y = -2 \cos \theta + 1$
- $y = \sin \theta - 3$
- $y = \tan \theta - 2$
- $y = 2 \cos \theta - 1$

14. Give the equations for each of the following graphs:

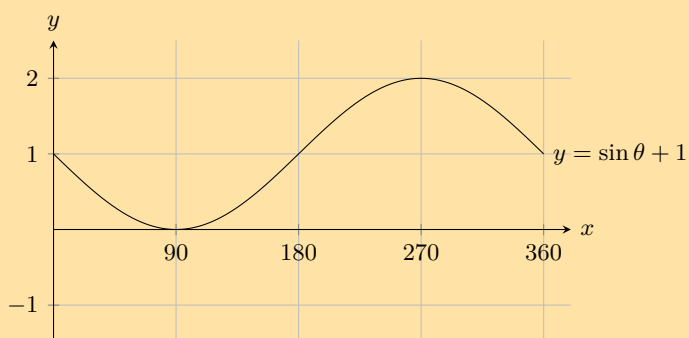
a)



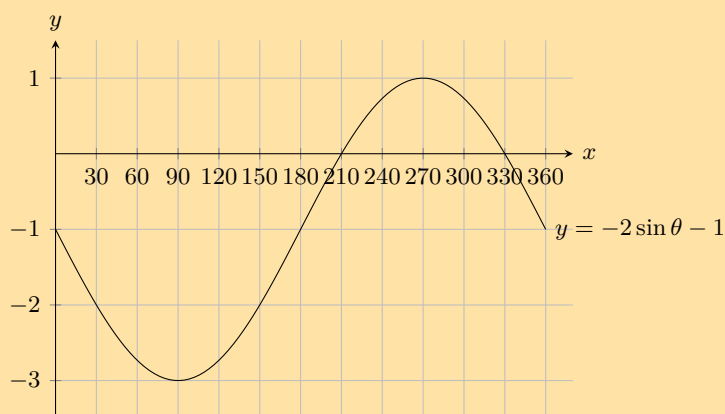
b)



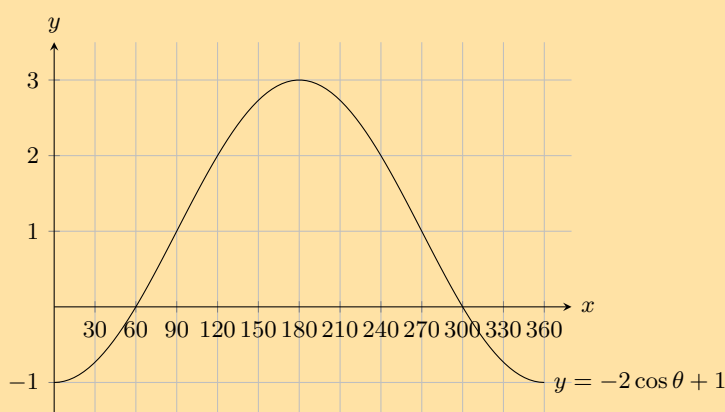
15. For which values of θ is the function increasing, in the interval shown?



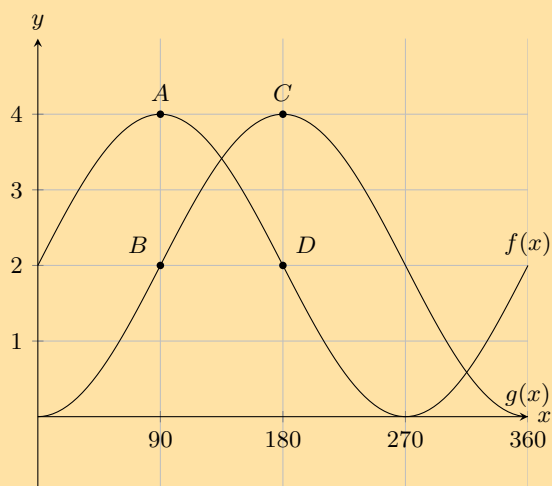
16. For which values of θ is the function negative, in the interval shown?



17. For which values of θ is the function positive, in the interval shown?

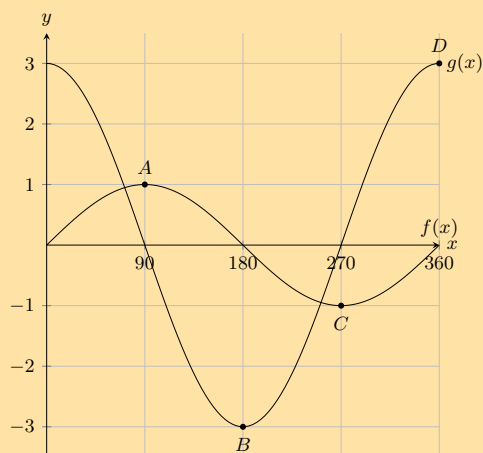


18. Given the following graph.



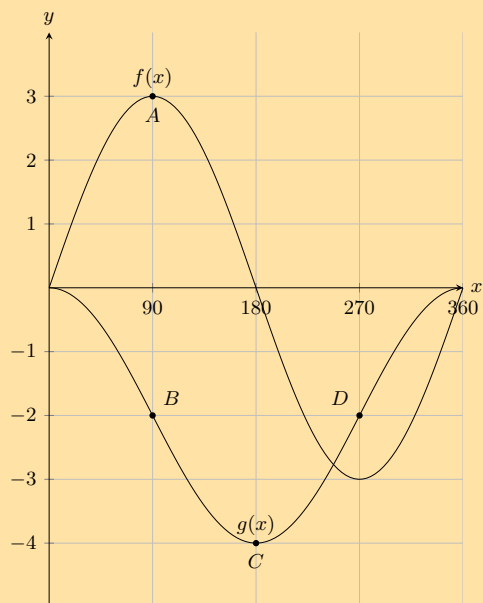
- State the coordinates at A , B , C and D .
- How many times in this interval does $f(x)$ intersect $g(x)$.
- What is the amplitude of $f(x)$.
- Evaluate: $f(360^\circ) - g(360^\circ)$.

19. Given the following graph.



- State the coordinates at A , B , C and D .
- How many times in this interval does $f(x)$ intersect $g(x)$.
- What is the amplitude of $g(x)$.
- Evaluate: $f(90^\circ) - g(90^\circ)$.

20. Given the following graph:



- State the coordinates at A , B , C and D .
- How many times in this interval does $f(x)$ intersect $g(x)$.
- What is the amplitude of $g(x)$.
- Evaluate: $f(270^\circ) - g(270^\circ)$.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 1. 2FZ6 | 2. 2FZ7 | 3. 2FZ8 | 4. 2FZ9 | 5. 2FZB | 6. 2FZC | 7. 2FZD | 8. 2FZF |
| 9. 2FZG | 10. 2FZH | 11a. 2FZJ | 11b. 2FZK | 11c. 2FZM | 11d. 2FZN | 12. 2FZP | 13a. 2FZQ |
| 13b. 2FZR | 13c. 2FZS | 13d. 2FZT | 13e. 2FZV | 13f. 2FZW | 14a. 2FZX | 14b. 2FZY | 15. 2FZZ |
| 16. 2G22 | 17. 2G23 | 18. 2G24 | 19. 2G25 | 20. 2G26 | | | |



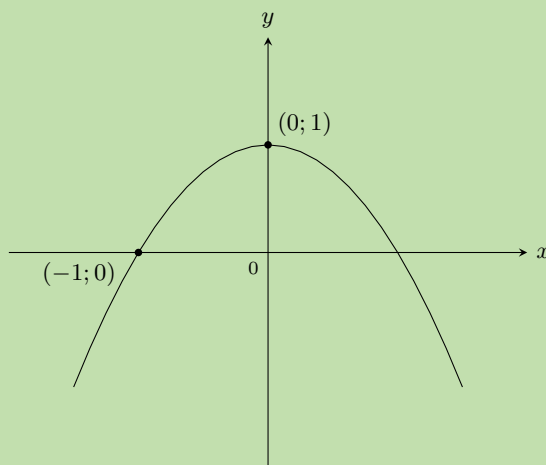
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Worked example 22: Determining the equation of a parabola**QUESTION**

Use the sketch below to determine the values of a and q for the parabola of the form $y = ax^2 + q$.

**SOLUTION****Step 1: Examine the sketch**

From the sketch we see that the shape of the graph is a “frown”, therefore $a < 0$. We also see that the graph has been shifted vertically upwards, therefore $q > 0$.

Step 2: Determine q using the y -intercept

The y -intercept is the point $(0; 1)$.

$$\begin{aligned}y &= ax^2 + q \\1 &= a(0)^2 + q \\\therefore q &= 1\end{aligned}$$

Step 3: Use the other given point to determine a

Substitute point $(-1; 0)$ into the equation:

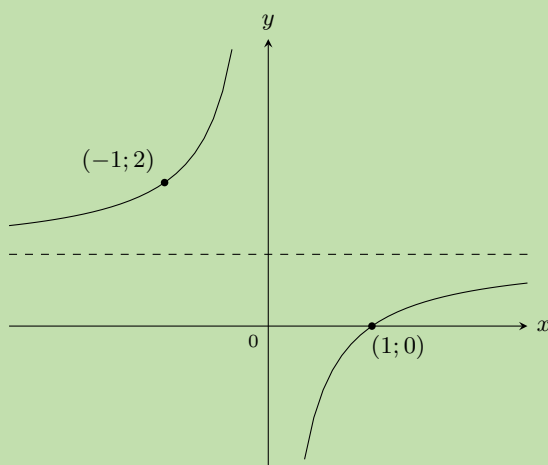
$$\begin{aligned}y &= ax^2 + q \\0 &= a(-1)^2 + 1 \\\therefore a &= -1\end{aligned}$$

Step 4: Write the final answer

$a = -1$ and $q = 1$, so the equation of the parabola is $y = -x^2 + 1$.

QUESTION

Use the sketch below to determine the values of a and q for the hyperbola of the form $y = \frac{a}{x} + q$.



SOLUTION

Step 1: Examine the sketch

The two curves of the hyperbola lie in the second and fourth quadrant, therefore $a < 0$. We also see that the graph has been shifted vertically upwards, therefore $q > 0$.

Step 2: Substitute the given points into the equation and solve

Substitute the point $(-1; 2)$:

$$\begin{aligned} y &= \frac{a}{x} + q \\ 2 &= \frac{a}{-1} + q \\ \therefore 2 &= -a + q \end{aligned}$$

Substitute the point $(1; 0)$:

$$\begin{aligned} y &= \frac{a}{x} + q \\ 0 &= \frac{a}{1} + q \\ \therefore a &= -q \end{aligned}$$

Step 3: Solve the equations simultaneously using substitution

$$\begin{aligned} 2 &= -a + q \\ &= q + q \\ &= 2q \\ \therefore q &= 1 \\ \therefore a &= -q \\ &= -1 \end{aligned}$$

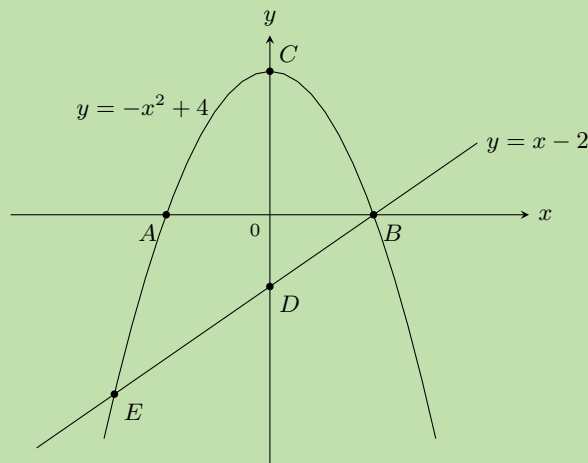
Step 4: Write the final answer

$a = -1$ and $q = 1$, therefore the equation of the hyperbola is $y = \frac{-1}{x} + 1$.

QUESTION

The graphs of $y = -x^2 + 4$ and $y = x - 2$ are given. Calculate the following:

1. coordinates of A, B, C, D
2. coordinates of E
3. distance CD



SOLUTION

Step 1: Calculate the intercepts

For the parabola, to calculate the y -intercept, let $x = 0$:

$$\begin{aligned} y &= -x^2 + 4 \\ &= -0^2 + 4 \\ &= 4 \end{aligned}$$

This gives the point $C(0; 4)$.

To calculate the x -intercept, let $y = 0$:

$$\begin{aligned} y &= -x^2 + 4 \\ 0 &= -x^2 + 4 \\ x^2 - 4 &= 0 \\ (x + 2)(x - 2) &= 0 \\ \therefore x &= \pm 2 \end{aligned}$$

This gives the points $A(-2; 0)$ and $B(2; 0)$.

For the straight line, to calculate the y -intercept, let $x = 0$:

$$\begin{aligned} y &= x - 2 \\ &= 0 - 2 \\ &= -2 \end{aligned}$$

This gives the point $D(0; -2)$.

Step 2: Calculate the point of intersection E

At E the two graphs intersect so we can equate the two expressions:

$$\begin{aligned}x - 2 &= -x^2 + 4 \\ \therefore x^2 + x - 6 &= 0 \\ \therefore (x - 2)(x + 3) &= 0 \\ \therefore x &= 2 \text{ or } -3\end{aligned}$$

At E , $x = -3$, therefore $y = x - 2 = -3 - 2 = -5$. This gives the point $E(-3; -5)$.

Step 3: Calculate distance CD

$$\begin{aligned}CD &= CO + OD \\ &= 4 + 2 \\ &= 6\end{aligned}$$

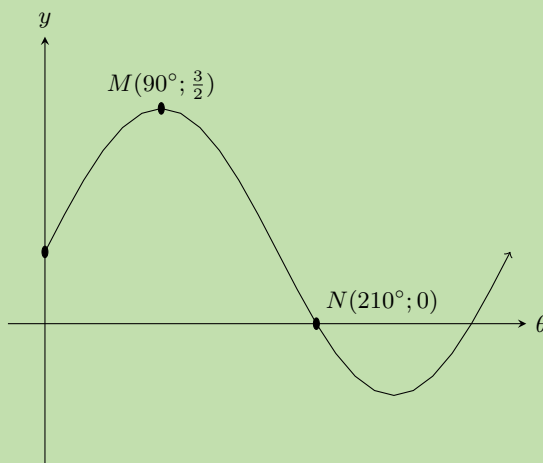
Distance CD is 6 units.

Step 4: Write the final answer

1. coordinates of $A(-2; 0)$, $B(2; 0)$, $C(0; 4)$, $D(0; -2)$
2. coordinates of $E(-3; -5)$
3. distance $CD = 6$ units

Worked example 25: Interpreting trigonometric graphs**QUESTION**

Use the sketch to determine the equation of the trigonometric function f of the form $y = af(\theta) + q$.



SOLUTION

Step 1: Examine the sketch

From the sketch we see that the graph is a sine graph that has been shifted vertically upwards. The general form of the equation is $y = a \sin \theta + q$.

Step 2: Substitute the given points into equation and solve

At N , $\theta = 210^\circ$ and $y = 0$:

$$\begin{aligned}y &= a \sin \theta + q \\0 &= a \sin 210^\circ + q \\&= a \left(-\frac{1}{2}\right) + q \\\therefore q &= \frac{a}{2}\end{aligned}$$

At M , $\theta = 90^\circ$ and $y = \frac{3}{2}$:

$$\begin{aligned}\frac{3}{2} &= a \sin 90^\circ + q \\&= a + q\end{aligned}$$

Step 3: Solve the equations simultaneously using substitution

$$\begin{aligned}\frac{3}{2} &= a + q \\&= a + \frac{a}{2} \\3 &= 2a + a \\3a &= 3 \\\therefore a &= 1 \\\therefore q &= \frac{a}{2} \\&= \frac{1}{2}\end{aligned}$$

Step 4: Write the final answer

$$y = \sin \theta + \frac{1}{2}$$

Exercise 6 – 7:

- Plot the following functions on the same set of axes and clearly label all the points at which the functions intersect.

a) $y = x^2 + 1$ and $y = 3^x$

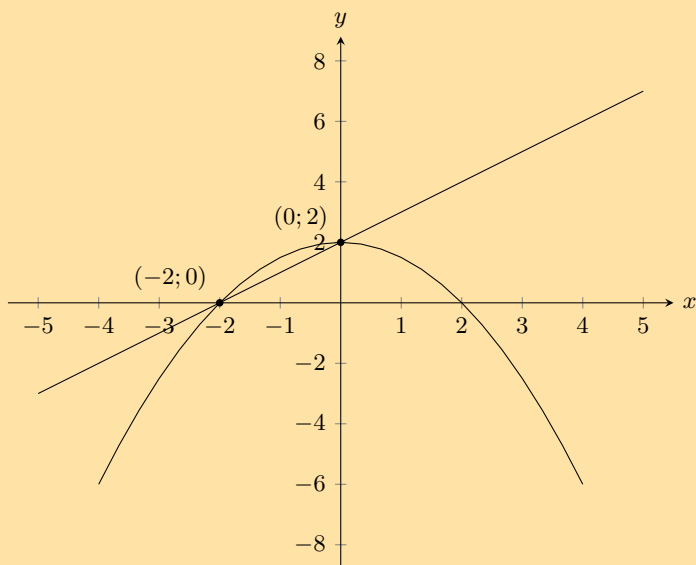
b) $y = x$ and $y = \frac{2}{x}$

c) $y = x^2 + 3$ and $y = 6$

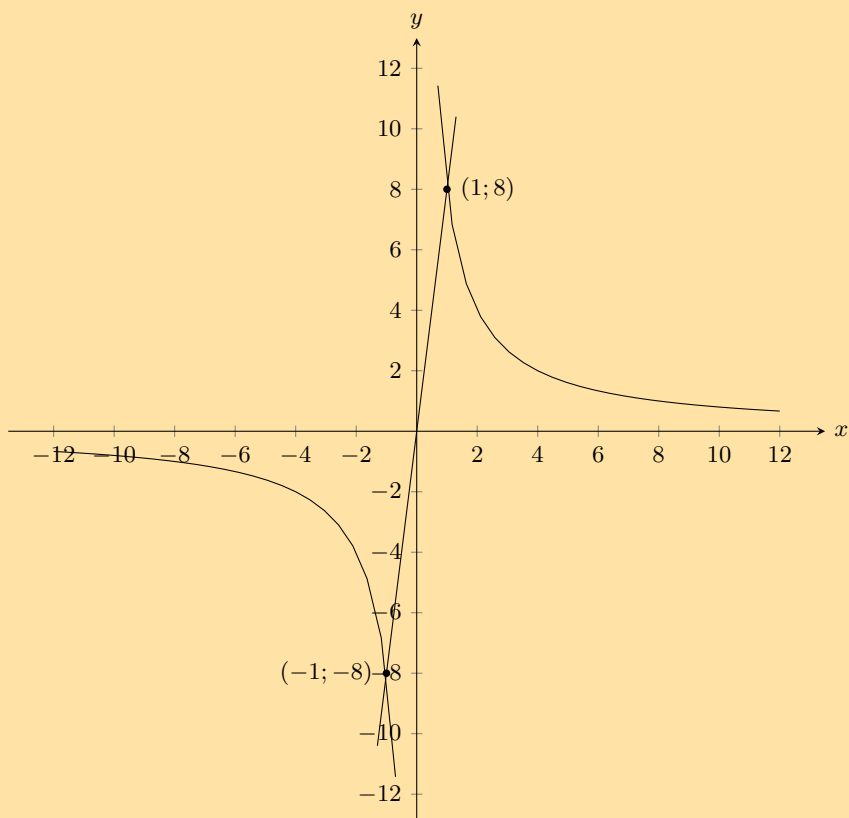
d) $y = -x^2$ and $y = \frac{8}{x}$

2. Determine the equations for the graphs given below.

a)



b)



3. Choose the correct answer:

a) The range of $y = 2 \sin \theta + 1$ is:

i. $1 \leq \theta \leq 2$

ii. $-2 \leq \theta \leq 2$

iii. $-1 \leq \theta \leq 3$

iv. $-2 \leq \theta \leq 3$

b) The range of $y = 2 \cos \theta - 4$ is:

i. $-6 \leq \theta \leq 2$

ii. $-4 \leq \theta \leq -2$

iii. $-6 \leq \theta \leq 1$

iv. $-6 \leq \theta \leq -2$

c) The y -intercept of $2^x + 1$ is:

i. 3

ii. 1

iii. 2

iv. 0

d) Which of the following passes through (1; 7)?

i. $y = \frac{7}{x}$

ii. $y = 2x + 3$

iii. $y = \frac{4}{x}$

iv. $y = x^2 + 1$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. [2G27](#)

1b. [2G28](#)

1c. [2G29](#)

1d. [2G2B](#)

2a. [2G2C](#)

2b. [2G2D](#)

3a. [2G2F](#)

3b. [2G2G](#)

3c. [2G2H](#)

3d. [2G2J](#)



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6.8 Chapter summary

EMA5K

► See presentation: [2G2K](#) at www.everythingmaths.co.za

- Characteristics of functions:

- The given x -value is known as the independent variable, because its value can be chosen freely. The calculated y -value is known as the dependent variable, because its value depends on the x -value.
- The domain of a function is the set of all x -values for which there exists at most one y -value according to that function. The range is the set of all y -values, which can be obtained using at least one x -value.
- An asymptote is a straight line, which the graph of a function will approach, but never touch.

- Special functions and their properties:

- Linear functions of the form $y = ax + q$.
- Parabolic functions of the form $y = ax^2 + q$.
- Hyperbolic functions of the form $y = \frac{a}{x} + q$.
- Exponential functions of the form $y = ab^x + q$.
- Trigonometric functions of the form $y = a \sin \theta + q$ $y = a \cos \theta + q$ $y = a \tan \theta + q$

End of chapter Exercise 6 – 8:

1. Complete the following tables and identify the function.

a)

x		2	3	4		6
y	3	6		12	15	

b)

x	1			4	5	6
y	-3	-2	-1		1	2

2. Plot the following points on a graph.

a)

x	1	2	3	4	5	6
y	1	2	3	4	5	6

b)

x	50	100	150	200	250	300
y	1	2	3	4	5	6

3. Create a table of values from the function given and then plot the function. Your table must have at least 5 ordered pairs.

a) $x^2 - 4$

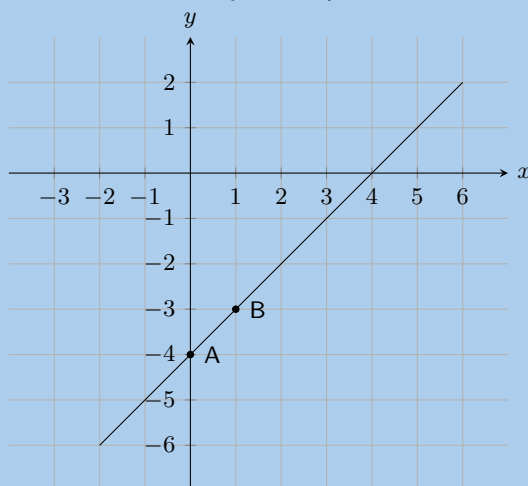
b) $y = 4x - 1$

4. Determine the y -intercept and the x -intercepts of the function.

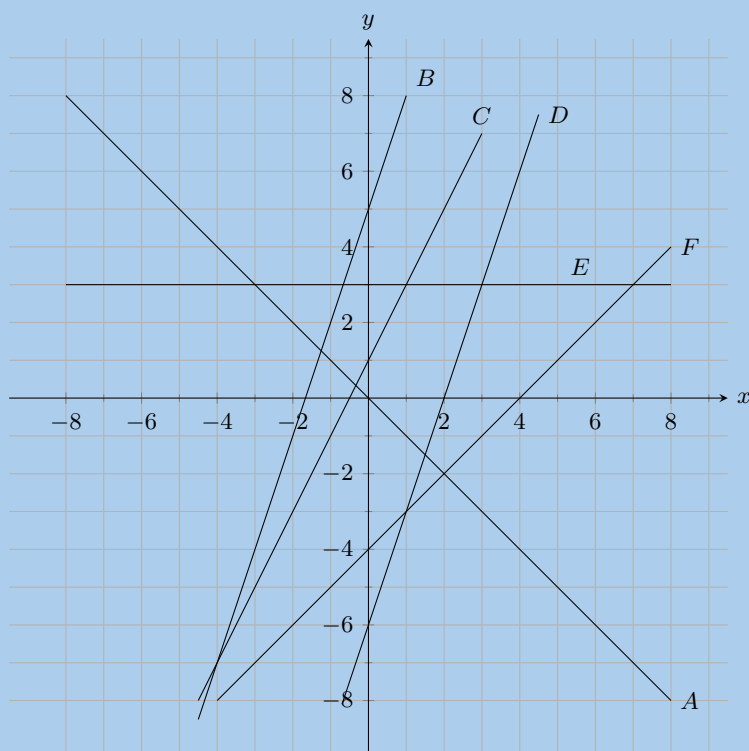
a) $y = -3x - 5$

b) $y = 2x + 4$

5. The graph below shows an equation, which has the form $y = mx + c$. Calculate or otherwise find the values of m (the gradient of the line) and c (the y -intercept of the line).



6. Look at the graphs below. Each graph is labelled with a letter. In the questions that follow, match any given equation with the label of a corresponding graph.



a) $y = 3$

b) $y = 3x + 5$

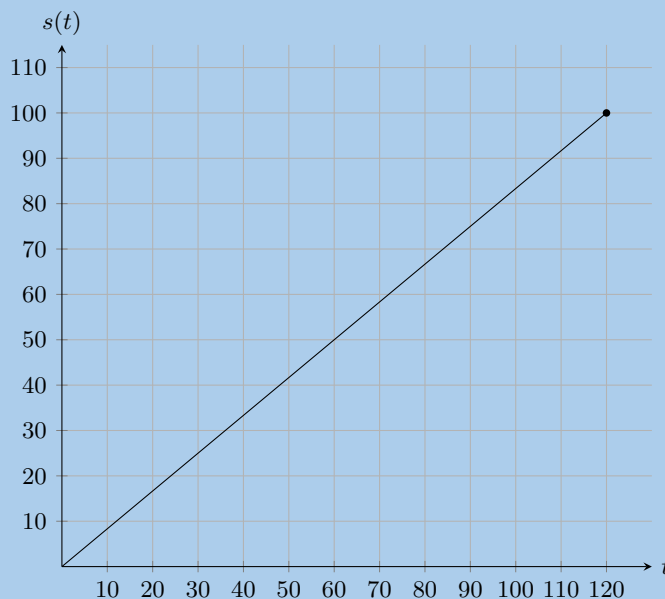
c) $y = -x$

d) $y = 2x + 1$

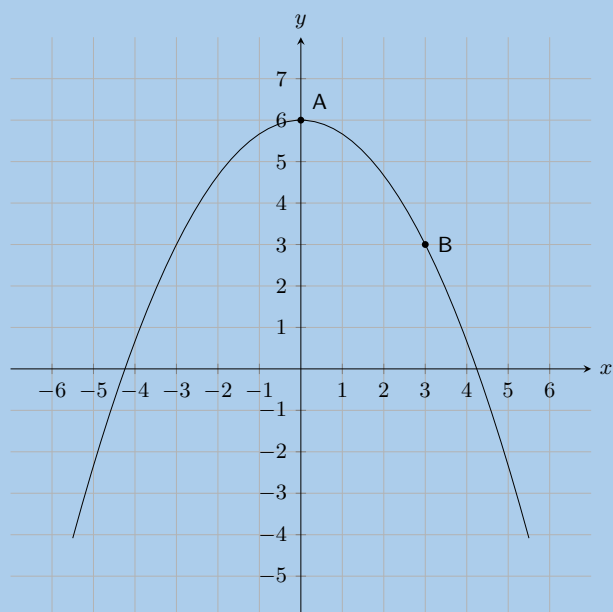
e) $y = x - 4$

f) $y = 3x - 6$

7. State whether the following is true or not
- The y -intercept of $y + 5 = x$ is -5 .
 - The gradient of $-y = x + 2$ is 1 .
 - The gradient of $-4y = 3$ is 1 .
8. Write the following in standard form:
- $2y - 5x = 6$
 - $6y - 3x = 5x + 1$
9. Sketch the graphs of the following:
- $y = 2x + 4$
 - $y - 3x = 0$
 - $2y = 4 - x$
10. The function for how much water a tap dispenses is given by: $V = 60t$, where x and V are in seconds and mL respectively.
Use this information to answer the following:
- Evaluate $V(2)$.
 - Evaluate $V(10)$.
 - How long will it take to fill a 2 L bottle of water?
 - How much water can the tap dispense in 4 s?
11. The graph below shows the distance travelled by a car over time, where $s(t)$ is distance in km and t is time in minutes.



- Use this information to answer the following:
- What distance did the car travel in an hour?
 - What is the domain of the function?
 - What is the range of the function? What does it represent?
12. On the graph here you see a function of the form: $y = ax^2 + q$.
Two points on the parabola are shown: **Point A**, the turning point of the parabola, at $(0; 6)$, and **Point B** is at $(3; 3)$. Calculate the values of a and q .



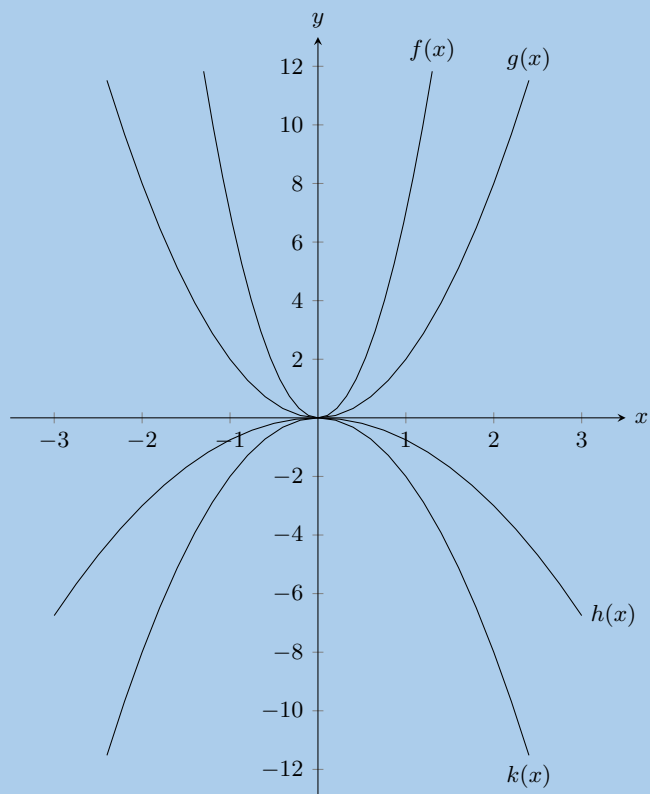
13. Given the following equation:

$$y = -5x^2 + 3$$

a) Calculate the y -coordinate of the y -intercept.

b) Now calculate the x -intercepts. Your answer must be correct to 2 decimal places.

14. Given the following graph, identify a function that matches each of the given equations:



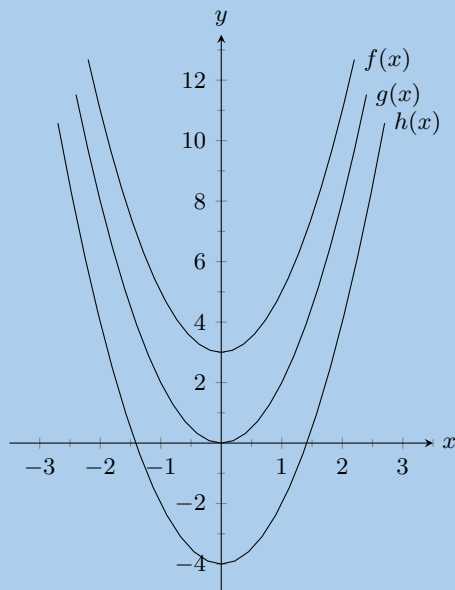
a) $y = -2x^2$

b) $y = 2x^2$

c) $y = -0,75x^2$

d) $y = 7x^2$

15. Given the following graph, identify a function that matches each of the given equations:



a) $y = 2x^2$

b) $y = 2x^2 + 3$

c) $y = 2x^2 - 4$

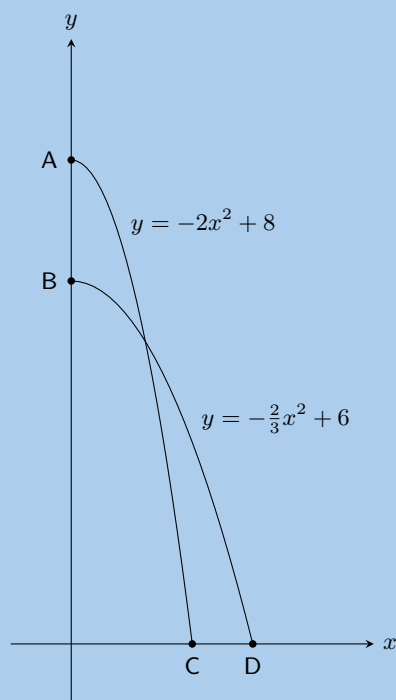
16. Sketch the following functions:

a) $y = x^2 + 3$

b) $y = \frac{1}{2}x^2 + 4$

c) $y = 2x^2 - 4$

17. Sebastian and Lucas dive into a pool from different heights. Their midair paths can be described by the following quadratic equations: $y = -2x^2 + 8$ for Sebastian and $y = -\frac{2}{3}x^2 + 6$ for Lucas.



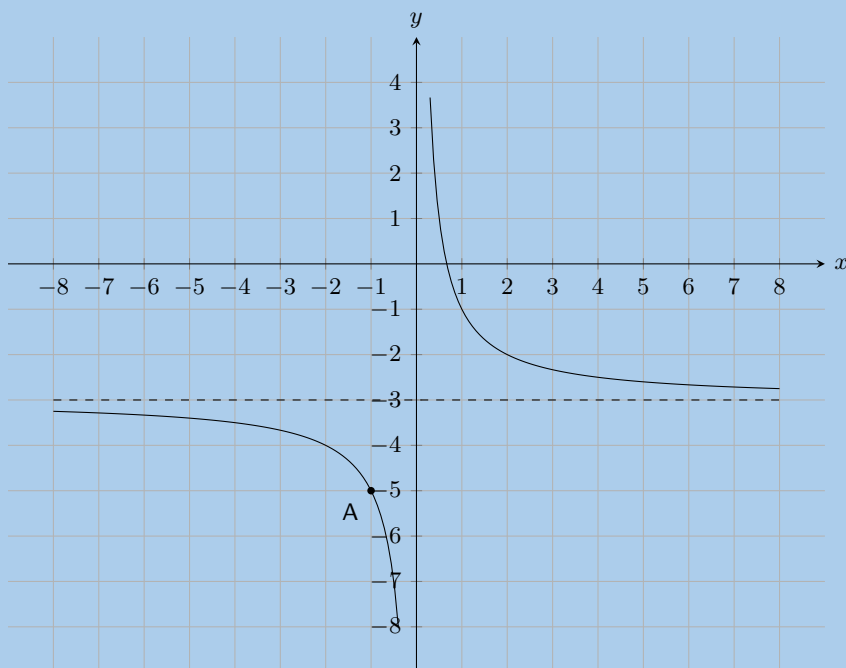
a) From what height did Sebastian dive?

b) From what height did Lucas dive?

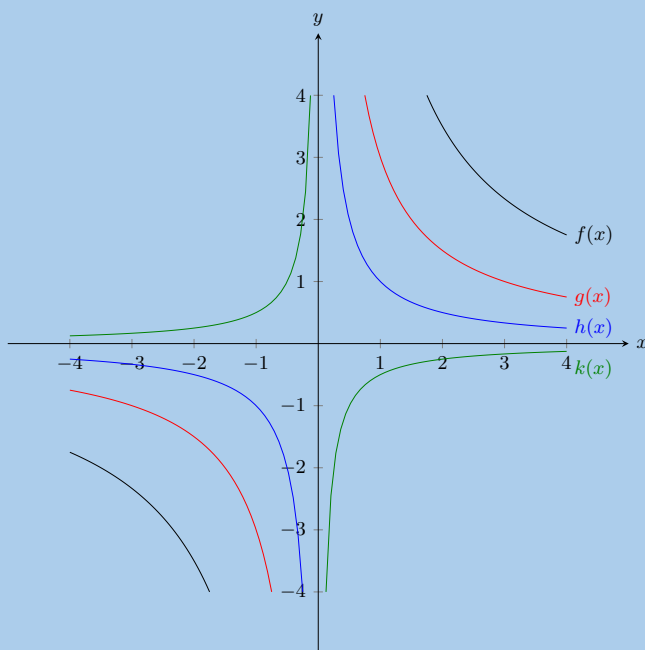
c) How far from the pool wall did Lucas land?

d) How much closer to the pool wall did Sebastian land compared to Lucas?

18. The following graph shows a hyperbolic equation of the form $y = \frac{a}{x} + q$. **Point A** is shown at $(-1; -5)$. Calculate the values of a and q .



19. Given the following equation: $y = -\frac{3}{x} + 4$
- Determine the location of the y -intercept.
 - Determine the location of the x -intercept.
20. Given the following graph, identify a function that matches each of the given equations:



a) $y = -\frac{1}{2x}$

b) $y = \frac{7}{x}$

c) $y = \frac{3}{x}$

d) $y = \frac{1}{x}$

21. Sketch the following functions and identify the asymptotes:

a) $y = \frac{3}{x} + 4$

b) $y = \frac{1}{x}$

c) $y = \frac{2}{x} - 2$

22. Sketch the functions given and describe the transformation used to obtain the second function. Show all asymptotes.

a) $y = \frac{2}{x}$ and $\frac{2}{x} + 2$

b) $y = \frac{2}{x}$ and $\frac{1}{2x}$

c) $y = \frac{3}{x}$ and $y = \frac{3x+3}{x}$

d) $y = \frac{3}{x}$ and $y = -\frac{3}{x}$

23. Given the following equation:

$$y = -\frac{1}{2} \cdot (4)^x + 3$$

a) Calculate the y -intercept. Your answer must be correct to 2 decimal places.

b) Now calculate the x -intercept. Estimate your answer to one decimal place if necessary.

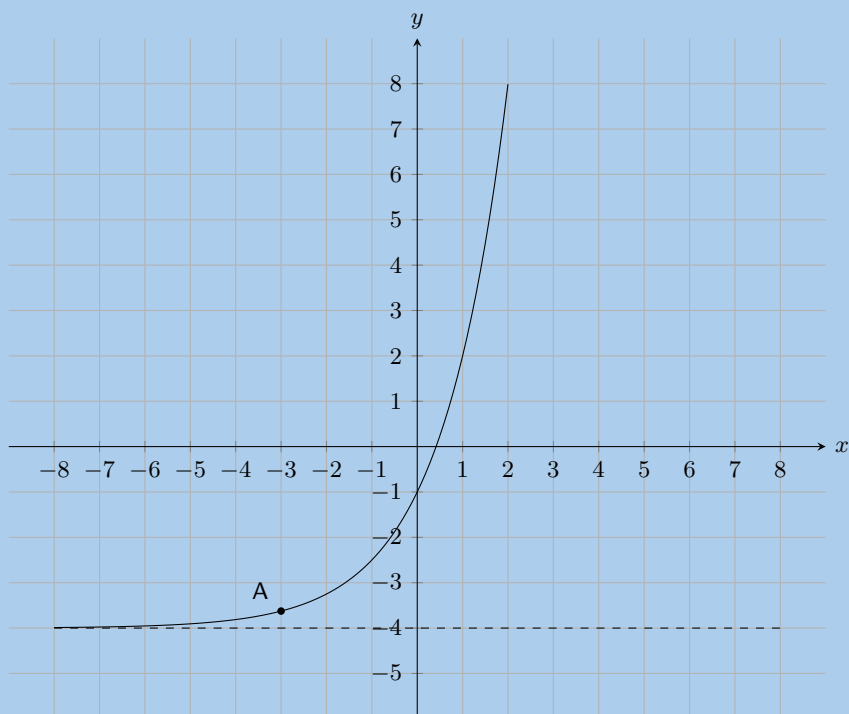
24. Sketch the following functions and identify the asymptotes:

a) $y = 3^x + 2$

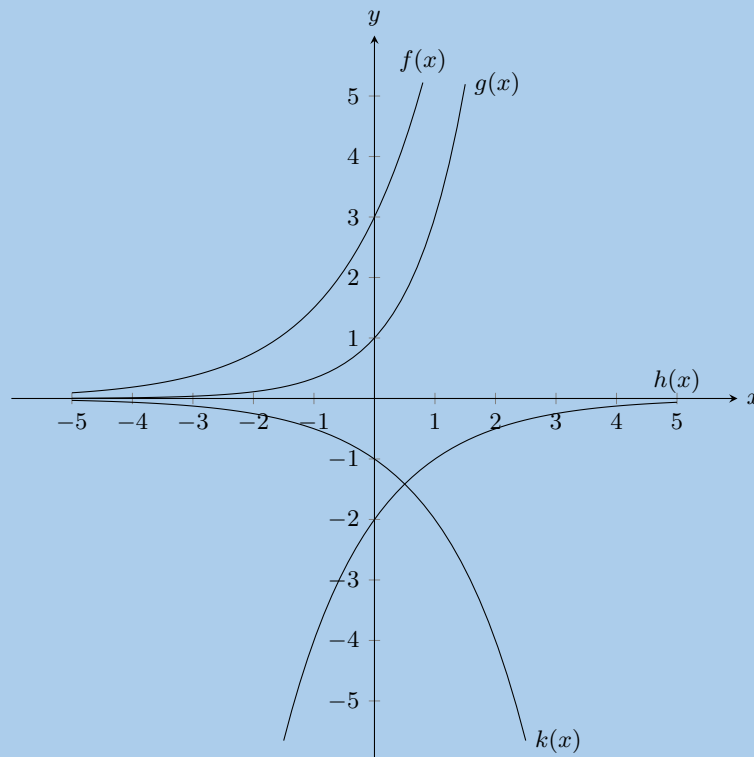
b) $y = -4 \times 2^x$

c) $y = \left(\frac{1}{3}\right)^x - 2$

25. The form of the curve graphed below is $y = a \cdot 2^x + q$. One point is given on the curve: **Point A** is at $(-3; -3,625)$. Find the values of a and q , correct to the nearest integer.



26. Given the following graph, identify a function that matches each of the given equations



- a) $y = -2\left(\frac{1}{2}\right)^x$ b) $y = 3,2^x$ c) $y = -2^x$ d) $y = 3^x$

27. Use the functions $f(x) = 3 - x$, $g(x) = 2x^2 - 4$; $h(x) = 3^x - 4$; $k(x) = \frac{3}{2x} - 1$, to find the value of the following:

- a) $f(7)$ b) $g(1)$ c) $h(-4)$ d) $k(5)$
 e) $f(-1) + h(-3)$ f) $h(g(-2))$ g) $k(f(6))$

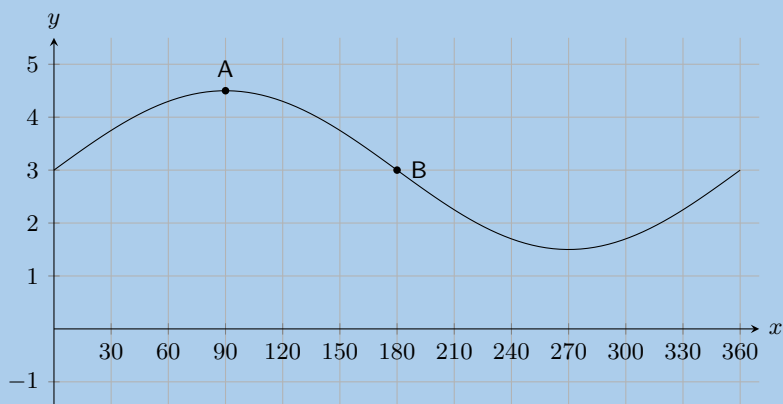
28. Determine whether the following statements are true or false. If the statement is false, give reasons why.

- The given or chosen y -value is known as the independent variable.
- A graph is said to be continuous if there are breaks in the graph.
- Functions of the form $y = ax + q$ are straight lines.
- Functions of the form $y = \frac{a}{x} + q$ are exponential functions.
- An asymptote is a straight line which a graph will intersect at least once.
- Given a function of the form $y = ax + q$, to find the y -intercept let $x = 0$ and solve for y .

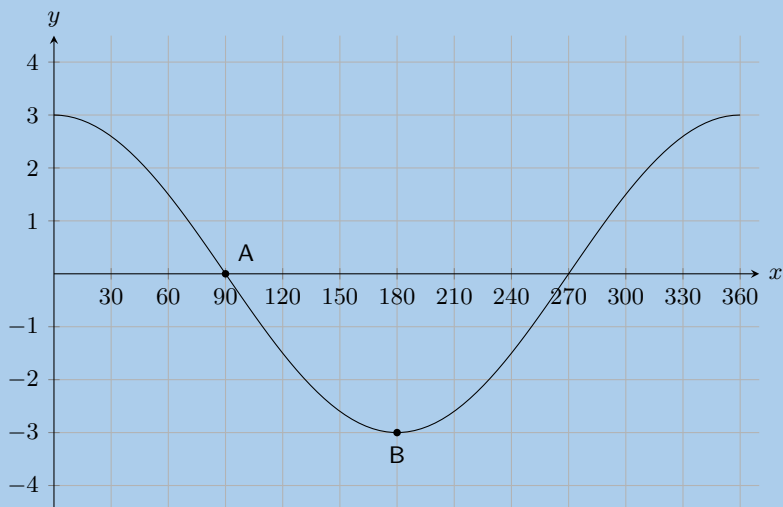
29. Given the functions $f(x) = 2x^2 - 6$ and $g(x) = -2x + 6$.

- Draw f and g on the same set of axes.
- Calculate the points of intersection of f and g .
- Use your graphs and the points of intersection to solve for x when:
 - $f(x) > 0$
 - $g(x) < 0$
 - $f(x) \leq g(x)$
- Give the equation of the reflection of f in the x -axis.

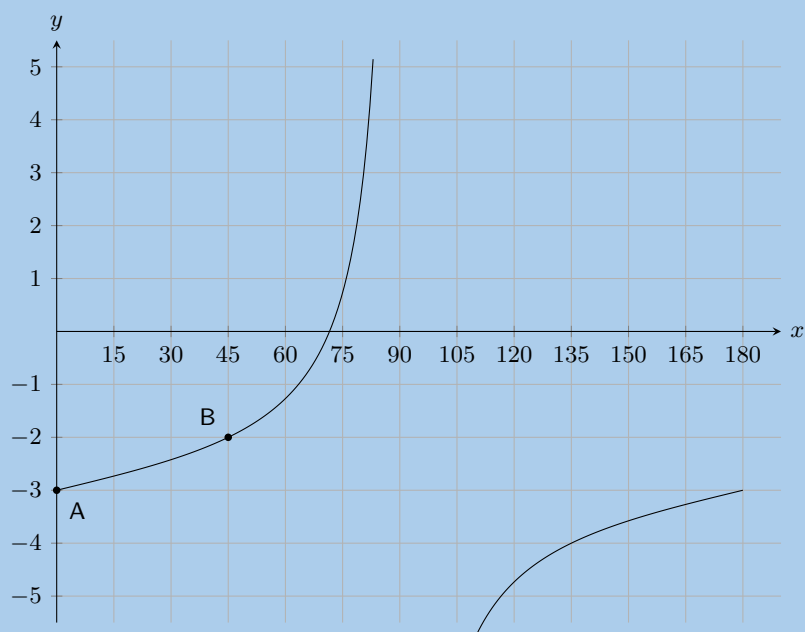
30. After a ball is dropped, the rebound height of each bounce decreases. The equation $y = 5(0,8)^x$ shows the relationship between the number of bounces x and the height of the bounce y for a certain ball. What is the approximate height of the fifth bounce of this ball to the nearest tenth of a unit?
31. Mark had 15 coins in R 5 and R 2 pieces. He had 3 more R 2 coins than R 5 coins. He wrote a system of equations to represent this situation, letting x represent the number of R 5 coins and y represent the number of R 2 coins. Then he solved the system by graphing.
- Write down the system of equations.
 - Draw their graphs on the same set of axes.
 - Use your sketch to determine how many R 5 and R 2 pieces Mark had.
32. Shown the following graph of the following form: $y = a \sin \theta + q$ where **Point A** is at $(90^\circ; 4,5)$, and **Point B** is at $(180^\circ; 3)$, determine the values of a and q .



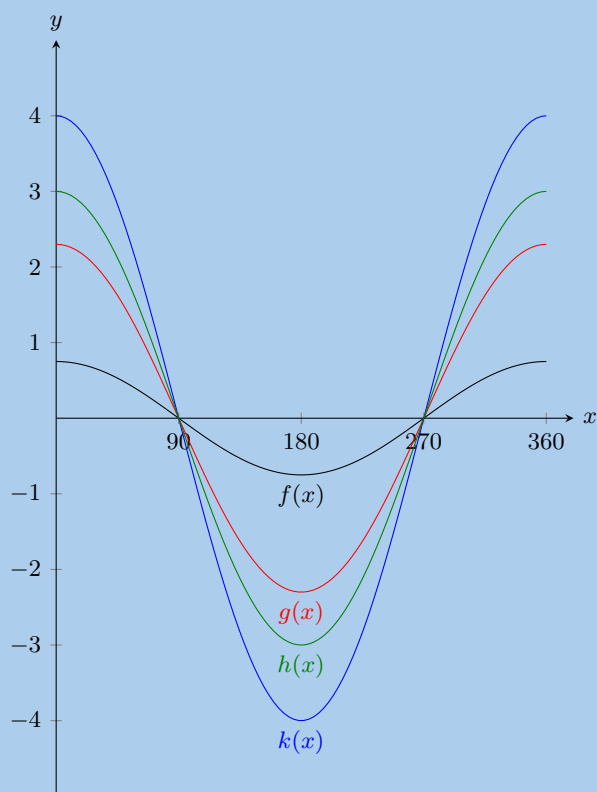
33. The graph below shows a trigonometric equation of the following form: $y = a \cos \theta + q$. Two points are shown on the graph: **Point A** at $(90^\circ; 0)$, and **Point B**: $(180^\circ; -3)$. Calculate the values of a (the amplitude of the graph) and q (the vertical shift of the graph).



34. On the graph below you see a tangent curve of the following form: $y = a \tan \theta + q$. Two points are labelled on the curve: **Point A** is at $(0^\circ; -3)$, and **Point B** is at $(45^\circ; -2)$. Calculate, or otherwise determine, the values of a and q .

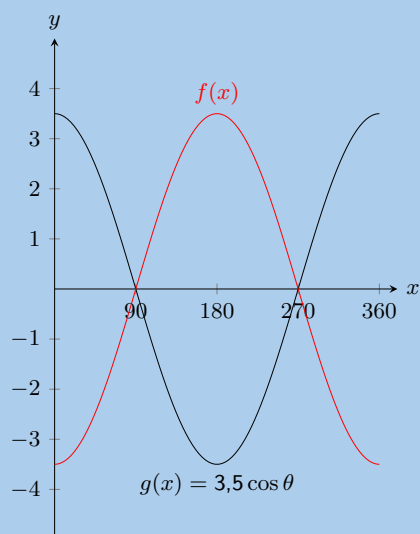


35. Given the following graph, identify a function that matches each of the given equations:



- a) $y = 2,3 \cos \theta$
- b) $y = 0,75 \cos \theta$
- c) $y = 4 \cos \theta$
- d) $y = 3 \cos \theta$

36. The graph below shows functions $f(x)$ and $g(x)$.



What is the equation for $f(x)$?

37. With the assistance of the table below sketch the three functions on the same set of axes.

θ	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\tan \theta$	0	1	undefined	-1	0	1	undefined	-1	0
$2 \tan \theta$	0	2	undefined	-2	0	2	undefined	-2	0
$\frac{1}{3} \tan \theta$	0	$\frac{1}{3}$	undefined	$-\frac{1}{3}$	0	$\frac{1}{3}$	undefined	$-\frac{1}{3}$	0

38. With the assistance of the table below sketch the three functions on the same set of axes.

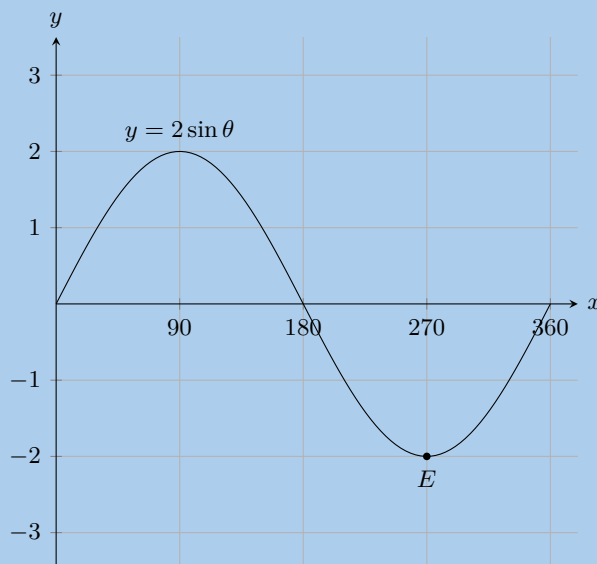
θ	0°	90°	180°	270°	360°
$\sin \theta + 1$	1	2	1	0	1
$\sin \theta + 2$	3	2	2	1	2
$\sin \theta - 2$	-2	-1	-2	-3	-2

39. Sketch graphs of the following trigonometric functions for $\theta \in [0^\circ; 360^\circ]$. Show intercepts and asymptotes.

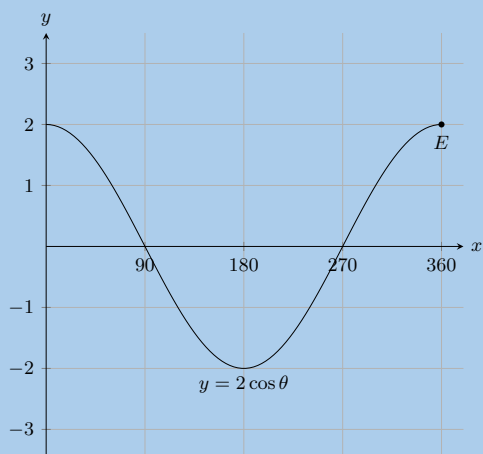
a) $y = -4 \cos \theta$ b) $y = \sin \theta - 2$ c) $y = -2 \sin \theta + 1$ d) $y = \tan \theta + 2$ e) $y = \frac{\cos \theta}{2}$

40. State the coordinates at E and the range of the function.

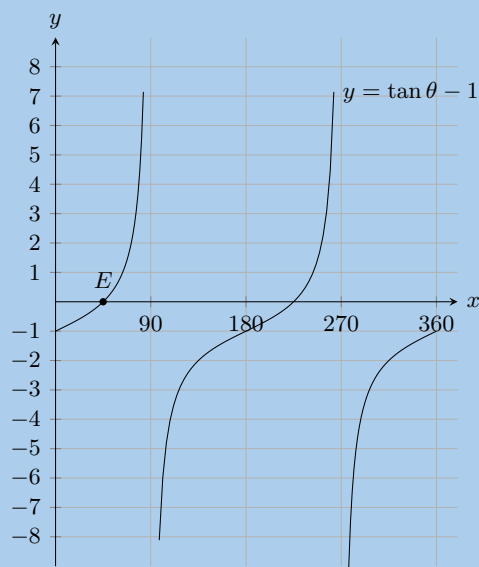
a)



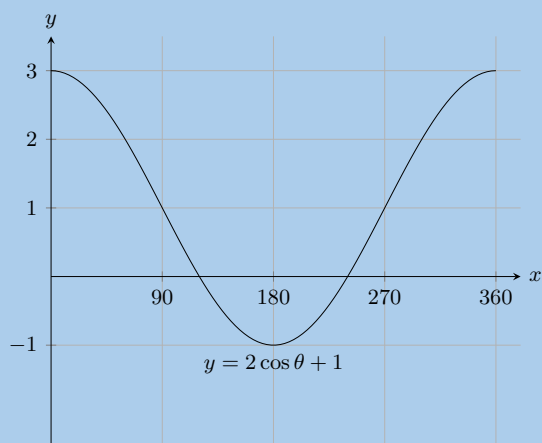
b)



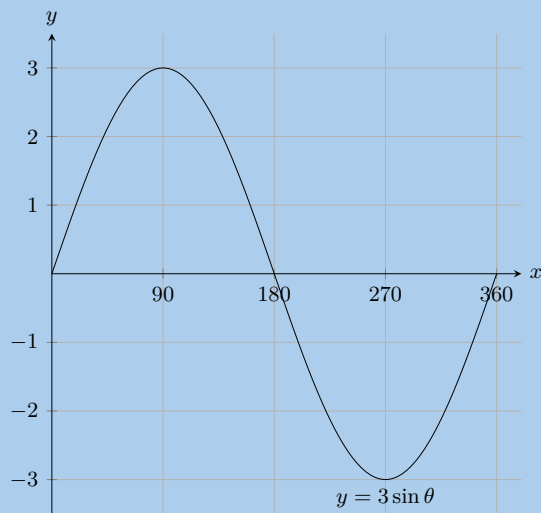
41. State the coordinates at E and the domain and range of the function in the interval shown.



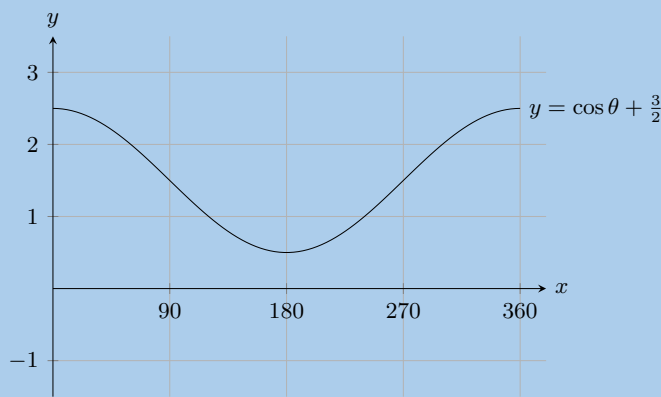
42. For which values of θ is the function decreasing, in the interval shown?



43. For which values of θ is the function increasing, in the interval shown?

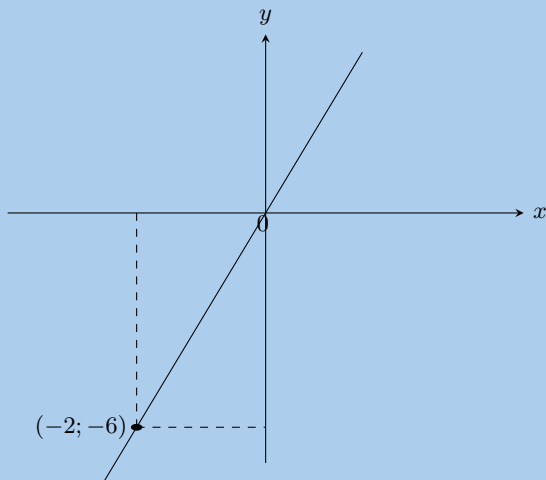


44. For which values of θ is the function positive, in the interval shown?

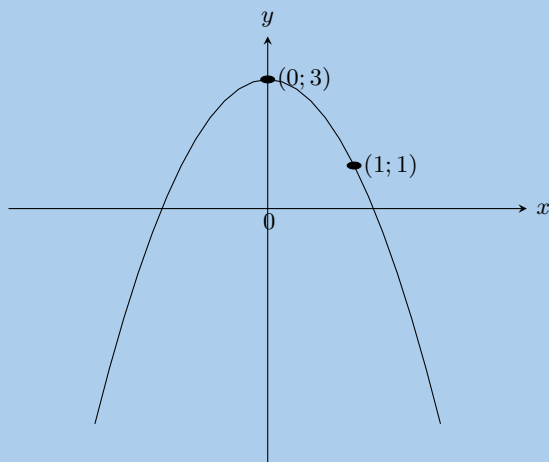


45. Given the general equations $y = mx + c$, $y = ax^2 + q$, $y = \frac{a}{x} + q$, $y = a \cdot b^x + q$, $y = a \sin \theta + q$, $y = a \cos \theta + q$ and $y = a \tan \theta$, determine the specific equations for each of the following graphs.

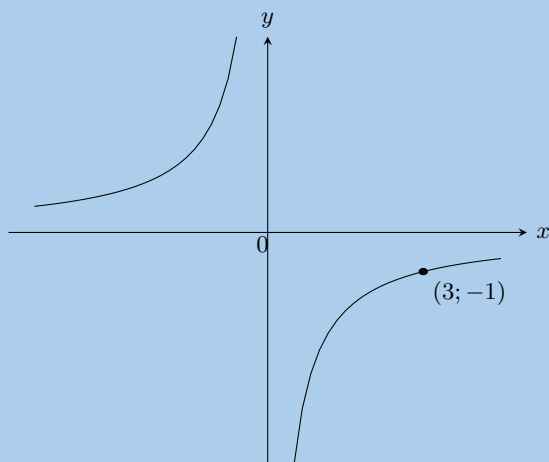
a)



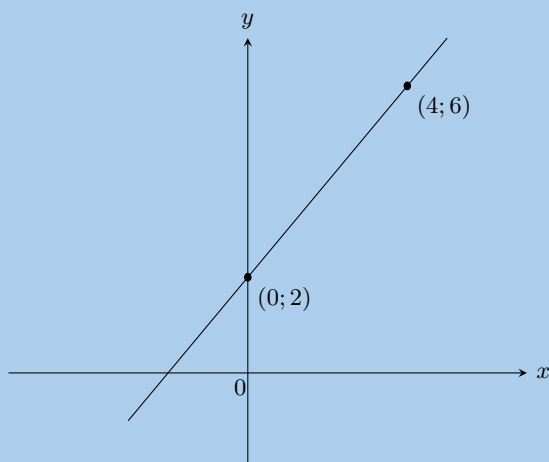
b)



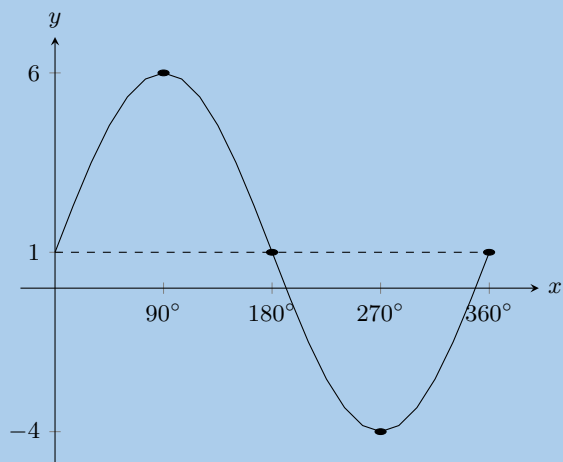
c)



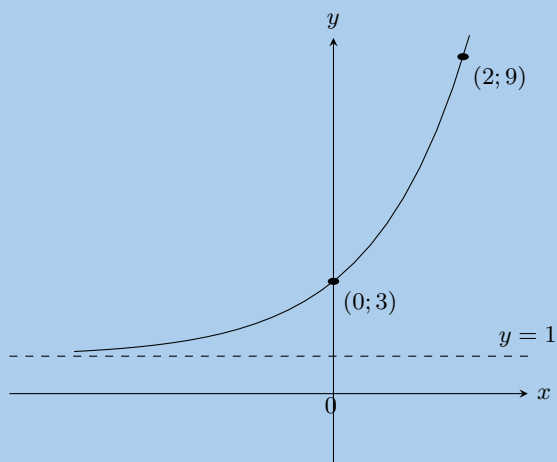
d)



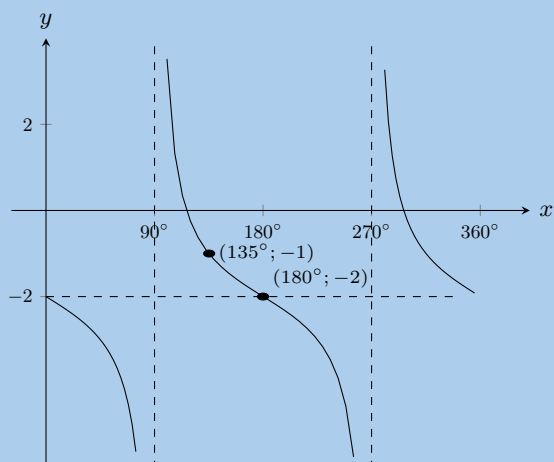
e)



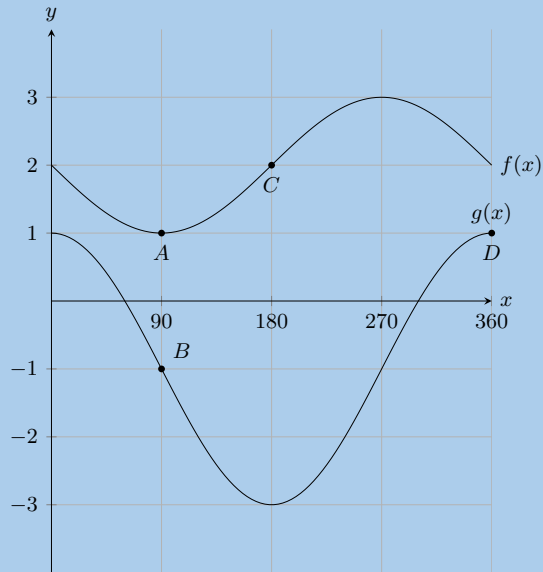
f)



g)

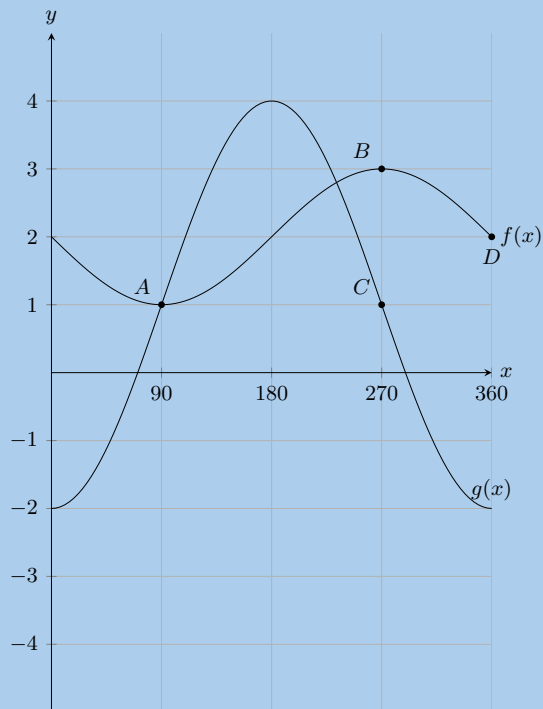


46.



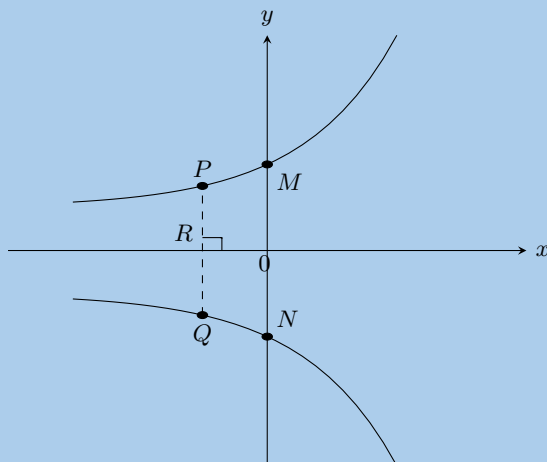
- State the coordinates at A , B , C and D .
- How many times in this interval does $f(x)$ intersect $g(x)$.
- What is the amplitude of $f(x)$.
- Evaluate: $f(180^\circ) - g(180^\circ)$.

47.

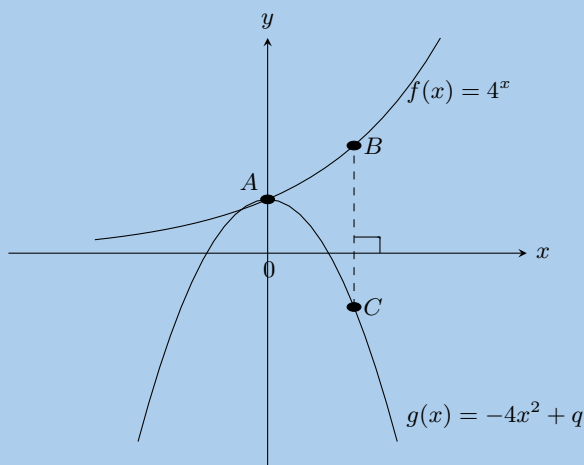


- State the coordinates at A , B , C and D .
- How many times in this interval does $f(x)$ intersect $g(x)$.
- What is the amplitude of $g(x)$.
- Evaluate: $g(180^\circ) - f(180^\circ)$.

48. $y = 2^x$ and $y = -2^x$ are sketched below. Answer the questions that follow.

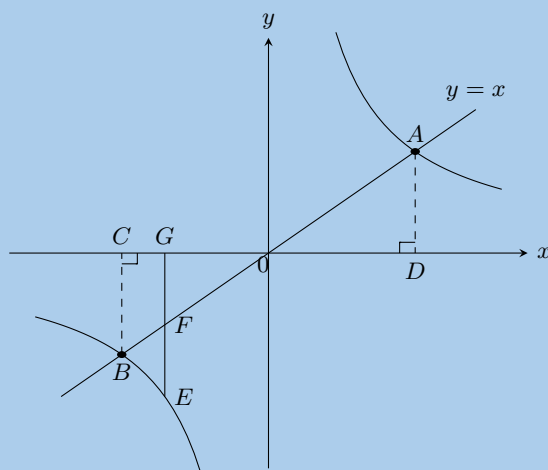


- Calculate the coordinates of M and N .
 - Calculate the length of MN .
 - Calculate the length of PQ if $OR = 1$ unit.
 - Give the equation of $y = 2^x$ reflected about the y -axis.
 - Give the range of both graphs.
49. Plot the following functions on the same set of axes and clearly label all points of intersection.
- $y = -2x^2 + 3$
 $y = 2x + 4$
 - $y = x^2 - 4$
 $y = 3x$
50. $f(x) = 4^x$ and $g(x) = -4x^2 + q$ are sketched below. The points $A(0; 1)$ and $B(1; 4)$ are given. Answer the questions that follow.



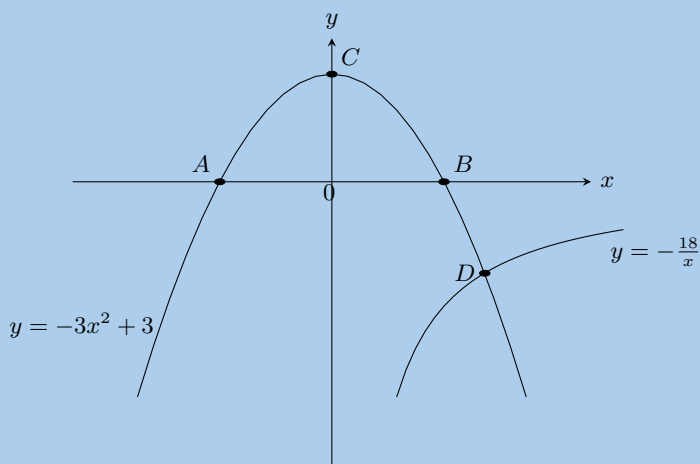
- Determine the value of q .
- Calculate the length of BC .
- Give the equation of $f(x)$ reflected about the x -axis.
- Give the equation of $f(x)$ shifted vertically upwards by 1 unit.
- Give the equation of the asymptote of $f(x)$.
- Give the ranges of $f(x)$ and $g(x)$.

51. Given $h(x) = x^2 - 4$ and $k(x) = -x^2 + 4$. Answer the questions that follow.
- Sketch both graphs on the same set of axes.
 - Describe the relationship between h and k .
 - Give the equation of $k(x)$ reflected about the line $y = 4$.
 - Give the domain and range of h .
52. Sketch the graphs of $f(\theta) = 2\sin \theta$ and $g(\theta) = \cos \theta - 1$ on the same set of axes. Use your sketch to determine:
- $f(180^\circ)$
 - $g(180^\circ)$
 - $g(270^\circ) - f(270^\circ)$
 - The domain and range of g .
 - The amplitude and period of f .
53. The graphs of $y = x$ and $y = \frac{8}{x}$ are shown in the following diagram.



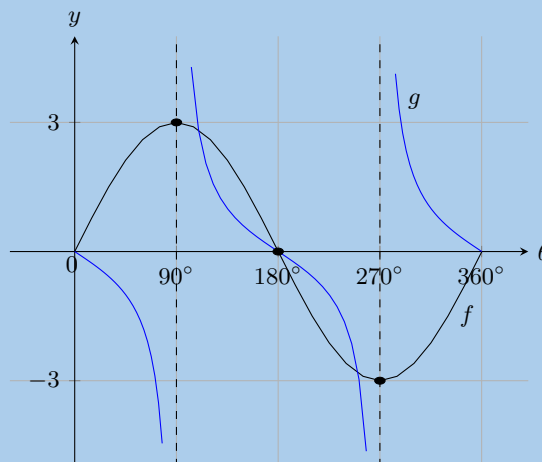
Calculate:

- The coordinates of points A and B .
 - The length of CD .
 - The length of AB .
 - The length of EF , given $G(-2; 0)$.
54. Given the diagram with $y = -3x^2 + 3$ and $y = -\frac{18}{x}$.



- Calculate the coordinates of A , B and C .
- Describe in words what happens at point D .
- Calculate the coordinates of D .
- Determine the equation of the straight line that would pass through points C and D .

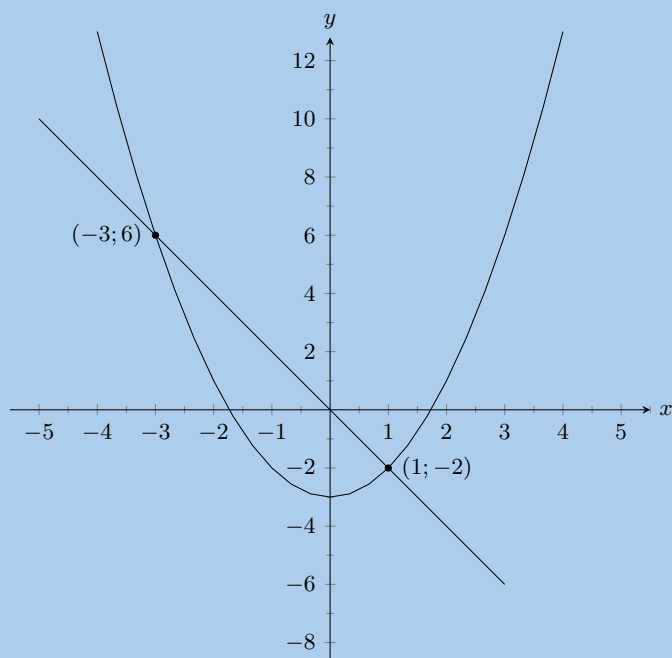
55. The diagram shows the graphs of $f(\theta) = 3 \sin \theta$ and $g(\theta) = -\tan \theta$



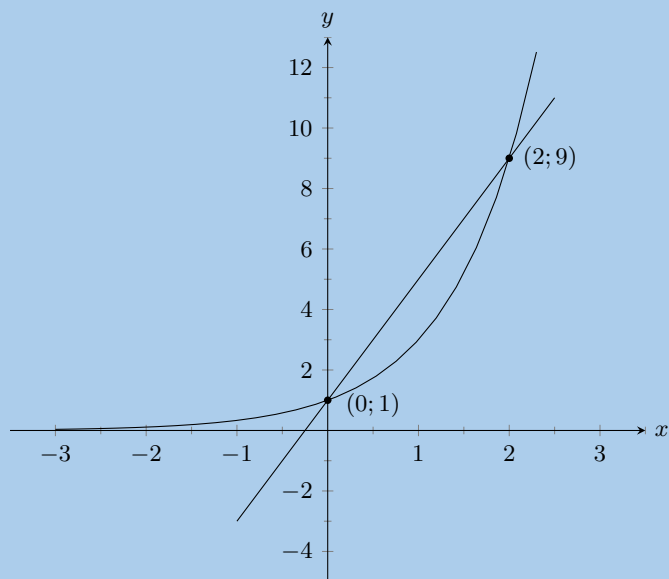
- Give the domain of g .
- What is the amplitude of f ?
- Determine for which values of θ :
 - $f(\theta) = 0 = g(\theta)$
 - $f(\theta) \times g(\theta) < 0$
 - $\frac{g(\theta)}{f(\theta)} > 0$
 - $f(\theta)$ is increasing

56. Determine the equations for the graphs given below.

a)



b)



57. Choose the correct answer:

- a) Which of the following does not have a gradient of 3?
- i. $y = 3x + 6$ ii. $3y = 9x - 1$ iii. $\frac{1}{3}(y - 1) = x$ iv. $\frac{1}{2}(y - 3) = 6x$
- b) The asymptote of $xy = 3 + x$ is:
- i. 3 ii. 1 iii. -3 iv. -1

58. Sketch the following

- a) $y = -1,5^x$
 b) $xy = 5 + 2x$
 c) $2y + 2x = 3$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 1a. 2G2M | 1b. 2G2N | 2a. 2G2P | 2b. 2G2Q | 3a. 2G2R | 3b. 2G2S |
| 4a. 2G2T | 4b. 2G2V | 5. 2G2W | 6. 2G2X | 7a. 2G2Y | 7b. 2G2Z |
| 7c. 2G32 | 8a. 2G33 | 8b. 2G34 | 9a. 2G35 | 9b. 2G36 | 9c. 2G37 |
| 10. 2G38 | 11. 2G39 | 12. 2G3B | 13. 2G3C | 14. 2G3D | 15. 2G3F |
| 16a. 2G3G | 16b. 2G3H | 16c. 2G3J | 17. 2G3K | 18. 2G3M | 19. 2G3N |
| 20. 2G3P | 21a. 2G3Q | 21b. 2G3R | 21c. 2G3S | 22a. 2G3T | 22b. 2G3V |
| 22c. 2G3W | 22d. 2G3X | 23. 2G3Y | 24a. 2G3Z | 24b. 2G42 | 24c. 2G43 |
| 25. 2G44 | 26. 2G45 | 27. 2G46 | 28a. 2G47 | 28b. 2G48 | 28c. 2G49 |
| 28d. 2G4B | 28e. 2G4C | 28f. 2G4D | 29. 2G4F | 30. 2G4G | 31. 2G4H |
| 32. 2G4J | 33. 2G4K | 34. 2G4M | 35. 2G4N | 36. 2G4P | 37. 2G4Q |
| 38. 2G4R | 39a. 2G4S | 39b. 2G4T | 39c. 2G4V | 39d. 2G4W | 39e. 2G4X |
| 40a. 2G4Y | 40b. 2G4Z | 41. 2G52 | 42. 2G53 | 43. 2G54 | 44. 2G55 |
| 45. 2G56 | 46. 2G57 | 47. 2G58 | 48. 2G59 | 49a. 2G5B | 49b. 2G5C |
| 50. 2G5D | 51. 2G5F | 52. 2G5G | 53. 2G5H | 54. 2G5J | 55. 2G5K |
| 56a. 2G5M | 56b. 2G5N | 57a. 2G5P | 57b. 2G5Q | 58a. 2G5R | 58b. 2G5S |
| 58c. 2G5T | | | | | |



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